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Method to use gene expression to determine likelihood of clinical outcome of renal cancer

Abstract

The present disclosure provides gene and gene sets, the expression of which is important in the classification and/or prognosis of cancer, in particular of renal cell carcinoma.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 16/952,566, filed Nov. 19, 2020, which is a continuation of U.S. application Ser. No. 15/368,979, filed Dec. 5, 2016, which is a continuation of U.S. application Ser. No. 12/987,023, filed Jan. 7, 2011, now U.S. Pat. No. 9,551,034, which claims priority benefit of U.S. provisional application Ser. No. 61/294,038, filed Jan. 11, 2010 and priority benefit of U.S. provisional application Ser. No. 61/346,230, filed May 19, 2010. All of the above applications are incorporated herein in their entireties.

SEQUENCE LISTING
(1) The present application contains a Sequence Listing which has been submitted electronically in XML format. Said XML copy, created on Aug. 10, 2023, is named “2023-08-10_ST26_GH1043USCON3.xml” and is 3,747,780 bytes in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure relates to molecular diagnostic assays that provide information concerning prognosis in renal cancer patients.
INTRODUCTION
(3) Each year in the United States there are approximately 51,000 cases of renal cell carcinoma (kidney cancer) and upper urinary tract cancer, resulting in more than 12,900 deaths. These tumors account for approximately 3% of adult malignancies. Renal cell carcinoma (RCC) represents about 3 percent of all cancers in the United States. Predictions for the United States for the year 2007 were that 40,000 new patients would be diagnosed with RCC and that 13,000 would die from this disease.
(4) The clinical outcome for a renal cell carcinoma patient depends largely on the aggressiveness of their particular cancer. Surgical resection is the most common treatment for this disease as systemic therapy has demonstrated only limited effectiveness. However, approximately 30% of patients with localized tumors will experience a relapse following surgery, and only 40% of all patients with renal cell carcinoma survive for 5 years.
(5) In the US, the number of adjuvant treatment decisions that will be made by patients with early stage renal cell carcinoma in 2005 exceeded 25,000. The rates in the European Union are expected to be similar. Physicians

require prognostic information to help them make informed treatment decisions for patients with renal cell carcinoma and recruit appropriate high-risk patients for clinical trials. Surgeons must decide how much kidney and surrounding tissue to remove based, in part, on predicting the aggressiveness of a particular tumor. Today, cancer tumors are generally classified based on clinical and pathological features, such as stage, grade, and the presence of necrosis. These designations are made by applying standardized criteria, the subjectivity of which has been demonstrated by a lack of concordance amongst pathology laboratories.

SUMMARY

(6) The present disclosure provides biomarkers, the expression of which has prognostic value in renal cancer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1***a*-**1***c*: Consistency between Stage I and III for exemplary genes associated with RFI

(2) FIG. **2**: Consistent results across endpoints (OS and RFI) for exemplary genes

(3) FIG. **3**: Kaplan-Meier curve: Recurrence Free Internal (RFI) by Cleveland Clinic Foundation (CCF) histologic necrosis

(4) FIG. **4**: Performance of Mayo prognostic tool applied to CCF data

(5) FIG. **5**: Example of using one gene to improve estimate: EMCN in addition to Mayo Criteria

(6) FIG. **6**: Example of using one gene to improve estimate: AQP1 in addition to Mayo Criteria

(7) FIG. **7**: Example of using one gene to improve estimate: PPAP2B in addition to Mayo Criteria

DETAILED DESCRIPTION

(8) Definitions

(9) Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Singleton et al., Dictionary of Microbiology and Molecular Biology 2nd ed., J. Wiley & Sons (New York, NY 1994), and March, Advanced Organic Chemistry Reactions, Mechanisms and Structure 4th ed., John Wiley & Sons (New York, NY 1992), provide one skilled in the art with a general guide to many of the terms used in the present application.

(10) One skilled in the art will recognize many methods and materials similar or equivalent to those described herein, which could be used in the practice of the present invention. Indeed, the present invention is in no way limited to the methods and materials described. For purposes of the present invention, the following terms are defined below.

(11) The term “tumor” is used herein to refer to all neoplastic cell growth and proliferation, and all pre-cancerous and cancerous cells and tissues. The term “primary tumor” is used herein to refer to a tumor that is at the original site where it first arose. For example, a primary renal cell carcinoma tumor is one that arose in the kidney. The term “metastatic tumor” is used herein to refer to a tumor that develops away from the site of origin. For example, renal cell carcinoma metastasis most commonly affects the spine, ribs, pelvis, and proximal long bones.

(12) The terms “cancer” and “carcinoma” refer to or describe the physiological condition in mammals that is typically characterized by unregulated cell growth. The pathology of cancer includes, for example, abnormal or uncontrollable cell growth, metastasis, interference with the normal functioning of neighboring cells, release of cytokines or other secretory products at abnormal levels, suppression or aggravation of inflammatory or immunological response, neoplasia, premalignancy, malignancy, invasion of surrounding or distant tissues or organs, such as lymph nodes, blood vessels, etc.

(13) As used herein, the terms “renal cancer” or “renal cell carcinoma” refer to cancer that has arisen from the kidney.

(14) The terms “renal cell cancer” or “renal cell carcinoma” (RCC), as used herein, refer to cancer which originates in the lining of the proximal convoluted tubule. More specifically, RCC encompasses several relatively common histologic subtypes: clear cell renal cell carcinoma, papillary (chromophil), chromophobe, collecting duct carcinoma, and medullary carcinoma. Further information about renal cell carcinoma may be found in Y. Thyavihally, et al., *Int Semin Surg Oncol* 2:18 (2005), the contents of which are incorporated by reference herein. Clear cell renal cell carcinoma (ccRCC) is the most common subtype of RCC. Incidence of ccRCC is increasing, comprising 80% of localized disease and more than 90% of metastatic disease.

(15) The staging system for renal cell carcinoma is based on the degree of tumor spread beyond the kidney. According to the tumor, node, metastasis (TNM) staging system of the American Joint Committee on Cancer (AJCC) (Greene, et al., AJCC Cancer Staging Manual, pp. 323-325 (6.sup.th Ed. 2002), the various stages of renal cell carcinoma are provided below. “Increased stage” as used herein refers to classification of a tumor at a

stage that is more advanced, e.g., Stage 4 is an increased stage relative to Stages 1, 2, and 3.
(16) Description of RCC Stages
(17) Stages for Renal Cell Carcinoma
(18) Stage 1: T1, N0, M0 Stage 2: T2, N0, M0 Stage 3: T1-T2, N1, M0; T3, N0-1, M0; T3a, N0-1, M0; T3b, N0-1, M0; and T3c, N0-1, M0 Stage 4: T4, N0-1, M0; Any T, N2, M0; and Any T, any N, M1
Primary Tumor (T) TX: Primary tumor cannot be assessed T0: No evidence of primary tumor T1: Tumor 7 cm or less in greatest dimension and limited to the kidney T1a: Tumor 4 cm or less in greatest dimension and limited to the kidney T1b: Tumor larger than 4 cm but 7 cm or less in greatest dimension and limited to the kidney T2: Tumor larger than 7 cm in greatest dimension and limited to the kidney T3: Tumor extends into major veins or invades adrenal gland or perinephric tissues but not beyond Gerota fascia T3a: Tumor directly invades adrenal gland or perirenal and/or renal sinus fat but not beyond Gerota fascia T3b: Tumor grossly extends into the renal vein or its segmental (i.e., muscle-containing) branches, or it extends into the vena cava below the diaphragm T3c: Tumor grossly extends into the vena cava above the diaphragm or invades the wall of the vena cava T4: Tumor invades beyond Gerota fascia
Rekional Lymph Nodes (N) NX: Regional lymph nodes cannot be assessed N0: No regional lymph node metastasis N1: Metastasis in a single regional lymph node N2: Metastasis in more than one regional lymph node
Distant Metastasis (M) MX: Distant metastasis cannot be assessed M0: No distant metastasis M1: Distant metastasis
(19) The term “early stage renal cancer”, as used herein, refers to Stages 1-3, as defined in the American Joint Committee on Cancer (AJCC) Cancer Staging Manual, pp. 323-325 (6.sup.th Ed. 2002).
(20) Reference to tumor “grade” for renal cell carcinoma as used herein refers to a grading system based on microscopic appearance of tumor cells. According to the TNM staging system of the AJCC, the various grades of renal cell carcinoma are: GX (grade of differentiation cannot be assessed); G1 (well differentiated); G2 (moderately differentiated); and G3-G4 (poorly differentiated/undifferentiated).
“Increased grade” as used herein refers to classification of a tumor at a grade that is more advanced, e.g., Grade 4 (G4) 4 is an increased grade relative to Grades 1, 2, and 3. Tumor grading is an important prognostic factor in renal cell carcinoma. H. Rauschmeier, et al., World J Urol 2:103-108 (1984).
(21) The terms “necrosis” or “histologic necrosis” as used herein refer to the death of living cells or tissues. The presence of necrosis may be a prognostic factor in cancer. For example, necrosis is commonly seen in renal cell carcinoma (RCC) and has been shown to be an adverse prognostic factor in certain RCC subtypes. V. Foria, et al., J Clin Pathol 58(1):39-43 (2005).
(22) The terms “nodal invasion” or “node-positive (N+)” as used herein refer to the presence of cancer cells in one or more lymph nodes associated with the organ (e.g., drain the organ) containing a primary tumor. Nodal invasion is part of tumor staging for most cancers, including renal cell carcinoma.
(23) The term “prognostic gene,” when used in the single or plural, refers to a gene, the expression level of which is correlated with a good or bad prognosis for a cancer patient. A gene may be both a prognostic and predictive gene, depending on the association of the gene expression level with the corresponding endpoint.
(24) The terms “correlated” and “associated” are used interchangeably herein to refer to the strength of association between two measurements (or measured entities). The disclosure provides genes and gene subsets, the expression levels of which are associated with a particular outcome measure, such as between the expression level of a gene and the likelihood of a recurrence event or relapse. For example, the increased expression level of a gene may be positively correlated (positively associated) with an increased likelihood of good clinical outcome for the patient, such as a decreased likelihood of recurrence of cancer. Such a positive correlation may be demonstrated statistically in various ways, e.g. by a hazard ratio less than 1.0. In another example, the increased expression level of a gene may be negatively correlated (negatively associated) with an increased likelihood of good clinical outcome for the patient. In that case, for example, a patient with a high expression level of a gene may have an increased likelihood of recurrence of the cancer. Such a negative correlation could indicate that the patient with a high expression level of a gene likely has a poor prognosis, or might respond poorly to a chemotherapy, and this may be demonstrated statistically in various ways, e.g., a hazard ratio greater than 1.0.
(25) “Co-expression” is used herein to refer to strength of association between the expression levels of two different genes that are biologically similar, such that expression level of a first gene may be substituted with an expression level of a second gene in a given analysis in view of their correlation of expression. Such co-expressed genes (or correlated expression) indicates that these two genes are substitutable in an expression algorithm, for example, if a first gene is highly correlated, positively or negatively, with increased likelihood of a good clinical outcome for renal cell carcinoma, then the second co-expressed gene also correlates, in the same

direction as the first gene, with the same outcome. Pairwise co-expression may be calculated by various methods known in the art, e.g., by calculating Pearson correlation coefficients or Spearman correlation coefficients or by clustering methods. The methods described herein may incorporate one or more genes that co-express, with a Pearson correlation co-efficient of at least 0.5. Co-expressed gene cliques may also be identified using graph theory. An analysis of co-expression may be calculated using normalized or standardized and normalized expression data.

(26) The terms “prognosis” and “clinical outcome” are used interchangeably herein to refer to an estimate of the likelihood of cancer-attributable death or progression, including recurrence, and metastatic spread of a neoplastic disease, such as renal cell carcinoma. The terms “good prognosis” or “positive clinical outcome” mean a desired clinical outcome. For example, in the context of renal cell carcinoma, a good prognosis may be an expectation of no local recurrences or metastasis within two, three, four, five or more years of the initial diagnosis of renal cell carcinoma. The terms “poor prognosis” or “negative clinical outcome” are used herein interchangeably to mean an undesired clinical outcome. For example, in the context of renal cell carcinoma, a poor prognosis may be an expectation of a local recurrence or metastasis within two, three, four, five or more years of the initial diagnosis of renal cell carcinoma.

(27) The term “predictive gene” is used herein to refer to a gene, the expression of which is correlated, positively or negatively, with likelihood of beneficial response to treatment.

(28) A “clinical outcome” can be assessed using any endpoint, including, without limitation, (1) aggressiveness of tumor growth (e.g., movement to higher stage); (2) metastasis; (3) local recurrence; (4) increase in the length of survival following treatment; and/or (5) decreased mortality at a given point of time following treatment. Clinical response may also be expressed in terms of various measures of clinical outcome. Clinical outcome can also be considered in the context of an individual's outcome relative to an outcome of a population of patients having a comparable clinical diagnosis, and can be assessed using various endpoints such as an increase in the duration of Recurrence-Free interval (RFI), an increase in the duration of Overall Survival (OS) in a population, an increase in the duration of Disease-Free Survival (DFS), an increase in the duration of Distant Recurrence-Free Interval (DRFI), and the like.

(29) The term “treatment”, as used herein, refers to therapeutic compounds administered to patients to cease or reduce proliferation of cancer cells, shrink the tumor, avoid progression and metastasis, or cause primary tumor or metastases regression. Examples of treatment include, for example, cytokine therapy, progestational agents, anti-angiogenic therapy, hormonal therapy, and chemotherapy (including small molecules and biologics).

(30) The terms “surgery” or “surgical resection” are used herein to refer to surgical removal of some or all of a tumor, and usually some of the surrounding tissue. Examples of surgical techniques include laproscopic procedures, biopsy, or tumor ablation, such as cryotherapy, radio frequency ablation, and high intensity ultrasound. In cancer patients, the extent of tissue removed during surgery depends on the state of the tumor as observed by a surgeon. For example, a partial nephrectomy indicates that part of one kidney is removed; a simple nephrectomy entails removal of all of one kidney; a radical nephrectomy, all of one kidney and neighboring tissue (e.g., adrenal gland, lymph nodes) removed; and bilateral nephrectomy, both kidneys removed.

(31) The terms “recurrence” and “relapse” are used herein, in the context of potential clinical outcomes of cancer, to refer to a local or distant metastases. Identification of a recurrence could be done by, for example, CT imaging, ultrasound, arteriogram, or X-ray, biopsy, urine or blood test, physical exam, or research center tumor registry.

(32) The term “recurrence-free interval” as used herein refers to the time from surgery to first recurrence or death due to recurrence of renal cancer. Losses to follow-up, second primary cancers, other primary cancers, and deaths prior to recurrence are considered censoring events.

(33) The term “overall survival” is defined as the time from surgery to death from any cause. Losses to follow-up are considered censoring events. Recurrences are ignored for the purposes of calculating overall survival (OS).

(34) The term “disease-free survival” is defined as the time from surgery to first recurrence or death from any cause, whichever occurs first. Losses to follow-up are considered censoring events.

(35) The term “Hazard Ratio (HR)” as used herein refers to the effect of an explanatory variable on the hazard or risk of an event (i.e. recurrence or death). In proportional hazards regression models, the HR is the ratio of the predicted hazard for two groups (e.g. patients with or without necrosis) or for a unit change in a continuous variable (e.g. one standard deviation change in gene expression).

(36) The term “Odds Ratio (OR)” as used herein refers to the effect of an explanatory variable on the odds of an event (e.g. presence of necrosis). In logistic regression models, the OR is the ratio of the predicted odds for a

unit change in a continuous variable (e.g. one standard deviation change in gene expression).
(37) The term "prognostic clinical and/or pathologic covariate" as used herein refers to clinical and/or prognostic covariates that are significantly associated ($p \leq 0.05$) with clinical outcome. For example, prognostic clinical and pathologic covariates in renal cell carcinoma include tumor stage (e.g. size, nodal invasion, etc.), and grade (e.g., Fuhrman grade), histologic necrosis, and gender.
(38) The term "proxy gene" refers to a gene, the expression of which is correlated (positively or negatively) with one or more prognostic clinical and/or pathologic covariates. The expression level(s) of one or more proxy genes may be used instead of, or in addition to, classification of a tumor by physical or mechanical examination in a pathology laboratory.
(39) The term "microarray" refers to an ordered arrangement of hybridizable array elements, preferably polynucleotide probes, on a substrate.
(40) The term "polynucleotide," when used in singular or plural, generally refers to any polyribonucleotide or polydeoxribonucleotide, which may be unmodified RNA or DNA or modified RNA or DNA. Thus, for instance, polynucleotides as defined herein include, without limitation, single- and double-stranded DNA, DNA including single- and double-stranded regions, single- and double-stranded RNA, and RNA including single- and double-stranded regions, hybrid molecules comprising DNA and RNA that may be single-stranded or, more typically, double-stranded or include single- and double-stranded regions. In addition, the term "polynucleotide" as used herein refers to triple-stranded regions comprising RNA or DNA or both RNA and DNA. The strands in such regions may be from the same molecule or from different molecules. The regions may include all of one or more of the molecules, but more typically involve only a region of some of the molecules. One of the molecules of a triple-helical region often is an oligonucleotide. The term "polynucleotide" specifically includes DNAs (e.g., cDNAs) and RNAs that contain one or more modified bases. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are "polynucleotides" as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritiated bases, are included within the term "polynucleotides" as defined herein. In general, the term "polynucleotide" embraces all chemically, enzymatically and/or metabolically modified forms of unmodified polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including simple and complex cells.
(41) The term "oligonucleotide" refers to a relatively short polynucleotide, including, without limitation, single-stranded deoxyribonucleotides, single- or double-stranded ribonucleotides, RNA:DNA hybrids and double-stranded DNAs. Oligonucleotides, such as single-stranded DNA probe oligonucleotides, are often synthesized by chemical methods, for example using automated oligonucleotide synthesizers that are commercially available. However, oligonucleotides can be made by a variety of other methods, including in vitro recombinant DNA-mediated techniques and by expression of DNAs in cells and organisms.
(42) The term "expression level" as applied to a gene refers to the normalized level of a gene product.
(43) The terms "gene product" or "expression product" are used herein interchangeably to refer to the RNA transcription products (RNA transcript) of a gene, including mRNA, and the polypeptide translation product of such RNA transcripts. A gene product can be, for example, an unspliced RNA, an mRNA, a splice variant mRNA, a microRNA, a fragmented RNA, a polypeptide, a post-translationally modified polypeptide, a splice variant polypeptide, etc.
(44) "Stringency" of hybridization reactions is readily determinable by one of ordinary skill in the art, and generally is an empirical calculation dependent upon probe length, washing temperature, and salt concentration. In general, longer probes require higher temperatures for proper annealing, while shorter probes need lower temperatures. Hybridization generally depends on the ability of denatured DNA to re-anneal when complementary strands are present in an environment below their melting temperature. The higher the degree of desired homology between the probe and hybridizable sequence, the higher the relative temperature that can be used. As a result, it follows that higher relative temperatures would tend to make the reaction conditions more stringent, while lower temperatures less so. For additional details and explanation of stringency of hybridization reactions, see Ausubel et al., *Current Protocols in Molecular Biology*, (Wiley Interscience, 1995).
(45) "Stringent conditions" or "high stringency conditions", as defined herein, typically: (1) employ low ionic strength solutions and high temperature for washing, for example 0.015 M sodium chloride/0.0015 M sodium citrate/0.1% sodium dodecyl sulfate at 50° C.; (2) employ during hybridization a denaturing agent, such as formamide, for example, 50% (v/v) formamide with 0.1% bovine serum albumin/0.1% Ficoll/0.1% polyvinylpyrrolidone/50 mM sodium phosphate buffer at pH 6.5 with 750 mM sodium chloride, 75 mM sodium citrate at 42° C.; or (3) employ 50% formamide, 5×SSC (0.75 M NaCl, 0.075 M sodium citrate), 50 mM sodium phosphate (pH 6.8), 0.1% sodium pyrophosphate, 5×Denhardt's solution, sonicated salmon sperm DNA (50 μg/ml), 0.1% SDS, and 10% dextran sulfate at 42° C., with washes at 42° C. in 0.2×SSC (sodium

chloride/sodium citrate) and 50% formamide at 55° C., followed by a high-stringency wash consisting of 0.1×SSC containing EDTA at 55° C.
(46) "Moderately stringent conditions" may be identified as described by Sambrook et al., *Molecular Cloning: A Laboratory Manual* (Cold Spring Harbor Press, 1989), and include the use of washing solution and hybridization conditions (e.g., temperature, ionic strength and % SDS) less stringent that those described above. An example of moderately stringent condition is overnight incubation at 37° C. in a solution comprising: 20% formamide, 5×SSC (150 mM NaCl, 15 mM trisodium citrate), 50 mM sodium phosphate (pH 7.6), 5×Denhardt's solution, 10% dextran sulfate, and 20 mg/ml denatured sheared salmon sperm DNA, followed by washing the filters in 1×SSC at about 37-50° C. The skilled artisan will recognize how to adjust the temperature, ionic strength, etc. as necessary to accommodate factors such as probe length and the like.
(47) In the context of the present invention, reference to "at least one," "at least two," "at least five," etc. of the genes listed in any particular gene set means any one or any and all combinations of the genes listed.
(48) The terms "splicing" and "RNA splicing" are used interchangeably and refer to RNA processing that removes introns and joins exons to produce mature mRNA with continuous coding sequence that moves into the cytoplasm of an eukaryotic cell.
(49) In theory, the term "exon" refers to any segment of an interrupted gene that is represented in a mature RNA product (B. Lewin, *Genes IV*(Cell Press, 1990)). In theory the term "intron" refers to any segment of DNA that is transcribed but removed from within the transcript by splicing together the exons on either side of it. Operationally, exon sequences occur in the mRNA sequence of a gene as defined by Ref. SEQ ID numbers. Operationally, intron sequences are the intervening sequences within the genomic DNA of a gene, bracketed by exon sequences and usually having GT and AG splice consensus sequences at their 5′ and 3′ boundaries.
(50) A "computer-based system" refers to a system of hardware, software, and data storage medium used to analyze information. The minimum hardware of a patient computer-based system comprises a central processing unit (CPU), and hardware for data input, data output (e.g., display), and data storage. An ordinarily skilled artisan can readily appreciate that any currently available computer-based systems and/or components thereof are suitable for use in connection with the methods of the present disclosure. The data storage medium may comprise any manufacture comprising a recording of the present information as described above, or a memory access device that can access such a manufacture.
(51) To "record" data, programming or other information on a computer readable medium refers to a process for storing information, using any such methods as known in the art. Any convenient data storage structure may be chosen, based on the means used to access the stored information. A variety of data processor programs and formats can be used for storage, e.g. word processing text file, database format, etc.
(52) A "processor" or "computing means" references any hardware and/or software combination that will perform the functions required of it. For example, a suitable processor may be a programmable digital microprocessor such as available in the form of an electronic controller, mainframe, server or personal computer (desktop or portable). Where the processor is programmable, suitable programming can be communicated from a remote location to the processor, or previously saved in a computer program product (such as a portable or fixed computer readable storage medium, whether magnetic, optical or solid state device based). For example, a magnetic medium or optical disk may carry the programming, and can be read by a suitable reader communicating with each processor at its corresponding station.
(53) The present disclosure provides methods for assessing a patient's risk of recurrence of cancer, which methods comprise assaying an expression level of at least one gene, or its gene product, in a biological sample obtained from the patient. In some embodiments, the biological sample can be a tumor sample obtained from the kidney, or surrounding tissues, of the patient. In other embodiments, the biological sample is obtained from a bodily fluid, such as blood or urine.
(54) The present disclosure provides genes useful in the methods disclosed herein. The genes are listed in Tables 3a and 3b, wherein increased expression of genes listed in Table 3a is significantly associated with a lower risk of cancer recurrence, and increased expression of genes listed in Table 3b is significantly associated with a higher risk of cancer recurrence. In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a gene listed in Tables 3a or 3b with which it co-expresses.
(55) The present disclosure further provides genes significantly associated, positively or negatively, with renal cancer recurrence after adjusting for clinical/pathologic covariates (stage, tumor grade, tumor size, nodal status, and presence of necrosis). For example, Table 8a lists genes wherein increased expression is significantly associated with lower risk of renal cancer recurrence after adjusting for clinical/pathologic covariates, and Table 8b lists genes wherein increased expression is significantly associated with a higher risk of renal cancer recurrence after adjusting for clinical/pathologic covariates. Of those genes listed in Tables 8a and 8b, 16 genes

with significant association, positively or negatively, with good prognosis after adjusting for clinical/pathologic covariates and controlling the false discovery rate at 10% are listed in Table 9. One or more of these genes may be used separately, or in addition to, at least one of the genes listed in Tables 3a and 3b, to provide information concerning a patient's prognosis.

(56) The present disclosure also provides proxy genes that are useful for assessing the status of clinical and/or pathologic covariates for a cancer patient. Proxy genes for tumor stage are listed in Tables 4a and 4b, wherein increased expression of genes listed in Table 4a is significantly associated with higher tumor stage, and increased expression of genes listed in Table 4b is significantly associated with lower tumor stage. Proxy genes for tumor grade are listed in Tables 5a and 5b wherein increased expression of genes listed in Table 5a is significantly associated with higher tumor grade, and increased expression of genes listed in Table 5b is significantly associated with lower tumor grade. Proxy genes for the presence of necrosis are listed in Tables 6a and 6b, wherein expression of genes listed in Table 6a is significantly associated with the presence of necrosis, and increased expression of genes listed in Table 6b is significantly associated with the absence of necrosis. Proxy genes for nodal involvement are listed in Tables 7a and 7b wherein higher expression of genes listed in Table 7a are significantly associated with the presence of nodal invasion, and increased expression of genes listed in Table 7b are significantly associated with the absence of nodal invasion. One or more proxy genes may be used separately, or in addition to, at least one of the genes listed in Tables 3a and 3b, to provide information concerning a patient's prognosis. In some embodiments, at least two of the following proxy genes are used to provide information concerning the patient's prognosis: TSPAN7, TEK, LDB2, TIMP3, SHANK3, RGS5, KDR, SDPR, EPAS1, ID1, TGFBR2, FLT4, SDPR, ENDRB, JAG1, DLC1, and KL. In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a proxy gene with which it co-expresses.

(57) The present disclosure also provides sets of genes in biological pathways that are useful for assessing the likelihood that a cancer patient is likely to have a positive clinical outcome, which sets of genes are referred to herein as “gene subsets”. The gene subsets include angiogenesis, immune response, transport, cell adhesion/extracellular matrix, cell cycle, and apoptosis. In some embodiments, the angiogenesis gene subset includes ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; the immune response gene subset includes CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1, and; the transport gene subset includes AQP1 and SGK1; the cell adhesion/extracellular matrix gene subset includes ITGB1, A2M, ITGB5, LAMB1, LOX, MMP14, TGFBR2, TIMP3, and TSPAN7; the cell cycle gene subset includes BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; the apoptosis gene subset includes CASP10; the early response gene subset includes EGR1 and CYR61; the metabolic signaling gene subset includes CA12, ENO2, UGCG, and SDPR; and the signaling gene subset includes ID1 and MAP2K3.

(58) The present disclosure also provides genes in biological pathways targeted by chemotherapy that are correlated, positively or negatively, to a risk of cancer recurrence. These genes include KIT, PDGFA, PDGFB, PDGFC, PDGFD, PDGFRb, KRAS, RAF1, MTOR, HIF1AN, VEGFA, VEGFB, and FLT4. In some embodiments, the chemotherapy is cytokine and/or anti-angiogenic therapy. In other embodiments, the chemotherapy is sunitinib, sorafenib, temsirolimus, bevacizumab, everolimus, and/or pazopanib.

(59) In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a gene with which it co-expresses.

(60) In some embodiments, the cancer is renal cell carcinoma. In other embodiments, the cancer is clear cell renal cell carcinoma (ccRCC), papillary, chromophobe, collecting duct carcinoma, and/or medullary carcinoma.

(61) Various technological approaches for determination of expression levels of the disclosed genes are set forth in this specification, including, without limitation, reverse-transciption polymerase chain reaction (RT-PCR), microarrays, high-throughput sequencing, serial analysis of gene expression (SAGE), and Digital Gene Expression (DGE), which will be discussed in detail below. In particular aspects, the expression level of each gene may be determined in relation to various features of the expression products of the gene, including exons, introns, protein epitopes, and protein activity.

(62) The expression levels of genes identified herein may be measured in tumor tissue. For example, the tumor tissue may be obtained upon surgical resection of the tumor, or by tumor biopsy. The expression level of the identified genes may also be measured in tumor cells recovered from sites distant from the tumor, including circulating tumor cells or body fluid (e.g., urine, blood, blood fraction, etc.).

(63) The expression product that is assayed can be, for example, RNA or a polypeptide. The expression product may be fragmented. For example, the assay may use primers that are complementary to target sequences of an expression product and could thus measure full transcripts as well as those fragmented expression products

containing the target sequence. Further information is provided in Tables A and B, which provide examples of sequences of forward primers, reverse primers, probes and amplicons generated by use of the primers.
(64) The RNA expression product may be assayed directly or by detection of a cDNA product resulting from a PCR-based amplification method, e.g., quantitative reverse transcription polymerase chain reaction (qRT-PCR). (See e.g., U.S. Pub. No. US2006-0008809A1.) Polypeptide expression product may be assayed using immunohistochemistry (IHC). Further, both RNA and polypeptide expression products may also be is assayed using microarrays.
(65) Clinical Utility
(66) Currently, of the expected clinical outcome for RCC patients is based on subjective determinations of a tumor's clinical and pathologic features. For example, physicians make decisions about the appropriate surgical procedures and adjuvant therapy based on a renal tumor's stage, grade, and the presence of necrosis. Although there are standardized measures to guide pathologists in making these decisions, the level of concordance between pathology laboratories is low. (See Al-Ayanti M et al. (2003) Arch Pathol Lab Med 127, 593-596) It would be useful to have a reproducible molecular assay for determining and/or confirming these tumor characteristics.
(67) In addition, standard clinical criteria, by themselves, have limited ability to accurately estimate a patient's prognosis. It would be useful to have a reproducible molecular assay to assess a patient's prognosis based on the biology of his or her tumor. Such information could be used for the purposes of patient counseling, selecting patients for clinical trials (e.g., adjuvant trials), and understanding the biology of renal cell carcinoma. In addition, such a test would assist physicians in making surgical and treatment recommendations based on the biology of each patient's tumor. For example, a genomic test could stratify RCC patients based on risk of recurrence and/or likelihood of long-term survival without recurrence (relapse, metastasis, etc.). There are several ongoing and planned clinical trials for RCC therapies, including adjuvant radiation and chemotherapies. It would be useful to have a genomic test able to identify high-risk patients more accurately than standard clinical criteria, thereby further enriching an adjuvant RCC population for study. This would reduce the number of patients needed for an adjuvant trial and the time needed for definitive testing of these new agents in the adjuvant setting.
(68) Finally, it would be useful to have a molecular assay that could predict a patient's likelihood to respond to treatment, such as chemotherapy. Again, this would facilitate individual treatment decisions and recruiting patients for clinical trials, and increase physician and patient confidence in making healthcare decisions after being diagnosed with cancer.
(69) Reporting Results
(70) The methods of the present disclosure are suited for the preparation of reports summarizing the expected clinical outcome resulting from the methods of the present disclosure. A “report,” as described herein, is an electronic or tangible document that includes report elements that provide information of interest relating to a likelihood assessment and its results. A subject report includes at least a likelihood assessment, e.g., an indication as to the risk of recurrence for a subject with renal cell carcinoma. A subject report can be completely or partially electronically generated, e.g., presented on an electronic display (e.g., computer monitor). A report can further include one or more of: 1) information regarding the testing facility; 2) service provider information; 3) patient data; 4) sample data; 5) an interpretive report, which can include various information including: a) indication; b) test data, where test data can include a normalized level of one or more genes of interest, and 6) other features.
(71) The present disclosure thus provides for methods of creating reports and the reports resulting therefrom. The report may include a summary of the expression levels of the RNA transcripts, or the expression products of such RNA transcripts, for certain genes in the cells obtained from the patient's tumor. The report can include information relating to prognostic covariates of the patient. The report may include an estimate that the patient has an increased risk of recurrence. That estimate may be in the form of a score or patient stratifier scheme (e.g., low, intermediate, or high risk of recurrence). The report may include information relevant to assist with decisions about the appropriate surgery (e.g., partial or total nephrectomy) or treatment for the patient.
(72) Thus, in some embodiments, the methods of the present disclosure further include generating a report that includes information regarding the patient's likely clinical outcome, e.g. risk of recurrence. For example, the methods disclosed herein can further include a step of generating or outputting a report providing the results of a subject risk assessment, which report can be provided in the form of an electronic medium (e.g., an electronic display on a computer monitor), or in the form of a tangible medium (e.g., a report printed on paper or other tangible medium).
(73) A report that includes information regarding the patient's likely prognosis (e.g., the likelihood that a patient

having renal cell carcinoma will have a good prognosis or positive clinical outcome in response to surgery and/or treatment) is provided to a user. An assessment as to the likelihood is referred to below as a "risk report" or, simply, "risk score." A person or entity that prepares a report ("report generator") may also perform the likelihood assessment. The report generator may also perform one or more of sample gathering, sample processing, and data generation, e.g., the report generator may also perform one or more of: a) sample gathering; b) sample processing; c) measuring a level of a risk gene; d) measuring a level of a reference gene; and e) determining a normalized level of a risk gene. Alternatively, an entity other than the report generator can perform one or more sample gathering, sample processing, and data generation.
(74) For clarity, it should be noted that the term "user," which is used interchangeably with "client," is meant to refer to a person or entity to whom a report is transmitted, and may be the same person or entity who does one or more of the following: a) collects a sample; b) processes a sample; c) provides a sample or a processed sample; and d) generates data (e.g., level of a risk gene; level of a reference gene product(s); normalized level of a risk gene for use in the likelihood assessment. In some cases, the person(s) or entity(ies) who provides sample collection and/or sample processing and/or data generation, and the person who receives the results and/or report may be different persons, but are both referred to as "users" or "clients" herein to avoid confusion. In certain embodiments, e.g., where the methods are completely executed on a single computer, the user or client provides for data input and review of data output. A "user" can be a health professional (e.g., a clinician, a laboratory technician, a physician (e.g., an oncologist, surgeon, pathologist), etc.).
(75) In embodiments where the user only executes a portion of the method, the individual who, after computerized data processing according to the methods of the present disclosure, reviews data output (e.g., results prior to release to provide a complete report, a complete, or reviews an "incomplete" report and provides for manual intervention and completion of an interpretive report) is referred to herein as a "reviewer." The reviewer may be located at a location remote to the user (e.g., at a service provided separate from a healthcare facility where a user may be located).
(76) Where government regulations or other restrictions apply (e.g., requirements by health, malpractice, or liability insurance), all results, whether generated wholly or partially electronically, are subjected to a quality control routine prior to release to the user.
(77) Methods of Assaying Expression Levels of a Gene Product
(78) Numerous assay methods for measuring an expression level of a gene product are known in the art, including assay methods for measuring an expression level of a nucleic acid gene product (e.g., an mRNA), and assay methods for measuring an expression level of a polypeptide gene product.
(79) Measuring a Level of a Nucleic Acid Gene Product
(80) In general, methods of measuring a level of a nucleic acid gene product (e.g., an mRNA) include methods involving hybridization analysis of polynucleotides, and methods involving amplification of polynucleotides. Commonly used methods known in the art for the quantification of mRNA expression in a sample include northern blotting and in situ hybridization (See for example, Parker & Barnes, *Methods in Molecular Biology* 106:247-283 (1999)); RNAse protection assays (Hod, *Biotechniques* 13:852-854 (1992)); and reverse transcription polymerase chain reaction (RT-PCR) (Weis et al., *Trends in Genetics* 8:263-264 (1992)). Alternatively, antibodies may be employed that can recognize specific duplexes, including DNA duplexes, RNA duplexes, and DNA-RNA hybrid duplexes or DNA-protein duplexes. Representative methods for sequencing-based gene expression analysis include Serial Analysis of Gene Expression (SAGE), and gene expression analysis by massively parallel signature sequencing (MPSS).
(81) Expression Methods Based on Hybridization
(82) The level of a target nucleic acid can be measured using a probe that hybridizes to the target nucleic acid. The target nucleic acid could be, for example, a RNA expression product of a response indicator gene associated with response to a VEGF/VEGFR Inhibitor, or a RNA expression product of a reference gene. In some embodiments, the target nucleic acid is first amplified, for example using a polymerase chain reaction (PCR) method.
(83) A number of methods are available for analyzing nucleic acid mixtures for the presence and/or level of a specific nucleic acid. mRNA may be assayed directly or reverse transcribed into cDNA for analysis.
(84) In some embodiments, the method involves contacting a sample (e.g., a sample derived from a cancer cell) under stringent hybridization conditions with a nucleic acid probe and detecting binding, if any, of the probe to a nucleic acid in the sample. A variety of nucleic acid hybridization methods are well known to those skilled in the art, and any known method can be used. In some embodiments, the nucleic acid probe will be detectably labeled.
(85) Expression Methods Based on Target Amplification
(86) Methods of amplifying (e.g., by PCR) nucleic acid, methods of performing primers extension, and methods

of assessing nucleic acids are generally well known in the art. (See e.g., Ausubel, et al, Short Protocols in Molecular Biology, 3rd ed., Wiley & Sons, 1995 and Sambrook, et al, Molecular Cloning: A Laboratory Manual, Third Edition, (2001) Cold Spring Harbor, N.Y.)

(87) A target mRNA can be amplified by reverse transcribing the mRNA into cDNA, and then performing PCR (reverse transcription-PCR or RT-PCR). Alternatively, a single enzyme may be used for both steps as described in U.S. Pat. No. 5,322,770.

(88) The fluorogenic 5′ nuclease assay, known as the TaqMan® assay (Roche Molecular Systems, Inc.), is a powerful and versatile PCR-based detection system for nucleic acid targets. For a detailed description of the TaqMan assay, reagents and conditions for use therein, see, e.g., Holland et al., Proc. Natl. Acad. Sci., U.S.A. (1991) 88:7276-7280; U.S. Pat. Nos. 5,538,848, 5,723,591, and 5,876,930, all incorporated herein by reference in their entireties. Hence, primers and probes derived from regions of a target nucleic acid as described herein can be used in TaqMan analyses to detect a level of target mRNA in a biological sample. Analysis is performed in conjunction with thermal cycling by monitoring the generation of fluorescence signals. (TaqMan is a registered trademark of Roche Molecular Systems.)

(89) The fluorogenic 5′ nuclease assay is conveniently performed using, for example, AmpliTaq Gold® DNA polymerase, which has endogenous 5′ nuclease activity, to digest an internal oligonucleotide probe labeled with both a fluorescent reporter dye and a quencher (see, Holland et al., Proc Nat Acad Sci USA (1991) 88:7276-7280; and Lee et al., Nucl. Acids Res. (1993) 21:3761-3766). Assay results are detected by measuring changes in fluorescence that occur during the amplification cycle as the fluorescent probe is digested, uncoupling the dye and quencher labels and causing an increase in the fluorescent signal that is proportional to the amplification of target nucleic acid. (AmpliTaq Gold is a registered trademark of Roche Molecular Systems.)

(90) The amplification products can be detected in solution or using solid supports. In this method, the TaqMan probe is designed to hybridize to a target sequence within the desired PCR product. The 5′ end of the TaqMan probe contains a fluorescent reporter dye. The 3′ end of the probe is blocked to prevent probe extension and contains a dye that will quench the fluorescence of the 5′ fluorophore. During subsequent amplification, the 5′ fluorescent label is cleaved off if a polymerase with 5′ exonuclease activity is present in the reaction. Excision of the 5′ fluorophore results in an increase in fluorescence that can be detected.

(91) The first step for gene expression analysis is the isolation of mRNA from a target sample. The starting material is typically total RNA isolated from human tumors or tumor cell lines, and corresponding normal tissues or cell lines, respectively. Thus RNA can be isolated from a variety of primary tumors, including breast, lung, colon, prostate, brain, liver, kidney, pancreas, spleen, thymus, testis, ovary, uterus, head and neck, etc., tumor, or tumor cell lines, with pooled DNA from healthy donors. If the source of mRNA is a primary tumor, mRNA can be extracted, for example, from frozen or archived paraffin-embedded and fixed (e.g., formalin-fixed) tissue samples.

(92) General methods for mRNA extraction are well known in the art and are disclosed in standard textbooks of molecular biology, including Ausubel et al., *Current Protocols of Molecular Biology* (Wiley and Sons, 1997). Methods for RNA extraction from paraffin embedded tissues are disclosed, for example, M. Cronin, Am J. Pathol 164(1):35-42 (2004), the contents of which are incorporated herein. In particular, RNA isolation can be performed using kits and reagents from commercial manufacturers according to the manufacturer's instructions. For example, total RNA from cells in culture can be isolated using RNeasy® mini-columns (Qiagen GmbH Corp.). Other commercially available RNA isolation kits include MasterPure™ Complete DNA and RNA Purification Kit (EPICENTRE® Biotechnologies, Madison, WI), mirVana (Applied Biosystems, Inc.), and Paraffin Block RNA Isolation Kit (Ambion, Inc.). Total RNA from tissue samples can be isolated using RNA STAT-60 TH (IsoTex Diagnostics, Inc., Friendswood TX). RNA prepared from tumor can be isolated, for example, by cesium chloride density gradient centrifugation. (RNeasy is a registered trademark of Qiagen GmbH Corp.; MasterPure is a trademark of EPICENTRE Biotechnologies; RNA STAT-60 is a trademark of Tel-Test Inc.)

(93) As RNA cannot serve as a template for PCR, the first step in gene expression profiling by RT-PCR is the reverse transcription of the RNA template into cDNA, followed by its exponential amplification in a PCR reaction. The two most commonly used reverse transcriptase enzymes are avilo myeloblastosis virus reverse transcriptase (AMV-RT) and Moloney murine leukemia virus reverse transcriptase (MMLV-RT). The reverse transcription step is typically primed using specific primers, random hexamers, or oligo-dT primers, depending on the circumstances and the goal of expression profiling. For example, extracted RNA can be reverse-transcribed using a GeneAmp® RNA PCR kit (Applied Biosystems Inc., Foster City, CA) according to the manufacturer's instructions. The derived cDNA can then be used as a template in a subsequent PCR reaction. (GeneAmp is a registered trademark of Applied Biosystems Inc.)

(94) Although the PCR step can use a variety of thermostable DNA-dependent DNA polymerases, it typically employs the Taq DNA polymerase, which has a 5′-3′ nuclease activity but lacks a 3′-5′ proofreading endonuclease activity. Thus, TaqMan PCR typically utilizes the 5′-nuclease activity of Taq or Tth polymerase to hydrolyze a hybridization probe bound to its target amplicon, but any enzyme with equivalent 5′ nuclease activity can be used. Two oligonucleotide primers are used to generate an amplicon. A third oligonucleotide, or probe, is designed to detect nucleotide sequence located between the two PCR primers. The probe is non-extendible by Taq DNA polymerase enzyme, and is labeled with a reporter fluorescent dye and a quencher fluorescent dye. Any laser-induced emission from the reporter dye is quenched by the quenching dye when the two dyes are located close together as they are on the probe. During the amplification reaction, the Taq DNA polymerase enzyme cleaves the probe in a template-dependent manner. The resultant probe fragments disassociate in solution, and signal from the released reporter dye is free from the quenching effect of the second fluorophore. One molecule of reporter dye is liberated for each new molecule synthesized, and detection of the unquenched reporter dye provides the basis for quantitative interpretation of the data. (TaqMan is a registered mark of Applied Biosystems.)
(95) TaqMan RT-PCR can be performed using commercially available equipment, such as, for example, the ABI PRISM 7700 ® Sequence Detection System (Applied Biosystems, Foster City, CA, USA), or the Lightcycler® (Roche Molecular Biochemicals, Mannheim, Germany). In a preferred embodiment, the 5′ nuclease procedure is run on a real-time quantitative PCR device such as the ABI PRISM 7900™ Sequence Detection System™. The system consists of a thermocycler, laser, charge-coupled device (CCD), camera and computer. The system amplifies samples in a 96-well format on a thermocycler. During amplification, laser-induced fluorescent signal is collected in real-time through fiber optics cables for all 96 wells, and detected at the CCD. The system includes software for running the instrument and for analyzing the data. (PRISM 7700 is a registered trademark of Applied Biosystems; Lightcycler is a registered trademark of Roche Diagnostics GmbH LLC.)
(96) 5′-Nuclease assay data are initially expressed as Cr, or the threshold cycle. As discussed above, fluorescence values are recorded during every cycle and represent the amount of product amplified to that point in the amplification reaction. The point when the fluorescent signal is first recorded as statistically significant is the threshold cycle (Ct).
(97) To minimize the effect of sample-to-sample variation, quantitative RT-PCR is usually performed using an internal standard, or one or more reference genes. The ideal internal standard is expressed at a constant level among different tissues, and is unaffected by the experimental treatment. RNAs that can be used to normalize patterns of gene expression include, e.g., mRNAs for the reference genes glyceraldehyde-3-phosphate-dehydrogenase (GAPDH) and β-actin.
(98) A more recent variation of the RT-PCR technique is the real time quantitative PCR, which measures PCR product accumulation through a dual-labeled fluorogenic probe (i.e., TaqMan® probe). Real time PCR is compatible both with quantitative competitive PCR, where internal competitor for each target sequence is used for normalization, and with quantitative comparative PCR using a normalization gene contained within the sample, or a reference gene for RT-PCR. For further details see, e.g., Held et al., Genome Research 6:986-994 (1996).
(99) Factors considered in PCR primer design include primer length, melting temperature (Tm), and G/C content, specificity, complementary primer sequences, and 3′-end sequence. In general, optimal PCR primers are generally 17-30 bases in length, and contain about 20-80%, such as, for example, about 50-60% G+C bases. Tm's between 50 and 80° C., e.g., about 50 to 70° C. can be used.
(100) For further guidelines for PCR primer and probe design see, e.g., Dieffenbach, C. W. et al., “General Concepts for PCR Primer Design” in: PCR Primer, A Laboratory Manual, Cold Spring Harbor Laboratory Press, New York, 1995, pp. 133-155; Innis and Gelfand, “Optimization of PCRs” in: *PCR Protocols, A Guide to Methods and Applications*, CRC Press, London, 1994, pp. 5-11; and Plasterer, T. N. PrimerSelect: Primer and probe design. *Methods Mol. Biol.* 70:520-527 (1997), the entire disclosures of which are hereby expressly incorporated by reference.
(101) Other suitable methods for assaying a level of a nucleic acid gene product include, e.g., microarrays; serial analysis of gene expression (SAGE); MassARRAY® analysis; digital gene expression (DGE) (J. *Marioni*, Genome Research 18(9):1509-1517 (2008), gene expression by massively parallel signature sequencing (see, e.g., Ding and Cantor, Proc. Nat'l Acad Sci 100:3059-3064 (2003); differential display (Liang and Pardee, Science 257:967-971 (1992)); amplified fragment length polymorphism (iAFLP) (Kawamoto et al., Genome Res. 12:1305-1312 (1999)); BeadArray™ technology (Illumina, San Diego, CA; Oliphant et al., Discovery of Markers for Disease (Supplement to Biotechniques), June 2002; Ferguson et al., Analytical Chemistry 72:5618 (2000)); BeadsArray for Detection of Gene Expression (BADGE), using the commercially available

Luminex100 LabMAP system and multiple color-coded microspheres (Luminex Corp., Austin, TX) in a rapid assay for gene expression (Yang et al., Genome Res. 11:1888-1898 (2001)); and high coverage expression profiling (HiCEP) analysis (Fukumura et al., Nucl. Acids. Res. 31(16) e94 (2003)).
(102) Introns
(103) Assays to measure the amount of an RNA gene expression product can be targeted to intron sequences or exon sequences of the primary transcript. The amount of a spliced intron that is measured in human tissue samples is generally indicative of the amount of a corresponding exon (i.e. an exon from the same gene) present in the samples. Polynucleotides that consist of or are complementary to intron sequences can be used, e.g., in hybridization methods or amplification methods to assay the expression level of response indicator genes.
(104) Measuring Levels of a Polypeptide Gene Product
(105) Methods of measuring a level of a polypeptide gene product are known in the art and include antibody-based methods such as enzyme-linked immunoabsorbent assay (ELISA), radioimmunoassay (RIA), protein blot analysis, immunohistochemical analysis and the like. The measure of a polypeptide gene product may also be measured in vivo in the subject using an antibody that specifically binds a target polypeptide, coupled to a paramagnetic label or other label used for in vivo imaging, and visualizing the distribution of the labeled antibody within the subject using an appropriate in vivo imaging method, such as magnetic resonance imaging. Such methods also include proteomics methods such as mass spectrometric methods, which are known in the art.
(106) Methods of Isolating RNA from Body Fluids
(107) Methods of isolating RNA for expression analysis from blood, plasma and serum (See for example, Tsui N B et al. (2002) 48, 1647-53 and references cited therein) and from urine (See for example, Boom R et al. (1990) J Clin Microbiol. 28, 495-503 and reference cited therein) have been described.
(108) Methods of Isolating RNA from Paraffin-Embedded Tissue
(109) The steps of a representative protocol for profiling gene expression using fixed, paraffin-embedded tissues as the RNA source, including mRNA isolation, purification primer extension and amplification are provided in various published journal articles. (See, e.g., T. E. Godfrey et al., J. Molec. Diagnostics 2: 84-91 (2000); K. Specht et al., Am. J. Pathol. 158: 419-29 (2001), M. Cronin, et al., Am J Pathol 164:35-42 (2004)).
(110) Manual and Computer-Assisted Methods and Products
(111) The methods and systems described herein can be implemented in numerous ways. In one embodiment of particular interest, the methods involve use of a communications infrastructure, for example the Internet. Several embodiments are discussed below. It is also to be understood that the present disclosure may be implemented in various forms of hardware, software, firmware, processors, or a combination thereof. The methods and systems described herein can be implemented as a combination of hardware and software. The software can be implemented as an application program tangibly embodied on a program storage device, or different portions of the software implemented in the user's computing environment (e.g., as an applet) and on the reviewer's computing environment, where the reviewer may be located at a remote site associated (e.g., at a service provider's facility).
(112) For example, during or after data input by the user, portions of the data processing can be performed in the user-side computing environment. For example, the user-side computing environment can be programmed to provide for defined test codes to denote a likelihood "risk score," where the score is transmitted as processed or partially processed responses to the reviewer's computing environment in the form of test code for subsequent execution of one or more algorithms to provide a results and/or generate a report in the reviewer's computing environment. The risk score can be a numerical score (representative of a numerical value) or a non-numerical score representative of a numerical value or range of numerical values (e.g., low, intermediate, or high).
(113) The application program for executing the algorithms described herein may be uploaded to, and executed by, a machine comprising any suitable architecture. In general, the machine involves a computer platform having hardware such as one or more central processing units (CPU), a random access memory (RAM), and input/output (I/O) interface(s). The computer platform also includes an operating system and microinstruction code. The various processes and functions described herein may either be part of the microinstruction code or part of the application program (or a combination thereof) that is executed via the operating system. In addition, various other peripheral devices may be connected to the computer platform such as an additional data storage device and a printing device.
(114) As a computer system, the system generally includes a processor unit. The processor unit operates to receive information, which can include test data (e.g., level of a risk gene, level of a reference gene product(s); normalized level of a gene; and may also include other data such as patient data. This information received can be stored at least temporarily in a database, and data analyzed to generate a report as described above.
(115) Part or all of the input and output data can also be sent electronically; certain output data (e.g., reports) can

be sent electronically or telephonically (e.g., by facsimile, e.g., using devices such as fax back). Exemplary output receiving devices can include a display element, a printer, a facsimile device and the like. Electronic forms of transmission and/or display can include email, interactive television, and the like. In an embodiment of particular interest, all or a portion of the input data and/or all or a portion of the output data (e.g., usually at least the final report) are maintained on a web server for access, preferably confidential access, with typical browsers. The data may be accessed or sent to health professionals as desired. The input and output data, including all or a portion of the final report, can be used to populate a patient's medical record which may exist in a confidential database at the healthcare facility.

(116) A system for use in the methods described herein generally includes at least one computer processor (e.g., where the method is carried out in its entirety at a single site) or at least two networked computer processors (e.g., where data is to be input by a user (also referred to herein as a “client”) and transmitted to a remote site to a second computer processor for analysis, where the first and second computer processors are connected by a network, e.g., via an intranet or internet). The system can also include a user component(s) for input; and a reviewer component(s) for review of data, generated reports, and manual intervention. Additional components of the system can include a server component(s); and a database(s) for storing data (e.g., as in a database of report elements, e.g., interpretive report elements, or a relational database (RDB) which can include data input by the user and data output. The computer processors can be processors that are typically found in personal desktop computers (e.g., IBM, Dell, Macintosh), portable computers, mainframes, minicomputers, or other computing devices.

(117) The networked client/server architecture can be selected as desired, and can be, for example, a classic two or three tier client server model. A relational database management system (RDMS), either as part of an application server component or as a separate component (RDB machine) provides the interface to the database.

(118) In one example, the architecture is provided as a database-centric client/server architecture, in which the client application generally requests services from the application server which makes requests to the database (or the database server) to populate the report with the various report elements as required, particularly the interpretive report elements, especially the interpretation text and alerts. The server(s) (e.g., either as part of the application server machine or a separate RDB/relational database machine) responds to the client's requests.

(119) The input client components can be complete, stand-alone personal computers offering a full range of power and features to run applications. The client component usually operates under any desired operating system and includes a communication element (e.g., a modem or other hardware for connecting to a network), one or more input devices (e.g., a keyboard, mouse, keypad, or other device used to transfer information or commands), a storage element (e.g., a hard drive or other computer-readable, computer-writable storage medium), and a display element (e.g., a monitor, television, LCD, LED, or other display device that conveys information to the user). The user enters input commands into the computer processor through an input device. Generally, the user interface is a graphical user interface (GUI) written for web browser applications.

(120) The server component(s) can be a personal computer, a minicomputer, or a mainframe and offers data management, information sharing between clients, network administration and security. The application and any databases used can be on the same or different servers.

(121) Other computing arrangements for the client and server(s), including processing on a single machine such as a mainframe, a collection of machines, or other suitable configuration are contemplated. In general, the client and server machines work together to accomplish the processing of the present disclosure.

(122) Where used, the database(s) is usually connected to the database server component and can be any device that will hold data. For example, the database can be a any magnetic or optical storing device for a computer (e.g., CDROM, internal hard drive, tape drive). The database can be located remote to the server component (with access via a network, modem, etc.) or locally to the server component.

(123) Where used in the system and methods, the database can be a relational database that is organized and accessed according to relationships between data items. The relational database is generally composed of a plurality of tables (entities). The rows of a table represent records (collections of information about separate items) and the columns represent fields (particular attributes of a record). In its simplest conception, the relational database is a collection of data entries that “relate” to each other through at least one common field.

(124) Additional workstations equipped with computers and printers may be used at point of service to enter data and, in some embodiments, generate appropriate reports, if desired. The computer(s) can have a shortcut (e.g., on the desktop) to launch the application to facilitate initiation of data entry, transmission, analysis, report receipt, etc. as desired.

(125) Computer-Readable Storage Media

(126) The present disclosure also contemplates a computer-readable storage medium (e.g. CD-ROM, memory

key, flash memory card, diskette, etc.) having stored thereon a program which, when executed in a computing environment, provides for implementation of algorithms to carry out all or a portion of the results of a response likelihood assessment as described herein. Where the computer-readable medium contains a complete program for carrying out the methods described herein, the program includes program instructions for collecting, analyzing and generating output, and generally includes computer readable code devices for interacting with a user as described herein, processing that data in conjunction with analytical information, and generating unique printed or electronic media for that user.

(127) Where the storage medium provides a program that provides for implementation of a portion of the methods described herein (e.g., the user-side aspect of the methods (e.g., data input, report receipt capabilities, etc.)), the program provides for transmission of data input by the user (e.g., via the internet, via an intranet, etc.) to a computing environment at a remote site. Processing or completion of processing of the data is carried out at the remote site to generate a report. After review of the report, and completion of any needed manual intervention, to provide a complete report, the complete report is then transmitted back to the user as an electronic document or printed document (e.g., fax or mailed paper report). The storage medium containing a program according to the present disclosure can be packaged with instructions (e.g., for program installation, use, etc.) recorded on a suitable substrate or a web address where such instructions may be obtained. The computer-readable storage medium can also be provided in combination with one or more reagents for carrying out response likelihood assessment (e.g., primers, probes, arrays, or other such kit components).

(128) Methods of Data Analysis

(129) Reference Normalization

(130) In order to minimize expression measurement variations due to non-biological variations in samples, e.g., the amount and quality of expression product to be measured, raw expression level data measured for a gene product (e.g., cycle threshold (CO measurements obtained by qRT-PCR) may be normalized relative to the mean expression level data obtained for one or more reference genes. In one approach to normalization, a small number of genes are used as reference genes; the genes chosen for reference genes typically show a minimal amount of variation in expression from sample to sample and the expression level of other genes is compared to the relatively stable expression of the reference genes. In the global normalization approach, the expression level of each gene in a sample is compared to an average expression level in the sample of all genes in order to compare the expression of a particular gene to the total amount of material.

(131) Unprocessed data from qRT-PCR is expressed as cycle threshold (CO, the number of amplification cycles required for the detectable signal to exceed a defined threshold. High C t is indicative of low expression since more cycles are required to detect the amplification product. Normalization may be carried out such that a one-unit increase in normalized expression level of a gene product generally reflects a 2-fold increase in quantity of expression product present in the sample. For further information on normalization techniques applicable to qRT-PCR data from tumor tissue, see, e.g., Silva S et al. (2006) BMC Cancer 6, 200; de Kok J et al. (2005) Laboratory Investigation 85, 154-159. Gene expression may then be standardized by dividing the normalized gene expression by the standard deviation of expression across all patients for that particular gene. By standardizing normalized gene expression the hazard ratios across genes are comparable and reflect the relative risk for each standard deviation of gene expression.

(132) Statistical Analysis

(133) One skilled in the art will recognize that variety of statistical methods are available that are suitable for comparing the expression level of a gene (or other variable) in two groups and determining the statistical significance of expression level differences that are found. (See e.g., H. Motulsky, *Intuitive Biostatistics*(Oxford University Press, 1995); D. Freedman, *Statistics* (W. W. Norton & Co, 4.sup.th Ed., 2007). For example, a Cox proportional hazards regression model may be fit to a particular clinical time-to-event endpoint (e.g., RFI, OS). One assumption of the Cox proportional hazards regression model is the proportional hazards assumption, i.e. the assumption that model effects multiply the underlying hazard. Assessments of model adequacy may be performed including, but not limited to, examination of the cumulative sum of martingale residuals. One skilled in the art would recognize that there are numerous statistical methods that may be used (e.g., Royston and Parmer (2002), smoothing spline, etc.) to fit a flexible parametric model using the hazard scale and the Weibull distribution with natural spline smoothing of the log cumulative hazards function, with effects allowed to be time-dependent. (See, P. Royston, M. Parmer, *Statistics in Medicine* 21(15:2175-2197 (2002).) The relationship between recurrence risk and (1) recurrence risk groups; and (2) clinical/pathologic covariates (e.g., tumor stage, tumor grade, presence of necrosis, lymphatic or vascular invasion, etc.) may also be tested for significance. Additional examples of models include logistic or ordinal logistic regression models in which the association between gene expression and dichotomous (for logistic) or ordinal (for ordinal logistic) clinical endpoints (i.e.

stage, necrosis, grade) may be evaluated. (See e.g., D. Hosmer and S. Lemeshow, *Applied Logistic Regression* (John Wiley and Sons, 1989).

(134) In an exemplary embodiment, results were adjusted for multiple hypothesis tests, and allowed for a 10% false discovery rate (FDR), using Storey's procedure, and using TDRAS with separate classes (M. Crager, Gene identification using true discovery rate degree of association sets and estimates corrected for regression to the mean, *Statistics in Medicine* (published online December 2009). In another embodiment, genes with significant association with RFI were identified through cross-validation techniques in which forward stepwise Cox PH regression was employed using a subset of factors identified through Principal Component Analysis (PCA).

(135) Methods for calculating correlation coefficients, particularly the Pearson product-moment correlation coefficient are known in the art. (See e.g., J. Rodgers and W. Nicewander, The American Statistician, 42, 59-66 (1988); H. Motulsky, H., Intuitive Biostatistics (Oxford University Press, 1995). To perform particular biological processes, genes often work together in a concerted way, i.e. they are co-expressed. Co-expressed gene groups identified for a disease process like cancer can serve as biomarkers for disease progression and response to treatment. Such co-expressed genes can be assayed in lieu of, or in addition to, assaying of the prognostic and/or predictive gene with which they are co-expressed.

(136) One skilled in the art will recognize that many co-expression analysis methods now known or later developed will fall within the scope and spirit of the present invention. These methods may incorporate, for example, correlation coefficients, co-expression network analysis, clique analysis, etc., and may be based on expression data from RT-PCR, microarrays, sequencing, and other similar technologies. For example, gene expression clusters can be identified using pair-wise analysis of correlation based on Pearson or Spearman correlation coefficients. (See, e.g., Pearson K. and Lee A., Biometrika 2, 357 (1902); C. Spearman, Amer. J. Psychol 15:72-101 (1904); J. Myers, A. Well, *Research Design and Statistical Analysis*, p. 508 (2nd Ed., 2003).) In general, a correlation coefficient of equal to or greater than 0.3 is considered to be statistically significant in a sample size of at least 20. (See, e.g., G. Norman, D. Streiner, *Biostatistics: The Bare Essentials*, 137-138 (3rd Ed. 2007).)

(137) All aspects of the present disclosure may also be practiced such that a limited number of additional genes that are co-expressed with the disclosed genes, for example as evidenced by high Pearson correlation coefficients, are included in a prognostic or predictive test in addition to and/or in place of disclosed genes.

(138) Having described the invention, the same will be more readily understood through reference to the following Examples, which are provided by way of illustration, and are not intended to limit the invention in any way. All citations throughout the disclosure are hereby expressly incorporated by reference.

EXAMPLES

(139) Two studies were performed to demonstrate the feasibility of gene expression profiling from renal tumors obtained from renal cell carcinoma patients. (See Abstract by M. Zhou, et al., Optimized RNA extraction and RT-PCR assays provide successful molecular analysis on a wide variety of archival fixed tissues, AACR Annual Meeting (2007)).

(140) Study Design

(141) Renal tumor tissue was obtained from approximately 1200 patients from the Cleveland Clinic Foundation (CCF) database. This database consists of patients who were diagnosed with renal carcinoma, clear cell type, stage I, II and III between the years of 1985 and 2003, who had available paraffin-embedded tumor (PET) blocks and adequate clinical follow-up, and who were not treated with adjuvant/neo-adjuvant systemic therapy. Patients with inherited VHL disease or bilateral tumors were also excluded. Tumors were graded using (1) Fuhrman grading system as noted in the World Health Organization Classification of Tumours: Pathology and Genetics: Tumours of the Urinary System and Male Genital Organs; and (2) the modified Fuhrman grading system (Table 1). In general, if no nodal involvement is expected or observed for patients, inspection of nodal involvement is not conducted and Nx is noted. In this study, Nx was treated as N0 for purposes of stage classification. The expression of 732 genes was quantitatively assessed for each patient tissue sample.

(142) TABLE-US-00001 TABLE 1 Fuhrman Grading Systems Fuhrman Grade (Modified) Fuhrman Grade (WHO) 1 Nuclei small as lymphocyte Small, round, uniform nuclei with condensed chromatin (~10 um); nucleoli absent or inconspicuous (at 400x) 2 Nuclei both small as Larger nuclei (~15 um) lymphocytes with condensed with irregular outline; small chromatin and other nuclei nucleoli present (at 400x) demonstrating enlarged, open chromatin 3 All nuclei enlarged with Larger nuclei (approaching 20 open chromatin um) with more irregular outline; prominent nucleoli present (at 100x) 4 Large bizarre nuclei Grade 3 features with pleomorphic or multilobed nuclei, with or without spindle cells

Inclusion Criteria (1) Patients who underwent nephrectomy at CCF and who have a minimum of 6 months clinical follow-up or have recurrent RCC, documented in the clinical chart, database or registry. (2) Diagnosed

with RCC, clear cell type, stage I, II, or III. (3) Renal blocks fixed in Formalin, Hollandes fluid or Zenkers fixative.
Exclusion Criteria (1) No tumor block available from initial diagnosis in the Cleveland Clinic archive. (2) No tumor or very little tumor (<5% of invasive cancer cell area) in block as assessed by examination of the Hollandes and/or hematoxylin and eosin stained (H&E) slide. (3) High cycle threshold (CO values of reference genes. All samples regardless of their RNA amount will be tested by RT-qPCR, but only plates where the average Ct of reference genes is less than 35 will be analyzed. (4) Patients with inherited VHL disease and/or bilateral tumors (5) Patients who received neo-adjuvant or adjuvant systemic therapy
Concordance for Clinical and Pathologic Factors
(143) Two separate pathology laboratories conducted analyses of several clinical and pathologic factors using the same standardized measures. The level of concordance, by covariate, is provided in Table 2, below. For purposes of the statistical analysis, the determination of only one of the central laboratories was used.
(144) TABLE-US-00002 TABLE 2 Concordance between two central laboratories for clinical/pathologic covariates Covariate Method of analysis Concordance Presence of necrosis Microscopic technique per 47% Leibovich B C et al. (2003) Cancer 97(3), 1663-1671. Tumor grade Fuhrman 65%
Expression Profile Gene Panel
(145) The RNA from paraffin embedded tissue (PET) samples obtained from 942 patients who met all inclusion/exclusion criteria was extracted using protocols optimized for fixed renal tissue and perform molecular assays of quantitative gene expression using TaqMan® RT-PCR. RT-PCR was performed with RNA input at 1.0 ng per 5 μL-reaction volume using two 384 well plates.
(146) RT-PCR analysis of PET samples was conducted using 732 genes. These genes were evaluated for association with the primary and secondary endpoints, recurrence-free interval (RFI), disease-free survival (DFS) and overall survival (OS).
(147) All primary and secondary analyses were conducted on reference normalized gene expression levels using the mean of the reference genes for normalization. Three normalization schemes were tested using AAMP, ARF1, ATP5E, EEF1A1, GPX1, RPS23, SDHA, UBB, and RPLP1. Some or all of the other genes in the test panel were used in analyses of alternative normalization schemes.
(148) Of the 732 genes, 647 were deemed evaluable for further consideration. The outcome analyses of the 647 evaluable genes were adjusted for multiple hypothesis tests, by allowing for a 10% false discovery rate (FDR), using Storey's procedure, and using True Discovery Rate Degree of Association Sets (TDRDAS) with separate classes (M. Crager, Gene identification using true discovery rate degree of association sets and estimates corrected for regression to the mean, *Statistics in Medicine* 29:33-45 (2009)). Unadjusted for baseline covariates and without controlling the FDR, a subset of 448 (69%) of genes were identified as significantly associated with RFI ($p \leq 0.05$). Additional analysis was conducted using a supervised principal component analysis (PCA) on a subset of 188 genes that have maximum lower bound (MLB)>1.2 using TDRDAS analysis, controlling the FDR but not taking into account the separate classes. The top 10 factors were modified by keeping the genes with high factor loadings (loading/max loading >0.7) were kept. The modified top 10 factors were put into a 5-fold cross validation to assess performance of the gene groups and identify factors that were appearing most frequently.
(149) Genes that had a significant association ($p \leq 0.05$) with risk of recurrence are listed in Tables 3a and 3b, wherein genes that are positively associated with a good prognosis (i.e., increased expression indicates a lower risk of recurrence) are listed in Table 3a. Genes that are negatively associated with a good prognosis (i.e., increased expression indicates a higher risk of recurrence) are listed in Table 3b. Genes that were associated, positively or negatively, with a good prognosis but not associated with clinical/pathologic covariates are BBC3, CCR7, CCR4, and VCAN. In addition, genes associated positively with a good prognosis after adjusting for clinical and pathologic covariates (stage, tumor grade, tumor size, nodal status, and presence of necrosis) are listed in Table 8a. Genes associated negatively with a good prognosis after adjusting for clinical/pathologic covariates are listed in Table 8b. Of those genes listed in Tables 8a and 8b, 16 genes with significant association, positively or negatively, with a good prognosis after adjusting for clinical and pathologic covariates and controlling the false discovery rate at 10% are listed in Table 9.
(150) For the majority of these genes significantly associated with RFI ($p \leq 0.05$) (82%), increased expression is associated with a good prognosis. Most of the genes significantly associated with RFI showed consistency between (1) between stages (I-III); (2) primary and secondary endpoints (RFI and OS). See, e.g., FIGS. **1** and **2**, respectively.
(151) From this analysis, certain gene subsets emerged as significantly associated with recurrence and overall survival. For example, increased expression of angiogenesis genes (e.g., EMCN, PPAP2B, NOS3, NUDT6,

PTPRB, SNRK, APOLD1, PRKCH, and CEACAM1), cell adhesion/extracellular matrix genes (e.g., ITGB5, ITGB1, A2M, TIMP3), immune response genes (e.g., CCL5, CCXL9, CCR7, IL8, IL6, and CX3CL1), cell cycle (e.g., BUB1, TPX2), apoptosis (e.g., CASP10), and transport genes (AQP1) were strongly associated, positively or negatively, with RFI.
(152) Also, certain genes that are associated with pathway targets for renal cancer drugs (sunitinib, sorafenib, temsirolimus, bevacizumab, everolimus, pazopanib) were identified as having a significant association with outcome, including: KIT, PDGFA, PDGFB, PDGFC, PDGFD, PDGFRb, KRAS, RAF1, MTOR, HIF1AN, VEGFA, VEGFB, and FLT4.
(153) It was determined that the presence of necrosis in these tumors was associated with a higher risk of recurrence, at least in the first 4 years after surgery. See FIG. **3**. However, the prognostic effect of necrosis after year 4 was negligible.
(154) It was also determined that expression of certain genes was correlated, positively or negatively, with pathologic and/or clinical factors ("proxy genes"). For example, increased expression of the proxy genes listed in Tables 4a-7b correlate, positively or negatively, with tumor stage, tumor grade, presence of necrosis, and nodal invasion, respectively. In Tables 4a-7b, gene expression was normalized and then standardized such that the odds ratio (OR) reflects a one standard deviation change in gene expression.
(155) From these, key genes were identified as good proxies for baseline covariates (stage, grade, necrosis), including TSPAN7, TEK, LDB2, TIMP3, SHANK3, RGS5, KDR, SDPR, EPAS1, ID1, TGFBR2, FLT4, SDPR, ENDRB, JAG1, DLC1, and KL. Several of these genes are in the hypoxia-induced pathway: SHANK3, RGS5, EPAS1, KDR, JAG1, TGFBR2, FLT4, SDPR, DLC1, EDNRB.
(156) FIGS. **4**-**7** provide a comparison of patient stratification (Low, Intermediate, or High Risk) obtained by applying the Mayo prognostic tool (described in Leibovich et al. "prediction of progression after radical nephrectomy for patients with clear cell renal cell carcinoma" (2003) Cancer 97:1663-1671) CCF expression data. As shown in FIG. **4**, the Mayo prognostic tool alone provides for stratification into three populations Low Risk (93% recurrence free at 5 years), Intermediate Risk (79% recurrence free at 5 years), and High Risk (36% recurrence free at 5 years). In contrast, use of expression data from even one gene (as exemplified by EMCN (FIG. **5**), AQP1 (FIG. **6**), or PPAP2B (FIG. **7**)) allowed for more detailed stratification of patients according to risk.
(157) TABLE-US-00003 TABLE 3a Genes for which increased expression is associated with lower risk of cancer recurrence (p-value ≤.05) Univariate Cox Analyses Univariate Cox Analyses (No Covariate (No Covariate Adjustment) with RFI Adjustment) with RFI Gene Official Symbol p-value for HR HR Gene Official Symbol p-value for HR HR YB-1.2 YBX1 <0.0001 0.75 IGF1R.3 IGF1R <0.0001 0.58 XIAP.1 XIAP 0.0009 0.81 IFI27.1 IFI27 0.0205 0.85 WWOX.5 WWOX <0.0001 0.71 ID3.1 ID3 <0.0001 0.69 VWF.1 VWF <0.0001 0.61 ID2.4 ID2 <0.0001 0.75 VEGF.1 VEGFA <0.0001 0.75 ID1.1 ID1 <0.0001 0.55 VCAM1.1 VCAM1 <0.0001 0.66 ICAM2.1 ICAM2 <0.0001 0.64 USP34.1 USP34 0.0003 0.79 HYAL2.1 HYAL2 <0.0001 0.60 UMOD.1 UMOD <0.0001 0.68 HYAL1.1 HYAL1 <0.0001 0.58 UGCG.1 UGCG <0.0001 0.71 HSPG2.1 HSPG2 <0.0001 0.60 UBB.1 UBB <0.0001 0.62 HSD11B2.1 HSD11B2 <0.0001 0.63 UBE1C.1 UBA3 0.0003 0.79 Hepsin.1 HPN 0.0001 0.76 TS.1 TYMS 0.0056 0.83 HPCAL1.1 HPCAL1 0.0031 0.82 tusc4.2 TUSC4 <0.0001 0.76 HMGB1.1 HMGB1 <0.0001 0.67 TSPAN7.2 TSPAN7 <0.0001 0.52 HLA-DPB1.1 HLA-DPB1 <0.0001 0.70 TSC2.1 TSC2 0.0043 0.82 HIF1AN.1 HIF1AN <0.0001 0.77 TSC1.1 TSC1 <0.0001 0.71 HDAC1.1 HDAC1 0.0003 0.78 P53.2 TP53 0.0008 0.81 HAVCR1.1 HAVCR1 0.0003 0.79 TOP2B.2 TOP2B 0.0001 0.80 HADH.1 HADH <0.0001 0.68 TNFSF12.1 TNFSF12 <0.0001 0.68 GZMA.1 GZMA 0.0125 0.84 TRAIL.1 TNFSF10 0.0065 0.83 GSTp.3 GSTP1 <0.0001 0.76 TNFRSF11B.1 TNFRSF11B 0.0002 0.78 GSTM3.2 GSTM3 <0.0001 0.70 TNFRSF10D.1 TNFRSF10D <0.0001 0.76 GSTM1.1 GSTM1 <0.0001 0.72 DR5.2 TNFRSF10B <0.0001 0.75 GRB7.2 GRB7 <0.0001 0.74 TNFAIP6.1 TNFAIP6 0.0005 0.79 GPX3.1 GPX3 <0.0001 0.75 TMEM47.1 TMEM47 <0.0001 0.57 GJA1.1 GJA1 0.0049 0.84 TMEM27.1 TMEM27 <0.0001 0.58 GFRA1.1 GFRA1 0.0003 0.77 TLR3.1 TLR3 <0.0001 0.72 GCLC.3 GCLC <0.0001 0.68 TIMP3.3 TIMP3 <0.0001 0.50 GBP2.2 GBP2 0.0156 0.85 TIMP2.1 TIMP2 <0.0001 0.73 GATM.1 GATM <0.0001 0.66 THBS1.1 THBS1 0.0002 0.79 GATA3.3 GATA3 0.0001 0.75 TGFBR2.3 TGFBR2 <0.0001 0.54 FOS.1 FOS <0.0001 0.74 TGFBR1.1 TGFBR1 <0.0001 0.71 FOLR1.1 FOLR1 0.0003 0.79 TGFB2.2 TGFB2 <0.0001 0.65 FLT4.1 FLT4 <0.0001 0.54 TGFA.2 TGFA <0.0001 0.70 FLT3LG.1 FLT3LG 0.0001 0.73 TEK.1 TEK <0.0001 0.47 FLT1.1 FLT1 <0.0001 0.59 TCF4.1 TCF4 <0.0001 0.62 FILIP1.1 FILIP1 <0.0001 0.72 TAP1.1 TAP1 0.0024 0.82 FIGF.1 FIGF 0.0014 0.76 TAGLN.1 TAGLN <0.0001 0.70 FHL1.1 FHL1 <0.0001 0.68 TACSTD2.1 TACSTD2 <0.0001 0.72 FHIT.1 FHIT <0.0001 0.73 SUCLG1.1 SUCLG1 <0.0001 0.72 FH.1 FH <0.0001 0.73 STK11.1 STK11 <0.0001 0.76 FGFR2 isoform 1.1 FGFR2 <0.0001 0.70 STAT5B.2 STAT5B <0.0001 0.71 FGFR1.3 FGFR1 0.0001 0.77 STAT5A.1 STAT5A <0.0001 0.72 FGF2.2 FGF2 <0.0001 0.73

STAT3.1 STAT3 0.0016 0.80 FGF1.1 FGF1 0.0015 0.78 SPRY1.1 SPRY1 <0.0001 0.68 FDPS.1 FDPS <0.0001 0.69 SPARCL1.1 SPARCL1 <0.0001 0.71 FBXW7.1 FBXW7 0.0001 0.75 SPARC.1 SPARC 0.0009 0.80 fas.1 FAS <0.0001 0.73 SOD1.1 SOD1 0.0114 0.84 FABP1.1 FABP1 0.0455 0.86 SNRK.1 SNRK <0.0001 0.54 ESRRG.3 ESRRG 0.0008 0.79 SNAI1.1 SNAI1 0.0055 0.82 ERG.1 ERG <0.0001 0.60 MADH4.1 SMAD4 <0.0001 0.60 ERCC1.2 ERCC1 <0.0001 0.78 MADH2.1 SMAD2 <0.0001 0.64 ErbB3.1 ERBB3 0.0001 0.77 SLC34A1.1 SLC34A1 0.0004 0.76 HER2.3 ERBB2 <0.0001 0.65 SLC22A6.1 SLC22A6 0.0003 0.77 EPHB4.1 EPHB4 <0.0001 0.64 SKIL.1 SKIL 0.0004 0.79 EPHA2.1 EPHA2 <0.0001 0.58 SHANK3.1 SHANK3 <0.0001 0.53 EPAS1.1 EPAS1 <0.0001 0.55 SGK.1 SGK1 <0.0001 0.67 ENPP2.1 ENPP2 0.0090 0.84 FRP1.3 SFRP1 0.0355 0.86 ENPEP.1 ENPEP <0.0001 0.74 SEMA3F.3 SEMA3F <0.0001 0.67 CD105.1 ENG <0.0001 0.60 SELENBP1.1 SELENBP1 0.0016 0.81 EMP1.1 EMP1 <0.0001 0.71 SDPR.1 SDPR <0.0001 0.51 EMCN.1 EMCN <0.0001 0.43 SDHA.1 SDHA 0.0003 0.77 ELTD1.1 ELTD1 <0.0001 0.76 SCNN1A.2 SCNN1A 0.0134 0.83 EIF2C1.1 EIF2C1 <0.0001 0.67 SCN4B.1 SCN4B <0.0001 0.68 EGR1.1 EGR1 <0.0001 0.72 KIAA1303 RPTOR <0.0001 0.77 EGLN3.1 EGLN3 0.0002 0.80 raptor.1 RPS6KB1.3 RPS6KB1 <0.0001 0.73 EGFR.2 EGFR 0.0005 0.80 RPS6KAI.1 RPS6KA1 0.0291 0.86 EFNB2.1 EFNB2 <0.0001 0.64 RPS23.1 RPS23 <0.0001 0.73 EFNB1.2 EFNB1 <0.0001 0.68 ROCK2.1 ROCK2 <0.0001 0.66 EEF1A1.1 EEF1A1 <0.0001 0.64 ROCK1.1 ROCK1 <0.0001 0.58 EDNRB.1 EDNRB <0.0001 0.56 RIPK1.1 RIPK1 <0.0001 0.64 EDN2.1 EDN2 0.0005 0.77 rhoC.1 RHOC 0.0213 0.86 EDN1 EDN1 <0.0001 0.62 endothelin.1 RhoB.1 RHOB <0.0001 0.63 EBAG9.1 EBAG9 0.0041 0.82 ARHA.1 RHOA <0.0001 0.64 DUSP1.1 DUSP1 0.0029 0.82 RGS5.1 RGS5 <0.0001 0.52 DPYS.1 DPYS 0.0112 0.84 FLJ22655.1 RERGL <0.0001 0.51 DPEP1.1 DPEP1 <0.0001 0.64 NFKBp65.3 RELA 0.0027 0.82 DLL4.1 DLL4 <0.0001 0.76 RB1.1 RB1 <0.0001 0.73 DLC1.1 DLC1 <0.0001 0.55 RASSF1.1 RASSF1 0.0040 0.83 DKFZP564O0823.1 DKFZP564O0823 <0.0001 0.62 RARB.2 RARB <0.0001 0.57 DICER1.2 DICER1 <0.0001 0.71 RALBP1.1 RALBP1 <0.0001 0.73 DIAPH1.1 DIAPH1 0.0009 0.80 RAF1.3 RAF1 0.0008 0.81 DIABLO.1 DIABLO 0.0134 0.84 PTPRG.1 PTPRG <0.0001 0.57 DHPS.3 DHPS <0.0001 0.70 PTPRB.1 PTPRB <0.0001 0.51 DET1.1 DET1 <0.0001 0.74 PTN.1 PTN <0.0001 0.71 DEFB1.1 DEFB1 0.0025 0.81 PTK2.1 PTK2 <0.0001 0.61 DDC.1 DDC <0.0001 0.68 PTHR1.1 PTH1R <0.0001 0.55 DCXR.1 DCXR 0.0081 0.83 PTEN.2 PTEN <0.0001 0.74 DAPK1.3 DAPK1 <0.0001 0.60 PSMB9.1 PSMB9 0.0025 0.82 CYR61.1 CYR61 <0.0001 0.70 PSMB8.1 PSMB8 0.0239 0.85 CYP3A4.2 CYP3A4 0.0380 0.86 PRSS8.1 PRSS8 <0.0001 0.71 CXCL9.1 CXCL9 0.0362 0.87 PRPS2.1 PRPS2 0.0156 0.85 CXCL12.1 CXCL12 <0.0001 0.69 PRKCH.1 PRKCH <0.0001 0.63 CX3CR1.1 CX3CR1 <0.0001 0.72 PPP2CA.1 PPP2CA <0.0001 0.77 CX3CL1.1 CX3CL1 <0.0001 0.58 PPARG.3 PPARG <0.0001 0.59 CUL1.1 CUL1 0.0003 0.80 PPAP2B.1 PPAP2B <0.0001 0.50 CUBN.1 CUBN <0.0001 0.61 PLG.1 PLG <0.0001 0.70 CTSS.1 CTSS 0.0016 0.82 PLAT.1 PLAT <0.0001 0.64 CTSH.2 CTSH <0.0001 0.78 PLA2G4C.1 PLA2G4C <0.0001 0.67 B-Catenin.3 CTNNB1 <0.0001 0.74 PIK3CA.1 PIK3CA <0.0001 0.75 A-Catenin.2 CTNNA1 <0.0001 0.77 PI3K.2 PIK3C2B <0.0001 0.56 CTGF.1 CTGF 0.0004 0.79 PFKP.1 PFKP 0.0067 0.84 CSF1R.2 CSF1R 0.0015 0.82 CD31.3 PECAM1 <0.0001 0.60 CSF1.1 CSF1 0.0016 0.81 PDZK3.1 PDZK3 0.0005 0.77 CRADD.1 CRADD 0.0021 0.81 PDZK1.1 PDZK1 <0.0001 0.69 COL4A2.1 COL4A2 0.0016 0.80 PDGFRb.3 PDGFRB <0.0001 0.74 COL18A1.1 COL18A1 0.0007 0.79 PDGFD.2 PDGFD <0.0001 0.58 CLU.3 CLU 0.0151 0.85 PDGFC.3 PDGFC <0.0001 0.66 CLDN7.2 CLDN7 0.0023 0.81 PDGFB.3 PDGFB <0.0001 0.63 CLDN10.1 CLDN10 <0.0001 0.69 PDGFA.3 PDGFA <0.0001 0.75 CLCNKB.1 CLCNKB 0.0002 0.74 PCK1.1 PCK1 <0.0001 0.65 CFLAR.1 CFLAR <0.0001 0.65 PCCA.1 PCCA <0.0001 0.66 CEACAM1.1 CEACAM1 <0.0001 0.59 PARD6A.1 PARD6A 0.0001 0.76 p27.3 CDKN1B 0.0018 0.83 Pak1.2 PAK1 0.0011 0.81 p21.3 CDKN1A 0.0027 0.81 PAH.1 PAH 0.0296 0.86 CDH6.1 CDH6 <0.0001 0.75 OGG1.1 OGG1 0.0051 0.84 CDH5.1 CDH5 <0.0001 0.59 BFGF.3 NUDT6 <0.0001 0.54 CDH16.1 CDH16 <0.0001 0.75 NRG1.3 NRG1 0.0049 0.81 CDH13.1 CDH13 <0.0001 0.66 NPR1.1 NPR1 <0.0001 0.73 CD4.1 CD4 0.0009 0.81 NPM1.2 NPM1 <0.0001 0.73 CD36.1 CD36 <0.0001 0.65 NOTCH3.1 NOTCH3 <0.0001 0.70 CD34.1 CD34 <0.0001 0.62 NOTCH2.1 NOTCH2 0.0040 0.84 CCR7.1 CCR7 0.0271 0.86 NOTCH1.1 NOTCH1 <0.0001 0.64 CCR4.2 CCR4 0.0106 0.83 NOS3.1 NOS3 <0.0001 0.55 CCND1.3 CCND1 <0.0001 0.70 NOS2A.3 NOS2 <0.0001 0.66 CCL4.2 CCL4 0.0012 0.80 NOL3.1 NOL3 0.0132 0.85 MCP1.1 CCL2 <0.0001 0.75 NFX1.1 NFX1 <0.0001 0.67 CAT.1 CAT <0.0001 0.72 NFKBp50.3 NFKB1 0.0001 0.77 CASP6.1 CASP6 0.0369 0.87 NFATC2.1 NFATC2 0.0003 0.78 CASP10.1 CASP10 <0.0001 0.69 NFAT5.1 NFAT5 0.0010 0.82 CALD1.2 CALD1 <0.0001 0.61 MYRIP.2 MYRIP 0.0002 0.75 CA9.3 CA9 0.0035 0.84 MYH11.1 MYH11 <0.0001 0.61 CA2.1 CA2 0.0006 0.79 cMYC.3 MYC 0.0002 0.79 C7.1 C7 0.0030 0.82 MVP.1 MVP 0.0052 0.83 ECRG4.1 C2orf40 <0.0001 0.57 MUC1.2 MUC1 0.0405 0.87 C13orf15.1 C13orf15 <0.0001 0.57 FRAP1.1 MTOR <0.0001 0.75 BUB3.1 BUB3 0.0002 0.77 MSH3.2 MSH3 <0.0001 0.75 BTRC.1 BTRC 0.0006 0.81 MSH2.3 MSH2 <0.0001 0.71 CIAP1.2 BIRC2 0.0030 0.82 GBL.1 MLST8 0.0246 0.87 BIN1.3 BIN1 0.0005 0.80 MIF.2 MIF 0.0012 0.80 BGN.1 BGN <0.0001 0.76 MICA.1 MICA <0.0001 0.71 BCL2L12.1 BCL2L12 0.0322 0.86

MGMT.1 MGMT <0.0001 0.68 Bclx.2 BCL2L1 <0.0001 0.74 MCM3.3 MCM3 0.0188 0.85 Bcl2.2 BCL2 <0.0001 0.57 MCAM.1 MCAM <0.0001 0.71 BBC3.2 BBC3 0.0449 0.87 MARCKS.1 MARCKS 0.0301 0.87 BAG1.2 BAG1 <0.0001 0.64 ERK1.3 MAPK3 <0.0001 0.65 BAD.1 BAD 0.0076 0.85 ERK2.3 MAPK1 0.0005 0.79 ATP6V1B1.1 ATP6V1B1 0.0001 0.71 MAP4.1 MAP4 <0.0001 0.65 ASS1.1 ASS1 <0.0001 0.75 MAP2K3.1 MAP2K3 <0.0001 0.69 ARRB1.1 ARRB1 <0.0001 0.62 MAP2K1.1 MAP2K1 0.0046 0.83 ARHGDIB.1 ARHGDIB <0.0001 0.66 MAL2.1 MAL2 0.0001 0.76 AQP1.1 AQP1 <0.0001 0.50 MAL.1 MAL <0.0001 0.66 APOLD1.1 APOLD1 <0.0001 0.57 LYZ.1 LYZ 0.0458 0.88 APC.4 APC <0.0001 0.73 LTF.1 LTF 0.0005 0.76 ANXA4.1 ANXA4 0.0018 0.81 LRP2.1 LRP2 <0.0001 0.67 ANXA1.2 ANXA1 0.0009 0.80 LMO2.1 LMO2 <0.0001 0.74 ANTXR1.1 ANTXR1 0.0043 0.82 LDB2.1 LDB2 <0.0001 0.52 ANGPTL4.1 ANGPTL4 0.0033 0.84 LDB1.2 LDB1 <0.0001 0.71 ANGPTL3.3 ANGPTL3 0.0003 0.75 LAMA4.1 LAMA4 0.0279 0.86 ANGPT1.1 ANGPT1 <0.0001 0.57 KRT7.1 KRT7 <0.0001 0.68 ALDOB.1 ALDOB <0.0001 0.62 K-ras.10 KRAS <0.0001 0.72 ALDH6A1.1 ALDH6A1 <0.0001 0.63 KL.1 KL <0.0001 0.55 ALDH4.2 ALDH4A1 0.0172 0.85 Kitlng.4 KITLG <0.0001 0.67 AKT3.2 AKT3 <0.0001 0.57 c-kit.2 KIT <0.0001 0.72 AKT2.3 AKT2 <0.0001 0.75 KDR.6 KDR <0.0001 0.54 AKT1.3 AKT1 <0.0001 0.71 KCNJ15.1 KCNJ15 <0.0001 0.63 AIF1.1 AIF1 0.0349 0.87 HTATIP.1 KAT5 <0.0001 0.64 AHR.1 AHR <0.0001 0.74 G-Catenin.1 JUP <0.0001 0.64 AGTR1.1 AGTR1 <0.0001 0.57 AP-1 (JUN JUN 0.0001 0.76 ADH1B.1 ADH1B 0.0002 0.77 official).2 JAG1.1 JAG1 <0.0001 0.55 ADFP.1 ADFP 0.0332 0.88 ITGB1.1 ITGB1 0.0085 0.83 ADD1.1 ADD1 <0.0001 0.58 ITGA7.1 ITGA7 <0.0001 0.66 ADAMTS5.1 ADAMTS5 0.0010 0.78 ITGA6.2 ITGA6 <0.0001 0.63 ADAMTS1.1 ADAMTS1 0.0056 0.83 ITGA4.2 ITGA4 <0.0001 0.74 ACE2.1 ACE2 <0.0001 0.62 ITGA3.2 ITGA3 0.0002 0.79 ACADSB.1 ACADSB <0.0001 0.71 IQGAP2.1 IQGAP2 <0.0001 0.69 BCRP.1 ABCG2 <0.0001 0.58 INSR.1 INSR <0.0001 0.67 MRP4.2 ABCC4 <0.0001 0.74 IMP3.1 IMP3 <0.0001 0.69 MRP3.1 ABCC3 0.0107 0.85 IL6ST.3 IL6ST <0.0001 0.66 MRP1.1 ABCC1 <0.0001 0.75 IL15.1 IL15 <0.0001 0.70 ABCB1.5 ABCB1 0.0093 0.84 IGFBP6.1 IGFBP6 0.0004 0.79 NPD009 (ABAT ABAT 0.0001 0.76 official).3 IGFBP3.1 IGFBP3 0.0394 0.87 AAMP.1 AAMP 0.0008 0.80 IGFBP2.1 IGFBP2 0.0134 0.84 A2M.1 A2M <0.0001 0.56

(158) TABLE-US-00004 TABLE 3b Genes for which increased expression is associated with higher risk of cancer recurrence (p-value ≤.05) Univariate Cox Analyses Univariate Cox Analyses (No Covariate (No Covariate Adjustment) with RFI Adjustment) with RFI Gene Official Symbol p-value for HR HR Gene Official Symbol p-value for HR HR WT1.1 WT1 0.0002 1.25 LGALS1.1 LGALS1 0.0017 1.25 VTN.1 VTN 0.0097 1.17 LAMB3.1 LAMB3 <0.0001 1.34 VDR.2 VDR 0.0031 1.22 LAMB1.1 LAMB1 0.0014 1.25 VCAN.1 VCAN 0.0036 1.22 L1CAM.1 L1CAM 0.0199 1.16 UBE2T.1 UBE2T <0.0001 1.38 IL-8.1 IL8 <0.0001 1.53 C20 orf1.1 TPX2 <0.0001 1.76 IL6.3 IL6 <0.0001 1.41 TOP2A.4 TOP2A <0.0001 1.39 ICAM1.1 ICAM1 0.0013 1.23 TK1.2 TK1 0.0018 1.22 HIST1H1D.1 HIST1H1D 0.0066 1.21 TIMP1.1 TIMP1 0.0259 1.16 FN1.1 FN1 0.0105 1.19 TGFBI.1 TGFBI 0.0004 1.26 F3.1 F3 <0.0001 1.31 SQSTM1.1 SQSTM1 0.0089 1.20 F2.1 F2 <0.0001 1.30 OPN, SPP1 <0.0001 1.43 ESPL1.3 ESPL1 0.0155 1.17 osteopontin.3 SPHK1.1 SPHK1 0.0025 1.22 EPHB2.1 EPHB2 0.0456 1.14 SLC7A5.2 SLC7A5 <0.0001 1.38 EPHB1.3 EPHB1 0.0007 1.22 SLC2A1.1 SLC2A1 0.0010 1.26 ENO2.1 ENO2 <0.0001 1.38 SLC16A3.1 SLC16A3 <0.0001 1.38 EIF4EBP1.1 EIF4EBP1 0.0098 1.19 SLC13A3.1 SLC13A3 0.0192 1.16 CXCR4.3 CXCR4 0.0066 1.21 SHC1.1 SHC1 0.0086 1.19 GRO1.2 CXCL1 <0.0001 1.30 SFN.1 SFN 0.0001 1.26 CTSB.1 CTSB 0.0233 1.17 SERPINA5.1 SERPINA5 0.0462 1.13 CRP.1 CRP 0.0314 1.13 SEMA3C.1 SEMA3C <0.0001 1.45 CP.1 CP 0.0002 1.32 SAA2.2 SAA2 <0.0001 1.59 COL7A1.1 COL7A1 0.0003 1.24 S100A1.1 S100A1 0.0348 1.16 COL1A1.1 COL1A1 0.0029 1.23 RRM2.1 RRM2 0.0002 1.27 Chk1.2 CHEK1 0.0002 1.26 RPLP1.1 RPLP1 0.0049 1.22 CENPF.1 CENPF <0.0001 1.36 PTTG1.2 PTTG1 <0.0001 1.45 CD82.3 CD82 0.0009 1.25 COX2.1 PTGS2 0.0013 1.22 CD44s.1 CD44_s 0.0065 1.21 PLAUR.3 PLAUR <0.0001 1.33 CCNE1.1 CCNE1 0.0098 1.17 PF4.1 PF4 0.0034 1.20 CCNB1.2 CCNB1 <0.0001 1.42 PCSK6.1 PCSK6 0.0269 1.17 CCL20.1 CCL20 0.0029 1.22 MYBL2.1 MYBL2 <0.0001 1.33 CA12.1 CA12 <0.0001 1.48 MT1X.1 MT1X 0.0070 1.20 C3.1 C3 0.0176 1.18 MMP9.1 MMP9 <0.0001 1.54 BUB1.1 BUB1 <0.0001 1.59 MMP7.1 MMP7 0.0312 1.15 SURV.2 BIRC5 <0.0001 1.37 MMP14.1 MMP14 <0.0001 1.47 cIAP2.2 BIRC3 0.0484 1.15 Ki-67.2 MKI67 <0.0001 1.33 BCL2A1.1 BCL2A1 0.0483 1.11 mGST1.2 MGST1 <0.0001 1.38 STK15.2 AURKA 0.0002 1.28 MDK.1 MDK 0.0001 1.31 ANXA2.2 ANXA2 0.0315 1.16 LOX.1 LOX <0.0001 1.42 ALOX5.1 ALOX5 0.0473 1.14 LMNB1.1 LMNB1 <0.0001 1.40 ADAM8.1 ADAM8 0.0002 1.29 LIMK1.1 LIMK1 <0.0001 1.43 MRP2.3 ABCC2 0.0004 1.28

(159) TABLE-US-00005 TABLE 4a Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤.05) Stage 3 vs. 1 Stage 3 vs. 1 Gene Official Symbol p-value OR Gene Official Symbol p-value OR WT1.1 WT1 <0.0001 1.41 LAMA3.1 LAMA3 0.0121 1.22 VTN.1 VTN 0.0007 1.29 L1CAM.1 L1CAM 0.0091 1.22 VDR.2 VDR 0.0065 1.25 ISG20.1 ISG20 0.0006 1.34 UBE2T.1 UBE2T <0.0001 1.61 IL-8.1 IL8

<0.0001 1.89 TSPAN8.1 TSPAN8 0.0072 1.23 IL6.3 IL6 <0.0001 1.68 C20 orf1.1 TPX2 <0.0001 1.89 IGF1.2 IGF1 0.0214 1.20 TOP2A.4 TOP2A <0.0001 1.55 ICAM1.1 ICAM1 <0.0001 1.42 TK1.2 TK1 0.0001 1.34 HIST1H1D.1 HIST1H1D 0.0005 1.33 TIMP1.1 TIMP1 0.0021 1.29 GPX2.2 GPX2 0.0129 1.22 TGFBI.1 TGFBI 0.0001 1.39 FN1.1 FN1 0.0002 1.36 OPN, SPP1 0.0001 1.38 FAP.1 FAP 0.0455 1.18 osteopontin.3 SLC7A5.2 SLC7A5 <0.0001 1.51 F3.1 F3 <0.0001 1.52 SLC2A1.1 SLC2A1 0.0081 1.24 F2.1 F2 <0.0001 1.79 SLC16A3.1 SLC16A3 <0.0001 1.46 ESPL1.3 ESPL1 0.0001 1.35 SFN.1 SFN 0.0001 1.36 EPB41L3.1 EPB41L3 0.0067 1.24 SEMA3C.1 SEMA3C <0.0001 1.42 ENO2.1 ENO2 0.0016 1.31 SELL.1 SELL 0.0313 1.19 EIF4EBP1.1 EIF4EBP1 0.0036 1.27 SAA2.2 SAA2 <0.0001 2.04 E2F1.3 E2F1 0.0017 1.27 RRM2.1 RRM2 <0.0001 1.47 DCN.1 DCN 0.0152 1.22 RPLP1.1 RPLP1 0.0007 1.33 CXCR6.1 CXCR6 0.0013 1.30 RAD51.1 RAD51 0.0010 1.31 BLR1.1 CXCR5 0.0232 1.19 PTTG1.2 PTTG1 <0.0001 1.61 CXCR4.3 CXCR4 0.0003 1.35 COX2.1 PTGS2 0.0011 1.29 GRO1.2 CXCL1 0.0005 1.31 PTGIS.1 PTGIS 0.0034 1.27 CTSB.1 CTSB 0.0110 1.24 PLAUR.3 PLAUR <0.0001 1.51 CRP.1 CRP 0.0002 1.31 PF4.1 PF4 0.0027 1.26 CP.1 CP 0.0008 1.34 PDGFRa.2 PDGFRA 0.0480 1.17 COL7A1.1 COL7A1 0.0010 1.28 PCSK6.1 PCSK6 0.0041 1.27 COL1A1.1 COL1A1 0.0001 1.40 NNMT.1 NNMT 0.0003 1.34 Chk2.3 CHEK2 0.0050 1.27 NME2.1 NME2 0.0028 1.28 Chk1.2 CHEK1 <0.0001 1.43 MYBL2.1 MYBL2 <0.0001 1.50 CENPF.1 CENPF <0.0001 1.55 MT1X.1 MT1X 0.0192 1.21 CD82.3 CD82 0.0001 1.38 MMP9.1 MMP9 <0.0001 1.79 CD44s.1 CD44_s 0.0060 1.25 MMP7.1 MMP7 0.0252 1.20 CCNE2.2 CCNE2_2 0.0229 1.19 MMP14.1 MMP14 <0.0001 1.88 CCNB1.2 CCNB1 <0.0001 1.60 Ki-67.2 MKI67 <0.0001 1.48 CCL20.1 CCL20 0.0010 1.30 mGST1.2 MGST1 0.0004 1.37 CA12.1 CA12 <0.0001 1.66 MDK.1 MDK <0.0001 1.42 C3.1 C3 0.0009 1.32 MDH2.1 MDH2 0.0321 1.19 BUB1.1 BUB1 <0.0001 1.82 LRRC2.1 LRRC2 0.0259 1.19 SURV.2 BIRC5 <0.0001 1.46 LOX.1 LOX <0.0001 1.78 BCL2A1.1 BCL2A1 <0.0001 1.44 LMNB1.1 LMNB1 <0.0001 1.82 STK15.2 AURKA 0.0002 1.36 LIMK1.1 LIMK1 <0.0001 1.46 APOL1.1 APOL1 0.0028 1.27 LAPTM5.1 LAPTM5 0.0102 1.23 ANXA2.2 ANXA2 0.0174 1.21 LAMB3.1 LAMB3 <0.0001 1.53 ADAM8.1 ADAM8 <0.0001 1.58 LAMB1.1 LAMB1 0.0452 1.18 MRP2.3 ABCC2 <0.0001 1.45

(160) TABLE-US-00006 TABLE 4b Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤.05) Stage 3 vs. 1 Stage 3 vs. 1 Gene Official Symbol p-value OR Gene Official Symbol p-value OR YB-1.2 YBX1 <0.0001 0.63 IL-7.1 IL7 0.0444 0.85 XIAP.1 XIAP <0.0001 0.62 IL6ST.3 IL6ST <0.0001 0.50 WWOX.5 WWOX 0.0042 0.79 IL15.1 IL15 <0.0001 0.67 WISP1.1 WISP1 0.0096 0.81 IGFBP6.1 IGFBP6 0.0001 0.73 VWF.1 VWF <0.0001 0.46 IGFBP3.1 IGFBP3 0.0191 0.83 VEGF.1 VEGFA <0.0001 0.63 IGF1R.3 IGF1R <0.0001 0.48 VCAM1.1 VCAM1 <0.0001 0.55 ID3.1 ID3 <0.0001 0.54 USP34.1 USP34 0.0001 0.72 ID2.4 ID2 0.0008 0.77 UMOD.1 UMOD <0.0001 0.47 ID1.1 ID1 <0.0001 0.34 UGCG.1 UGCG <0.0001 0.68 ICAM2.1 ICAM2 <0.0001 0.58 UBB.1 UBB <0.0001 0.54 HYAL2.1 HYAL2 <0.0001 0.41 UBE1C.1 UBA3 <0.0001 0.62 HYAL1.1 HYAL1 <0.0001 0.44 TS.1 TYMS 0.0010 0.76 HSPG2.1 HSPG2 <0.0001 0.44 tusc4.2 TUSC4 <0.0001 0.57 HSD11B2.1 HSD11B2 <0.0001 0.47 TUSC2.1 TUSC2 0.0295 0.83 Hepsin.1 HPN 0.0031 0.79 TSPAN7.2 TSPAN7 <0.0001 0.35 HPCAL1.1 HPCAL1 0.0004 0.75 TSC2.1 TSC2 <0.0001 0.66 HNRPAB.3 HNRNPAB 0.0039 0.78 TSC1.1 TSC1 <0.0001 0.56 HMGB1.1 HMGB1 <0.0001 0.46 P53.2 TP53 <0.0001 0.67 HLA-DPB1.1 HLA-DPB1 <0.0001 0.55 TOP2B.2 TOP2B <0.0001 0.69 HIF1AN.1 HIF1AN <0.0001 0.57 TNIP2.1 TNIP2 0.0359 0.85 HIF1A.3 HIF1A 0.0076 0.80 TNFSF12.1 TNFSF12 <0.0001 0.50 HGF.4 HGF 0.0067 0.80 TRAIL.1 TNFSF10 0.0011 0.77 HDAC1.1 HDAC1 <0.0001 0.61 TNFRSF11B.1 TNFRSF11B <0.0001 0.63 HAVCR1.1 HAVCR1 0.0001 0.73 TNFRSF10D.1 TNFRSF10D <0.0001 0.57 HADH.1 HADH <0.0001 0.62 DR5.2 TNFRSF10B <0.0001 0.64 GSTT1.3 GSTT1 0.0112 0.82 TNFAIP6.1 TNFAIP6 0.0001 0.74 GSTp.3 GSTP1 <0.0001 0.64 TNF.1 TNF 0.0138 0.80 GSTM3.2 GSTM3 <0.0001 0.57 TMEM47.1 TMEM47 <0.0001 0.41 GSTM1.1 GSTM1 <0.0001 0.55 TMEM27.1 TMEM27 <0.0001 0.53 GRB7.2 GRB7 <0.0001 0.66 TLR3.1 TLR3 <0.0001 0.68 GRB14.1 GRB14 0.0123 0.81 TIMP3.3 TIMP3 <0.0001 0.39 GPX3.1 GPX3 0.0120 0.82 TIMP2.1 TIMP2 <0.0001 0.68 GNAS.1 GNAS 0.0003 0.74 THBS1.1 THBS1 <0.0001 0.65 GJA1.1 GJA1 0.0034 0.79 TGFBR2.3 TGFBR2 <0.0001 0.41 GFRA1.1 GFRA1 0.0164 0.82 TGFBR1.1 TGFBR1 <0.0001 0.59 GCLM.2 GCLM 0.0056 0.80 TGFB2.2 TGFB2 <0.0001 0.50 GCLC.3 GCLC <0.0001 0.49 TGFb1.1 TGFB1 <0.0001 0.67 GBP2.2 GBP2 0.0388 0.84 TGFA.2 TGFA <0.0001 0.63 GATM.1 GATM <0.0001 0.59 TEK.1 TEK <0.0001 0.34 GATA3.3 GATA3 0.0002 0.72 TCF4.1 TCF4 <0.0001 0.47 FOS.1 FOS <0.0001 0.65 TAP1.1 TAP1 0.0017 0.77 FOLR1.1 FOLR1 <0.0001 0.65 TAGLN.1 TAGLN 0.0001 0.72 FLT4.1 FLT4 <0.0001 0.37 TACSTD2.1 TACSTD2 <0.0001 0.58 FLT3LG.1 FLT3LG <0.0001 0.61 SUCLG1.1 SUCLG1 <0.0001 0.56 FLT1.1 FLT1 <0.0001 0.40 STK11.1 STK11 <0.0001 0.58 FILIP1.1 FILIP1 <0.0001 0.47 STAT5B.2 STAT5B <0.0001 0.50 FIGF.1 FIGF <0.0001 0.53 STAT5A.1 STAT5A <0.0001 0.60 FHL1.1 FHL1 <0.0001 0.52 STAT3.1 STAT3 0.0001 0.72 FHIT.1 FHIT <0.0001 0.54 STAT1.3 STAT1 0.0344 0.84 FH.1 FH <0.0001 0.67 SPRY1.1 SPRY1 <0.0001 0.57 FGFR2 isoform 1.1 FGFR2 <0.0001 0.62 SPAST.1 SPAST 0.0142 0.82 FGFR1.3 FGFR1 <0.0001 0.63 SPARCL1.1

SPARCL1 <0.0001 0.59 FGF2.2 FGF2 <0.0001 0.58 SPARC.1 SPARC <0.0001 0.69 FGF1.1 FGF1 <0.0001 0.66 SOD1.1 SOD1 0.0245 0.83 FDPS.1 FDPS <0.0001 0.47 SNRK.1 SNRK <0.0001 0.35 FBXW7.1 FBXW7 <0.0001 0.66 SNAI1.1 SNAI1 <0.0001 0.70 fas.1 FAS <0.0001 0.66 MADH4.1 SMAD4 <0.0001 0.47 ESRRG.3 ESRRG 0.0001 0.72 MADH2.1 SMAD2 <0.0001 0.50 ERG.1 ERG <0.0001 0.44 SLC9A1.1 SLC9A1 0.0207 0.82 ERCC1.2 ERCC1 <0.0001 0.60 SLC34A1.1 SLC34A1 <0.0001 0.63 ERBB4.3 ERBB4 0.0018 0.74 SLC22A6.1 SLC22A6 0.0010 0.76 ErbB3.1 ERBB3 0.0031 0.79 SKIL.1 SKIL <0.0001 0.64 HER2.3 ERBB2 <0.0001 0.53 PTPNS1.1 SIRPA 0.0075 0.81 EPHB4.1 EPHB4 <0.0001 0.51 SHANK3.1 SHANK3 <0.0001 0.37 EPHA2.1 EPHA2 <0.0001 0.40 SGK.1 SGK1 <0.0001 0.55 EPAS1.1 EPAS1 <0.0001 0.38 FRP1.3 SFRP1 0.0156 0.82 ENPP2.1 ENPP2 0.0001 0.72 SEMA3F.3 SEMA3F <0.0001 0.50 ENPEP.1 ENPEP <0.0001 0.65 SELPLG.1 SELPLG 0.0022 0.78 CD105.1 ENG <0.0001 0.38 SELENBP1.1 SELENBP1 0.0003 0.75 EMP1.1 EMP1 <0.0001 0.64 SDPR.1 SDPR <0.0001 0.42 EMCN.1 EMCN <0.0001 0.27 SDHA.1 SDHA <0.0001 0.65 ELTD1.1 ELTD1 <0.0001 0.67 SCNN1A.2 SCNN1A 0.0024 0.77 EIF2C1.1 EIF2C1 <0.0001 0.51 SCN4B.1 SCN4B <0.0001 0.50 EGR1.1 EGR1 <0.0001 0.59 S100A2.1 S100A2 0.0008 0.75 EGLN3.1 EGLN3 0.0002 0.75 KIAA1303 RPTOR <0.0001 0.60 EGFR.2 EGFR 0.0072 0.81 raptor.1 RPS6KB1.3 RPS6KB1 <0.0001 0.60 EGF.3 EGF 0.0051 0.77 RPS6KAI.1 RPS6KA1 0.0002 0.71 EFNB2.1 EFNB2 <0.0001 0.45 RPS23.1 RPS23 0.0002 0.73 EFNB1.2 EFNB1 <0.0001 0.55 ROCK2.1 ROCK2 <0.0001 0.52 EEF1A1.1 EEF1A1 <0.0001 0.55 ROCK1.1 ROCK1 <0.0001 0.37 EDNRB.1 EDNRB <0.0001 0.44 RIPK1.1 RIPK1 <0.0001 0.55 EDN2.1 EDN2 0.0012 0.74 rhoC.1 RHOC <0.0001 0.66 EDN1 EDN1 <0.0001 0.53 endothelin.1 RhoB.1 RHOB <0.0001 0.57 EBAG9.1 EBAG9 0.0240 0.83 ARHA.1 RHOA <0.0001 0.50 DUSP1.1 DUSP1 0.0130 0.82 RHEB.2 RHEB <0.0001 0.61 DPYS.1 DPYS 0.0355 0.85 RGS5.1 RGS5 <0.0001 0.37 DPEP1.1 DPEP1 <0.0001 0.66 FLJ22655.1 RERGL <0.0001 0.33 DLL4.1 DLL4 <0.0001 0.66 RB1.1 RB1 <0.0001 0.61 DLC1.1 DLC1 <0.0001 0.42 RASSF1.1 RASSF1 0.0002 0.75 DKFZP564O0823.1 DKFZP564O0823 <0.0001 0.51 RARB.2 RARB <0.0001 0.37 DICER1.2 DICER1 <0.0001 0.50 RALBP1.1 RALBP1 <0.0001 0.43 DIAPH1.1 DIAPH1 0.0219 0.83 RAF1.3 RAF1 <0.0001 0.59 DIABLO.1 DIABLO 0.0022 0.78 RAC1.3 RAC1 0.0356 0.84 DHPS.3 DHPS <0.0001 0.55 PTPRG.1 PTPRG <0.0001 0.35 DET1.1 DET1 0.0005 0.74 PTPRB.1 PTPRB <0.0001 0.31 DEFB1.1 DEFB1 0.0002 0.73 PTN.1 PTN <0.0001 0.56 DDC.1 DDC <0.0001 0.72 PTK2.1 PTK2 <0.0001 0.45 DAPK1.3 DAPK1 <0.0001 0.42 PTHR1.1 PTH1R <0.0001 0.45 CYR61.1 CYR61 <0.0001 0.59 PTEN.2 PTEN <0.0001 0.51 CXCL12.1 CXCL12 <0.0001 0.62 PSMB9.1 PSMB9 0.0139 0.82 CX3CR1.1 CX3CR1 <0.0001 0.46 PSMB8.1 PSMB8 0.0243 0.83 CX3CL1.1 CX3CL1 <0.0001 0.47 PSMA7.1 PSMA7 0.0101 0.81 CUL1.1 CUL1 <0.0001 0.64 PRSS8.1 PRSS8 0.0025 0.78 CUBN.1 CUBN <0.0001 0.55 PRPS2.1 PRPS2 0.0190 0.82 CTSS.1 CTSS 0.0007 0.76 PRKCH.1 PRKCH <0.0001 0.48 CTSH.2 CTSH <0.0001 0.64 PRKCD.2 PRKCD 0.0001 0.72 B-Catenin.3 CTNNB1 <0.0001 0.54 PPP2CA.1 PPP2CA <0.0001 0.58 A-Catenin.2 CTNNA1 <0.0001 0.65 PPARG.3 PPARG <0.0001 0.40 CTGF.1 CTGF <0.0001 0.71 PPAP2B.1 PPAP2B <0.0001 0.31 CSF2RA.2 CSF2RA 0.0037 0.78 PLG.1 PLG <0.0001 0.53 CSF1R.2 CSF1R <0.0001 0.67 PLAT.1 PLAT <0.0001 0.54 CSF1.1 CSF1 <0.0001 0.65 PLA2G4C.1 PLA2G4C <0.0001 0.65 CRADD.1 CRADD 0.0032 0.79 PIK3CA.1 PIK3CA <0.0001 0.53 COL4A2.1 COL4A2 <0.0001 0.65 PI3K.2 PIK3C2B <0.0001 0.40 COL4A1.1 COL4A1 0.0067 0.80 PFKP.1 PFKP 0.0124 0.82 COL18A1.1 COL18A1 0.0001 0.72 CD31.3 PECAM1 <0.0001 0.42 CLU.3 CLU 0.0004 0.75 PDZK3.1 PDZK3 0.0003 0.72 CLDN10.1 CLDN10 <0.0001 0.65 PDZK1.1 PDZK1 <0.0001 0.58 CLCNKB.1 CLCNKB <0.0001 0.52 PDGFRb.3 PDGFRB <0.0001 0.62 CFLAR.1 CFLAR <0.0001 0.60 PDGFD.2 PDGFD <0.0001 0.43 CEACAM1.1 CEACAM1 <0.0001 0.55 PDGFC.3 PDGFC <0.0001 0.51 p21.3 CDKN1A 0.0002 0.73 PDGFB.3 PDGFB <0.0001 0.40 CDH6.1 CDH6 <0.0001 0.72 PDGFA.3 PDGFA <0.0001 0.57 CDH5.1 CDH5 <0.0001 0.44 PCK1.1 PCK1 <0.0001 0.60 CDH2.1 CDH2 0.0392 0.85 PCCA.1 PCCA <0.0001 0.58 CDH16.1 CDH16 <0.0001 0.70 PARD6A.1 PARD6A 0.0042 0.79 CDH13.1 CDH13 <0.0001 0.53 Pak1.2 PAK1 <0.0001 0.69 CDC25B.1 CDC25B 0.0037 0.79 PAH.1 PAH 0.0309 0.84 CD4.1 CD4 <0.0001 0.72 OGG1.1 OGG1 0.0024 0.78 CD36.1 CD36 <0.0001 0.51 BFGF.3 NUDT6 <0.0001 0.46 CD34.1 CD34 <0.0001 0.48 NPR1.1 NPR1 <0.0001 0.58 CD24.1 CD24 0.0206 0.83 NPM1.2 NPM1 <0.0001 0.65 CD14.1 CD14 0.0152 0.82 NOTCH3.1 NOTCH3 <0.0001 0.58 CCND1.3 CCND1 <0.0001 0.51 NOTCH2.1 NOTCH2 <0.0001 0.64 CCL4.2 CCL4 0.0017 0.77 NOTCH1.1 NOTCH1 <0.0001 0.44 MCP1.1 CCL2 <0.0001 0.66 NOS3.1 NOS3 <0.0001 0.44 CAT.1 CAT <0.0001 0.53 NOS2A.3 NOS2 <0.0001 0.49 CASP10.1 CASP10 0.0001 0.72 NOL3.1 NOL3 0.0003 0.76 CALD1.2 CALD1 <0.0001 0.55 NFX1.1 NFX1 <0.0001 0.50 CACNA2D1.1 CACNA2D1 0.0352 0.84 NFKBp50.3 NFKB1 <0.0001 0.59 CA9.3 CA9 0.0299 0.84 NFATC2.1 NFATC2 <0.0001 0.64 CA2.1 CA2 <0.0001 0.58 NFAT5.1 NFAT5 <0.0001 0.65 C3AR1.1 C3AR1 0.0010 0.77 MYRIP.2 MYRIP 0.0004 0.72 ECRG4.1 C2orf40 <0.0001 0.47 MYH11.1 MYH11 <0.0001 0.50 C1QA.1 C1QA 0.0119 0.82 cMYC.3 MYC <0.0001 0.70 C13orf15.1 C13orf15 <0.0001 0.37 MX1.1 MX1 0.0103 0.81 BUB3.1 BUB3 <0.0001 0.67 MVP.1 MVP 0.0002 0.74 BTRC.1 BTRC <0.0001 0.65 MUC1.2 MUC1 0.0005 0.75 CIAP1.2

BIRC2 <0.0001 0.64 FRAP1.1 MTOR <0.0001 0.61 BIN1.3 BIN1 0.0001 0.73 MSH3.2 MSH3 <0.0001 0.60 BGN.1 BGN <0.0001 0.63 MSH2.3 MSH2 <0.0001 0.53 Bclx.2 BCL2L1 <0.0001 0.58 STMY3.3 MMP11 0.0034 0.79 Bcl2.2 BCL2 <0.0001 0.37 GBL.1 MLST8 0.0011 0.77 Bax.1 BAX 0.0035 0.78 MIF.2 MIF 0.0008 0.76 Bak.2 BAK1 0.0215 0.83 MICA.1 MICA <0.0001 0.70 BAG1.2 BAG1 <0.0001 0.40 MGMT.1 MGMT <0.0001 0.60 ATP6V1B1.1 ATP6V1B1 <0.0001 0.54 MCM3.3 MCM3 <0.0001 0.68 ATP1A1.1 ATP1A1 0.0037 0.78 MCAM.1 MCAM <0.0001 0.52 ASS1.1 ASS1 <0.0001 0.60 MARCKS.1 MARCKS 0.0001 0.73 ARRB1.1 ARRB1 <0.0001 0.45 ERK1.3 MAPK3 <0.0001 0.48 ARHGDIB.1 ARHGDIB <0.0001 0.50 ERK2.3 MAPK1 0.0221 0.83 AQP1.1 AQP1 <0.0001 0.40 MAP4.1 MAP4 <0.0001 0.63 APOLD1.1 APOLD1 <0.0001 0.54 MAP2K3.1 MAP2K3 <0.0001 0.59 APC.4 APC <0.0001 0.57 MAP2K1.1 MAP2K1 0.0002 0.74 ANXA4.1 ANXA4 0.0003 0.74 MAL2.1 MAL2 <0.0001 0.64 ANXA1.2 ANXA1 0.0001 0.73 MAL.1 MAL <0.0001 0.49 ANTXR1.1 ANTXR1 0.0051 0.80 LYZ.1 LYZ 0.0318 0.84 ANGPTL4.1 ANGPTL4 0.0041 0.80 LTF.1 LTF 0.0131 0.80 ANGPTL3.3 ANGPTL3 0.0258 0.82 LRP2.1 LRP2 <0.0001 0.63 ANGPTL2.1 ANGPTL2 0.0015 0.77 LMO2.1 LMO2 <0.0001 0.56 ANGPT2.1 ANGPT2 <0.0001 0.72 LDB2.1 LDB2 <0.0001 0.41 ANGPT1.1 ANGPT1 <0.0001 0.45 LDB1.2 LDB1 <0.0001 0.54 ALDOB.1 ALDOB <0.0001 0.60 LAMA4.1 LAMA4 0.0004 0.75 ALDH6A1.1 ALDH6A1 <0.0001 0.55 KRT7.1 KRT7 <0.0001 0.60 ALDH4.2 ALDH4A1 0.0124 0.82 K-ras.10 KRAS <0.0001 0.68 AKT3.2 AKT3 <0.0001 0.43 KL.1 KL <0.0001 0.49 AKT2.3 AKT2 <0.0001 0.64 Kitlng.4 KITLG <0.0001 0.43 AKT1.3 AKT1 <0.0001 0.58 c-kit.2 KIT <0.0001 0.60 AIF1.1 AIF1 0.0002 0.74 KDR.6 KDR <0.0001 0.36 AHR.1 AHR <0.0001 0.59 KCNJ15.1 KCNJ15 <0.0001 0.54 AGTR1.1 AGTR1 <0.0001 0.36 HTATIP.1 KAT5 <0.0001 0.40 ADH1B.1 ADH1B 0.0015 0.77 G-Catenin.1 JUP <0.0001 0.42 ADD1.1 ADD1 <0.0001 0.40 AP-1 (JUN JUN 0.0001 0.73 ADAMTS5.1 ADAMTS5 0.0006 0.73 official).2 JAG1.1 JAG1 <0.0001 0.42 ADAM17.1 ADAM17 <0.0001 0.72 ITGB5.1 ITGB5 0.0115 0.81 ACE2.1 ACE2 <0.0001 0.61 ITGB1.1 ITGB1 <0.0001 0.64 ACADSB.1 ACADSB <0.0001 0.54 ITGA7.1 ITGA7 <0.0001 0.54 BCRP.1 ABCG2 <0.0001 0.41 ITGA6.2 ITGA6 <0.0001 0.51 MRP4.2 ABCC4 <0.0001 0.65 ITGA5.1 ITGA5 0.0325 0.84 MRP3.1 ABCC3 0.0005 0.76 ITGA4.2 ITGA4 <0.0001 0.54 MRP1.1 ABCC1 0.0017 0.78 ITGA3.2 ITGA3 <0.0001 0.61 ABCB1.5 ABCB1 0.0003 0.75 IQGAP2.1 IQGAP2 <0.0001 0.63 NPD009 (ABAT ABAT <0.0001 0.70 official).3 INSR.1 INSR <0.0001 0.59 AAMP.1 AAMP 0.0292 0.84 IMP3.1 IMP3 <0.0001 0.54 A2M.1 A2M <0.0001 0.36

(161) TABLE-US-00007 TABLE 5a Proxy genes for which increased expression is associated with higher tumor grade (p-value ≤.05) Official CCF Grade Official CCF Grade Gene Symbol p-value OR Gene Symbol p-value OR WT1.1 WT1 <0.0001 1.39 HPD.1 HPD 0.0054 1.20 VTN.1 VTN 0.0359 1.16 HIST1H1D.1 HIST1H1D 0.0248 1.16 VDR.2 VDR 0.0001 1.29 HGD.1 HGD <0.0001 1.41 UBE2T.1 UBE2T <0.0001 1.81 GZMA.1 GZMA 0.0006 1.26 TP.3 TYMP <0.0001 1.35 GPX2.2 GPX2 0.0182 1.18 C20 orf1.1 TPX2 <0.0001 2.14 GPX1.2 GPX1 0.0008 1.25 TOP2A.4 TOP2A <0.0001 2.07 FCGR3A.1 FCGR3A 0.0003 1.27 TNFSF13B.1 TNFSF13B 0.0062 1.20 fasl.2 FASLG 0.0045 1.21 TK1.2 TK1 <0.0001 1.66 FABP1.1 FABP1 <0.0001 1.32 TGFBI.1 TGFBI 0.0452 1.14 F2.1 F2 <0.0001 1.77 STAT1.3 STAT1 <0.0001 1.31 ESPL1.3 ESPL1 <0.0001 1.60 SQSTM1.1 SQSTM1 0.0003 1.27 E2F1.3 E2F1 <0.0001 1.36 OPN, SPP1 0.0002 1.29 CXCR6.1 CXCR6 <0.0001 1.50 osteopontin.3 BLR1.1 CXCR5 0.0338 1.15 SLC7A5.2 SLC7A5 0.0002 1.28 CXCL9.1 CXCL9 0.0001 1.29 SLC16A3.1 SLC16A3 0.0052 1.21 CXCL10.1 CXCL10 0.0078 1.19 SLC13A3.1 SLC13A3 0.0003 1.27 GRO1.2 CXCL1 <0.0001 1.39 SFN.1 SFN 0.0066 1.20 CTSD.2 CTSD 0.0183 1.17 SEMA3C.1 SEMA3C <0.0001 1.32 CTSB.1 CTSB 0.0006 1.26 SAA2.2 SAA2 <0.0001 2.13 CRP.1 CRP 0.0342 1.15 S100A1.1 S100A1 <0.0001 1.40 CP.1 CP <0.0001 1.37 RRM2.1 RRM2 <0.0001 1.71 Chk2.3 CHEK2 <0.0001 1.37 RPLP1.1 RPLP1 0.0007 1.25 Chk1.2 CHEK1 <0.0001 1.37 RAD51.1 RAD51 <0.0001 1.53 CENPF.1 CENPF <0.0001 1.78 PTTG1.2 PTTG1 <0.0001 1.89 CD8A.1 CD8A <0.0001 1.35 PSMB9.1 PSMB9 0.0010 1.25 CD82.3 CD82 <0.0001 1.50 PSMB8.1 PSMB8 0.0181 1.17 TNFSF7.1 CD70 <0.0001 1.43 PRKCB1.1 PRKCB 0.0218 1.16 CCNE1.1 CCNE1 0.0002 1.29 PDCD1.1 PDCD1 <0.0001 1.43 CCNB1.2 CCNB1 <0.0001 1.75 PCSK6.1 PCSK6 0.0009 1.25 CCL5.2 CCL5 <0.0001 1.63 PCNA.2 PCNA 0.0041 1.21 CCL20.1 CCL20 0.0082 1.19 NME2.1 NME2 0.0106 1.19 CAV2.1 CAV2 0.0210 1.17 MYBL2.1 MYBL2 <0.0001 1.70 CA12.1 CA12 <0.0001 1.41 MMP9.1 MMP9 <0.0001 1.46 C3.1 C3 <0.0001 1.34 MMP14.1 MMP14 0.0003 1.28 C1QB.1 C1QB 0.0201 1.17 Ki-67.2 MKI67 <0.0001 1.70 BUB1.1 BUB1 <0.0001 2.16 mGST1.2 MGST1 <0.0001 2.13 BRCA1.2 BRCA1 0.0004 1.26 cMet.2 MET <0.0001 1.57 SURV.2 BIRC5 <0.0001 1.93 MDK.1 MDK <0.0001 1.55 cIAP2.2 BIRC3 <0.0001 1.44 MDH2.1 MDH2 <0.0001 1.35 STK15.2 AURKA <0.0001 1.38 MCM2.2 MCM2 <0.0001 1.33 ATP5E.1 ATP5E <0.0001 1.45 LOX.1 LOX <0.0001 1.35 APOL1.1 APOL1 <0.0001 1.47 LMNB1.1 LMNB1 <0.0001 1.76 APOE.1 APOE <0.0001 1.40 LIMK1.1 LIMK1 <0.0001 1.43 APOC1.3 APOC1 <0.0001 1.42 LAPTM5.1 LAPTM5 0.0044 1.21 ANXA2.2 ANXA2 0.0020 1.23 LAMB3.1 LAMB3 <0.0001 1.49 ANGPTL3.3 ANGPTL3 <0.0001 1.32 L1CAM.1 L1CAM 0.0338 1.15 AMACR1.1 AMACR 0.0382 1.15 KLRK1.2 KLRK1 0.0412 1.15 ALOX5.1 ALOX5 0.0001 1.29 CD18.2

ITGB2 0.0069 1.21 ALDH4.2 ALDH4A1 0.0002 1.28 IL-8.1 IL8 0.0088 1.19 ADAM8.1 ADAM8 0.0006 1.26 IL6.3 IL6 0.0091 1.19 MRP2.3 ABCC2 <0.0001 1.97 ICAM1.1 ICAM1 0.0014 1.24 HSPA8.1 HSPA8 <0.0001 1.39

(162) TABLE-US-00008 TABLE 5b Proxy genes for which increased expression is associated with lower tumor grade (p-value ≤.05) Official CCF Grade Official CCF Grade Gene Symbol p-value OR Gene Symbol p-value OR ZHX2.1 ZHX2 0.0061 0.83 ITGA4.2 ITGA4 <0.0001 0.75 YB-1.2 YBX1 <0.0001 0.62 ITGA3.2 ITGA3 0.0128 0.85 XPNPEP2.2 XPNPEP2 0.0040 0.82 IQGAP2.1 IQGAP2 <0.0001 0.68 XIAP.1 XIAP <0.0001 0.64 INSR.1 INSR <0.0001 0.46 WISP1.1 WISP1 <0.0001 0.58 INHBA.1 INHBA 0.0049 0.83 VWF.1 VWF <0.0001 0.34 IMP3.1 IMP3 <0.0001 0.67 VHL.1 VHL 0.0088 0.84 IL6ST.3 IL6ST <0.0001 0.43 VEGF.1 VEGFA <0.0001 0.48 IL1B.1 IL1B 0.0102 0.84 VCAN.1 VCAN 0.0023 0.82 IL15.1 IL15 0.0001 0.76 VCAM1.1 VCAM1 0.0049 0.83 IL10.3 IL10 0.0239 0.86 USP34.1 USP34 <0.0001 0.62 IGFBP6.1 IGFBP6 <0.0001 0.75 UMOD.1 UMOD <0.0001 0.69 IGFBP5.1 IGFBP5 <0.0001 0.74 UGCG.1 UGCG <0.0001 0.58 IGFBP2.1 IGFBP2 <0.0001 0.76 UBB.1 UBB <0.0001 0.62 IGF2.2 IGF2 <0.0001 0.69 UBE1C.1 UBA3 <0.0001 0.67 IGF1R.3 IGF1R <0.0001 0.43 tusc4.2 TUSC4 <0.0001 0.61 ID3.1 ID3 <0.0001 0.48 TUSC2.1 TUSC2 0.0481 0.88 ID2.4 ID2 <0.0001 0.64 TSPAN7.2 TSPAN7 <0.0001 0.29 ID1.1 ID1 <0.0001 0.37 TSC2.1 TSC2 <0.0001 0.60 ICAM2.1 ICAM2 <0.0001 0.42 TSC1.1 TSC1 <0.0001 0.52 HYAL2.1 HYAL2 <0.0001 0.32 P53.2 TP53 <0.0001 0.66 HYAL1.1 HYAL1 <0.0001 0.52 TOP2B.2 TOP2B <0.0001 0.68 HSPG2.1 HSPG2 <0.0001 0.33 TNIP2.1 TNIP2 0.0001 0.76 HSPA1A.1 HSPA1A 0.0022 0.81 TNFSF12.1 TNFSF12 <0.0001 0.54 HSP90AB1.1 HSP90AB1 <0.0001 0.68 TNFRSF11B.1 TNFRSF11B <0.0001 0.73 HSD11B2.1 HSD11B2 <0.0001 0.43 TNFRSF10D.1 TNFRSF10D <0.0001 0.51 HPCAL1.1 HPCAL1 <0.0001 0.69 TNFRSF10C.3 TNFRSF10C 0.0003 0.78 HNRPAB.3 HNRNPAB 0.0432 0.87 DR5.2 TNFRSF10B <0.0001 0.69 HMGB1.1 HMGB1 <0.0001 0.47 TNFAIP6.1 TNFAIP6 0.0338 0.87 HIF1AN.1 HIF1AN <0.0001 0.64 TNFAIP3.1 TNFAIP3 0.0083 0.84 HIF1A.3 HIF1A <0.0001 0.55 TNF.1 TNF 0.0392 0.87 HGF.4 HGF 0.0022 0.81 TMEM47.1 TMEM47 <0.0001 0.33 HDAC1.1 HDAC1 <0.0001 0.48 TMEM27.1 TMEM27 <0.0001 0.73 HADH.1 HADH <0.0001 0.63 TIMP3.3 TIMP3 <0.0001 0.29 GSTp.3 GSTP1 <0.0001 0.66 TIMP2.1 TIMP2 <0.0001 0.54 GSTM3.2 GSTM3 <0.0001 0.53 THBS1.1 THBS1 <0.0001 0.58 GSTM1.1 GSTM1 <0.0001 0.61 THBD.1 THBD <0.0001 0.66 GRB7.2 GRB7 <0.0001 0.73 TGFBR2.3 TGFBR2 <0.0001 0.32 GRB14.1 GRB14 0.0001 0.76 TGFBR1.1 TGFBR1 <0.0001 0.59 GPC3.1 GPC3 0.0012 0.80 TGFB2.2 TGFB2 <0.0001 0.54 GNAS.1 GNAS <0.0001 0.70 TGFb1.1 TGFB1 <0.0001 0.66 GMNN.1 GMNN 0.0006 0.80 TGFA.2 TGFA <0.0001 0.68 GJA1.1 GJA1 <0.0001 0.66 TEK.1 TEK <0.0001 0.34 GCLM.2 GCLM 0.0319 0.87 cripto (TDGF1 TDGF1 0.0328 0.86 GCLC.3 GCLC <0.0001 0.57 official).1 GATM.1 GATM 0.0006 0.79 TCF4.1 TCF4 <0.0001 0.33 GATA3.3 GATA3 0.0029 0.82 TAGLN.1 TAGLN <0.0001 0.58 GAS2.1 GAS2 0.0136 0.84 TACSTD2.1 TACSTD2 <0.0001 0.61 GADD45B.1 GADD45B <0.0001 0.54 SUCLG1.1 SUCLG1 <0.0001 0.76 FST.1 FST 0.0197 0.85 STK11.1 STK11 <0.0001 0.53 FOS.1 FOS <0.0001 0.44 STC2.1 STC2 0.0085 0.84 FOLR1.1 FOLR1 <0.0001 0.75 STAT5B.2 STAT5B <0.0001 0.40 FLT4.1 FLT4 <0.0001 0.34 STAT5A.1 STAT5A <0.0001 0.60 FLT3LG.1 FLT3LG <0.0001 0.73 STAT3.1 STAT3 <0.0001 0.48 FLT1.1 FLT1 <0.0001 0.29 SPRY1.1 SPRY1 <0.0001 0.40 FILIP1.1 FILIP1 <0.0001 0.47 SPAST.1 SPAST 0.0002 0.78 FIGF.1 FIGF <0.0001 0.70 SPARCL1.1 SPARCL1 <0.0001 0.41 FHL1.1 FHL1 <0.0001 0.54 SPARC.1 SPARC <0.0001 0.50 FHIT.1 FHIT <0.0001 0.72 SNRK.1 SNRK <0.0001 0.29 FH.1 FH 0.0203 0.86 SNAI1.1 SNAI1 <0.0001 0.54 FGFR2 isoform FGFR2 <0.0001 0.62 MADH4.1 SMAD4 <0.0001 0.39 1.1 MADH2.1 SMAD2 <0.0001 0.40 FGFR1.3 FGFR1 <0.0001 0.48 SLC9A1.1 SLC9A1 <0.0001 0.75 FGF2.2 FGF2 <0.0001 0.62 SLC34A1.1 SLC34A1 0.0001 0.76 FGF1.1 FGF1 <0.0001 0.61 SKIL.1 SKIL <0.0001 0.52 FDPS.1 FDPS <0.0001 0.52 SHC1.1 SHC1 0.0063 0.83 FBXW7.1 FBXW7 <0.0001 0.57 SHANK3.1 SHANK3 <0.0001 0.27 FAP.1 FAP 0.0440 0.87 SGK.1 SGK1 <0.0001 0.60 ESRRG.3 ESRRG 0.0340 0.87 FRP1.3 SFRP1 0.0003 0.78 ERG.1 ERG <0.0001 0.36 PAI1.3 SERPINE1 0.0115 0.85 ERCC4.1 ERCC4 0.0337 0.87 SEMA3F.3 SEMA3F <0.0001 0.45 ERCC1.2 ERCC1 <0.0001 0.59 SELENBP1.1 SELENBP1 <0.0001 0.63 ERBB4.3 ERBB4 <0.0001 0.66 SELE.1 SELE <0.0001 0.74 HER2.3 ERBB2 <0.0001 0.57 SDPR.1 SDPR <0.0001 0.35 EPHB4.1 EPHB4 <0.0001 0.43 SDHA.1 SDHA <0.0001 0.69 EPHA2.1 EPHA2 <0.0001 0.44 SCNN1A.2 SCNN1A 0.0011 0.80 EPAS1.1 EPAS1 <0.0001 0.26 SCN4B.1 SCN4B <0.0001 0.47 ENPP2.1 ENPP2 <0.0001 0.62 S100A2.1 S100A2 <0.0001 0.72 ENPEP.1 ENPEP <0.0001 0.75 RUNX1.1 RUNX1 0.0001 0.77 ENO2.1 ENO2 0.0449 0.88 RRM1.2 RRM1 0.0438 0.87 CD105.1 ENG <0.0001 0.30 KIAA1303 RPTOR <0.0001 0.53 EMP1.1 EMP1 <0.0001 0.42 raptor.1 EMCN.1 EMCN <0.0001 0.31 RPS6KB1.3 RPS6KB1 <0.0001 0.49 ELTD1.1 ELTD1 <0.0001 0.59 RPS23.1 RPS23 <0.0001 0.59 EIF2C1.1 EIF2C1 <0.0001 0.63 ROCK2.1 ROCK2 <0.0001 0.42 EGR1.1 EGR1 <0.0001 0.50 ROCK1.1 ROCK1 <0.0001 0.31 EGLN3.1 EGLN3 <0.0001 0.69 RIPK1.1 RIPK1 <0.0001 0.50 EGF.3 EGF 0.0200 0.85 rhoC.1 RHOC <0.0001 0.70 EFNB2.1 EFNB2 <0.0001 0.36 RhoB.1 RHOB <0.0001 0.36 EFNB1.2 EFNB1 <0.0001 0.46 ARHA.1 RHOA <0.0001 0.34

EEF1A1.1 EEF1A1 <0.0001 0.39 RHEB.2 RHEB <0.0001 0.72 EDNRB.1 EDNRB <0.0001 0.33 RGS5.1 RGS5 <0.0001 0.30 EDN2.1 EDN2 <0.0001 0.67 FLJ22655.1 RERGL <0.0001 0.42 EDN1 EDN1 <0.0001 0.41 NFKBp65.3 RELA <0.0001 0.69 endothelin.1 RB1.1 RB1 <0.0001 0.74 EBAG9.1 EBAG9 0.0057 0.83 RASSF1.1 RASSF1 <0.0001 0.60 DUSP1.1 DUSP1 <0.0001 0.54 RARB.2 RARB <0.0001 0.41 DPEP1.1 DPEP1 0.0001 0.76 RALBP1.1 RALBP1 <0.0001 0.47 DLL4.1 DLL4 <0.0001 0.49 RAF1.3 RAF1 <0.0001 0.48 DLC1.1 DLC1 <0.0001 0.33 RAC1.3 RAC1 <0.0001 0.75 DKFZP564O0823.1 DKFZP564O0823 <0.0001 0.46 PXDN.1 PXDN <0.0001 0.74 DICER1.2 DICER1 <0.0001 0.41 PTPRG.1 PTPRG <0.0001 0.29 DIAPH1.1 DIAPH1 0.0022 0.81 PTPRB.1 PTPRB <0.0001 0.27 DIABLO.1 DIABLO <0.0001 0.73 PTN.1 PTN <0.0001 0.58 DHPS.3 DHPS <0.0001 0.41 PTK2.1 PTK2 <0.0001 0.36 DET1.1 DET1 <0.0001 0.63 PTHR1.1 PTH1R <0.0001 0.50 DEFB1.1 DEFB1 0.0001 0.77 PTEN.2 PTEN <0.0001 0.40 DAPK1.3 DAPK1 <0.0001 0.48 PSMA7.1 PSMA7 0.0002 0.78 DAG1.1 DAG1 0.0150 0.85 PRPS2.1 PRPS2 0.0012 0.80 CYR61.1 CYR61 <0.0001 0.49 PROM2.1 PROM2 0.0326 0.87 CXCL12.1 CXCL12 <0.0001 0.64 PRKCH.1 PRKCH <0.0001 0.45 CX3CR1.1 CX3CR1 0.0008 0.80 PRKCD.2 PRKCD <0.0001 0.76 CX3CL1.1 CX3CL1 <0.0001 0.55 PPP2CA.1 PPP2CA <0.0001 0.68 CUL1.1 CUL1 <0.0001 0.62 PPARG.3 PPARG <0.0001 0.42 CUBN.1 CUBN 0.0081 0.84 PPAP2B.1 PPAP2B <0.0001 0.32 CTSL.2 CTSL1 0.0328 0.87 PMP22.1 PMP22 <0.0001 0.61 B-Catenin.3 CTNNB1 <0.0001 0.36 PLG.1 PLG <0.0001 0.75 A-Catenin.2 CTNNA1 <0.0001 0.73 PLAT.1 PLAT <0.0001 0.42 CTGF.1 CTGF <0.0001 0.61 PLA2G4C.1 PLA2G4C 0.0002 0.78 CSF1.1 CSF1 <0.0001 0.66 PIK3CA.1 PIK3CA <0.0001 0.48 CRADD.1 CRADD 0.0151 0.85 PI3K.2 PIK3C2B <0.0001 0.53 COL5A2.2 COL5A2 0.0155 0.85 PGF.1 PGF <0.0001 0.74 COL4A2.1 COL4A2 <0.0001 0.51 PFKP.1 PFKP 0.0008 0.80 COL4A1.1 COL4A1 <0.0001 0.59 CD31.3 PECAM1 <0.0001 0.32 COL1A2.1 COL1A2 0.0005 0.79 PDZK3.1 PDZK3 <0.0001 0.72 COL18A1.1 COL18A1 <0.0001 0.53 PDZK1.1 PDZK1 <0.0001 0.76 CLDN10.1 CLDN10 0.0351 0.87 PDGFRb.3 PDGFRB <0.0001 0.42 CLCNKB.1 CLCNKB <0.0001 0.73 PDGFRa.2 PDGFRA 0.0022 0.82 CFLAR.1 CFLAR <0.0001 0.45 PDGFD.2 PDGFD <0.0001 0.39 CEACAM1.1 CEACAM1 <0.0001 0.64 PDGFC.3 PDGFC <0.0001 0.57 p27.3 CDKN1B <0.0001 0.65 PDGFB.3 PDGFB <0.0001 0.33 p21.3 CDKN1A <0.0001 0.52 PDGFA.3 PDGFA <0.0001 0.42 CDK4.1 CDK4 0.0200 0.86 PCK1.1 PCK1 <0.0001 0.71 CDH5.1 CDH5 <0.0001 0.31 PCCA.1 PCCA <0.0001 0.75 CDH16.1 CDH16 0.0005 0.79 PARD6A.1 PARD6A 0.0088 0.84 CDH13.1 CDH13 <0.0001 0.40 Pak1.2 PAK1 0.0014 0.81 CD99.1 CD99 0.0001 0.77 PAH.1 PAH 0.0160 0.85 CD44.1 CD44_1 0.0216 0.86 OGG1.1 OGG1 0.0139 0.85 CD36.1 CD36 <0.0001 0.42 BFGF.3 NUDT6 <0.0001 0.63 CD34.1 CD34 <0.0001 0.39 NPR1.1 NPR1 <0.0001 0.42 CD14.1 CD14 0.0021 0.81 NPM1.2 NPM1 <0.0001 0.61 CCND1.3 CCND1 <0.0001 0.63 NOTCH3.1 NOTCH3 <0.0001 0.40 MCP1.1 CCL2 <0.0001 0.69 NOTCH2.1 NOTCH2 <0.0001 0.58 CAT.1 CAT 0.0014 0.81 NOTCH1.1 NOTCH1 <0.0001 0.38 CASP10.1 CASP10 <0.0001 0.65 NOS3.1 NOS3 <0.0001 0.42 CALD1.2 CALD1 <0.0001 0.44 NOS2A.3 NOS2 <0.0001 0.56 CACNA2D1.1 CACNA2D1 0.0003 0.78 NOL3.1 NOL3 <0.0001 0.61 CA9.3 CA9 0.0269 0.86 NFX1.1 NFX1 <0.0001 0.53 CA2.1 CA2 0.0001 0.77 NFKBp50.3 NFKB1 <0.0001 0.59 C7.1 C7 <0.0001 0.71 NFATC2.1 NFATC2 <0.0001 0.73 C3AR1.1 C3AR1 0.0032 0.82 NFAT5.1 NFAT5 <0.0001 0.58 ECRG4.1 C2orf40 <0.0001 0.50 MYRIP.2 MYRIP <0.0001 0.63 C13orf15.1 C13orf15 <0.0001 0.37 MYH11.1 MYH11 <0.0001 0.48 BUB3.1 BUB3 <0.0001 0.65 cMYC.3 MYC <0.0001 0.68 BTRC.1 BTRC <0.0001 0.61 MX1.1 MX1 0.0087 0.84 BNIP3.1 BNIP3 0.0018 0.81 MVP.1 MVP 0.0291 0.87 CIAP1.2 BIRC2 <0.0001 0.61 MUC1.2 MUC1 0.0043 0.83 BGN.1 BGN <0.0001 0.43 FRAP1.1 MTOR <0.0001 0.61 Bclx.2 BCL2L1 <0.0001 0.76 MT1X.1 MT1X 0.0276 0.86 Bcl2.2 BCL2 <0.0001 0.45 MSH3.2 MSH3 0.0001 0.76 BAG1.2 BAG1 <0.0001 0.47 MSH2.3 MSH2 <0.0001 0.60 AXL.1 AXL <0.0001 0.74 MMP2.2 MMP2 <0.0001 0.74 ATP6V1B1.1 ATP6V1B1 <0.0001 0.65 STMY3.3 MMP11 <0.0001 0.70 ASS1.1 ASS1 <0.0001 0.67 MIF.2 MIF 0.0004 0.79 ARRB1.1 ARRB1 <0.0001 0.49 MICA.1 MICA <0.0001 0.60 ARHGDIB.1 ARHGDIB <0.0001 0.50 MGMT.1 MGMT <0.0001 0.57 ARF1.1 ARF1 <0.0001 0.69 MCM3.3 MCM3 <0.0001 0.73 AREG.2 AREG 0.0007 0.80 MCAM.1 MCAM <0.0001 0.42 AQP1.1 AQP1 <0.0001 0.49 MARCKS.1 MARCKS <0.0001 0.63 APOLD1.1 APOLD1 <0.0001 0.40 ERK1.3 MAPK3 <0.0001 0.35 APC.4 APC <0.0001 0.62 ERK2.3 MAPK1 <0.0001 0.71 APAF1.2 APAF1 <0.0001 0.76 MAP4.1 MAP4 <0.0001 0.58 ANXA1.2 ANXA1 <0.0001 0.65 MAP2K3.1 MAP2K3 <0.0001 0.60 ANTXR1.1 ANTXR1 <0.0001 0.61 MAP2K1.1 MAP2K1 <0.0001 0.62 ANGPTL4.1 ANGPTL4 <0.0001 0.75 MAL.1 MAL <0.0001 0.63 ANGPTL2.1 ANGPTL2 <0.0001 0.60 LRP2.1 LRP2 0.0275 0.86 ANGPT2.1 ANGPT2 <0.0001 0.61 LMO2.1 LMO2 <0.0001 0.68 ANGPT1.1 ANGPT1 <0.0001 0.33 LDB2.1 LDB2 <0.0001 0.28 ALDOB.1 ALDOB 0.0348 0.87 LDB1.2 LDB1 <0.0001 0.52 ALDH6A1.1 ALDH6A1 <0.0001 0.61 LAMB1.1 LAMB1 <0.0001 0.73 AKT3.2 AKT3 <0.0001 0.30 LAMA4.1 LAMA4 <0.0001 0.55 AKT2.3 AKT2 <0.0001 0.65 KRT7.1 KRT7 <0.0001 0.61 AKT1.3 AKT1 <0.0001 0.50 K-ras.10 KRAS <0.0001 0.54 AHR.1 AHR <0.0001 0.55 KL.1 KL <0.0001 0.62 AGTR1.1 AGTR1 <0.0001 0.44 Kitlng.4 KITLG <0.0001 0.46 ADH1B.1 ADH1B <0.0001 0.70 c-kit.2 KIT <0.0001 0.48 ADD1.1 ADD1 <0.0001 0.40 KDR.6 KDR <0.0001 0.33

ADAMTS9.1 ADAMTS9 <0.0001 0.64 KCNJ15.1 KCNJ15 <0.0001 0.67 ADAMTS5.1 ADAMTS5 <0.0001 0.56 HTATIP.1 KAT5 <0.0001 0.33 ADAMTS4.1 ADAMTS4 <0.0001 0.66 G-Catenin.1 JUP <0.0001 0.39 ADAMTS2.1 ADAMTS2 <0.0001 0.69 AP-1 (JUN JUN <0.0001 0.54 ADAMTS1.1 ADAMTS1 <0.0001 0.51 official).2 ADAM17.1 ADAM17 <0.0001 0.67 JAG1.1 JAG1 <0.0001 0.28 ACADSB.1 ACADSB <0.0001 0.63 ITGB5.1 ITGB5 <0.0001 0.60 BCRP.1 ABCG2 <0.0001 0.43 ITGB3.1 ITGB3 <0.0001 0.68 MRP4.2 ABCC4 0.0337 0.87 ITGB1.1 ITGB1 <0.0001 0.46 AAMP.1 AAMP <0.0001 0.71 ITGA7.1 ITGA7 <0.0001 0.40 A2M.1 A2M <0.0001 0.29 ITGA6.2 ITGA6 <0.0001 0.51 ITGA5.1 ITGA5 <0.0001 0.62

(163) TABLE-US-00009 TABLE 6a Proxy genes for which increased expression is associated with the presence of necrosis (p-value ≤.05) Official CCF Necrosis Official CCF Necrosis Gene Symbol p-value OR Gene Symbol p-value OR WT1.1 WT1 <0.0001 1.57 KRT19.3 KRT19 0.0001 1.43 VTN.1 VTN <0.0001 1.38 ITGB4.2 ITGB4 0.0492 1.19 VDR.2 VDR 0.0013 1.34 ISG20.1 ISG20 0.0006 1.38 UBE2T.1 UBE2T <0.0001 2.08 IL-8.1 IL8 <0.0001 2.40 TP.3 TYMP 0.0008 1.37 IL6.3 IL6 <0.0001 2.02 TSPAN8.1 TSPAN8 0.0016 1.29 ICAM1.1 ICAM1 <0.0001 1.75 C20 orf1.1 TPX2 <0.0001 2.64 HSPA8.1 HSPA8 0.0004 1.38 TOP2A.4 TOP2A <0.0001 2.02 HIST1H1D.1 HIST1H1D 0.0017 1.33 TNFSF13B.1 TNFSF13B 0.0002 1.38 GPX1.2 GPX1 <0.0001 1.46 TK1.2 TK1 <0.0001 1.71 FZD2.2 FZD2 0.0031 1.27 TIMP1.1 TIMP1 0.0076 1.27 FN1.1 FN1 <0.0001 1.45 TGFBI.1 TGFBI <0.0001 1.69 FCGR3A.1 FCGR3A 0.0001 1.43 OPN, SPP1 <0.0001 1.81 FCER1G.2 FCER1G <0.0001 1.50 osteopontin.3 FAP.1 FAP 0.0395 1.20 SPHK1.1 SPHK1 <0.0001 1.44 F3.1 F3 <0.0001 1.62 SLC7A5.2 SLC7A5 <0.0001 2.12 F2.1 F2 <0.0001 1.76 SLC2A1.1 SLC2A1 <0.0001 1.46 ESPL1.3 ESPL1 0.0008 1.33 SLC16A3.1 SLC16A3 0.0001 1.48 EPHB2.1 EPHB2 0.0037 1.28 SLC13A3.1 SLC13A3 0.0019 1.29 EPHB1.3 EPHB1 0.0041 1.25 SHC1.1 SHC1 0.0156 1.24 EPB41L3.1 EPB41L3 0.0410 1.19 SFN.1 SFN <0.0001 1.57 ENO2.1 ENO2 0.0001 1.46 PAI1.3 SERPINE1 0.0164 1.25 EIF4EBP1.1 EIF4EBP1 <0.0001 1.60 SERPINA5.1 SERPINA5 0.0016 1.27 E2F1.3 E2F1 <0.0001 1.65 SEMA3C.1 SEMA3C <0.0001 2.14 CXCR6.1 CXCR6 <0.0001 1.48 SELL.1 SELL 0.0096 1.26 CXCR4.3 CXCR4 0.0118 1.26 SAA2.2 SAA2 <0.0001 2.50 GRO1.2 CXCL1 <0.0001 1.83 RRM2.1 RRM2 <0.0001 1.86 CTSB.1 CTSB <0.0001 1.70 RPLP1.1 RPLP1 0.0025 1.32 CRP.1 CRP 0.0494 1.16 RND3.1 RND3 0.0002 1.41 CP.1 CP <0.0001 1.98 RAD51.1 RAD51 <0.0001 1.51 COL7A1.1 COL7A1 <0.0001 1.47 PTTG1.2 PTTG1 <0.0001 2.52 COL1A1.1 COL1A1 0.0059 1.28 COX2.1 PTGS2 0.0002 1.36 Chk2.3 CHEK2 0.0010 1.33 PRKCB1.1 PRKCB 0.0188 1.23 Chk1.2 CHEK1 <0.0001 1.49 PRKCA.1 PRKCA 0.0339 1.21 CENPF.1 CENPF <0.0001 2.08 PLAUR.3 PLAUR <0.0001 1.60 CD82.3 CD82 <0.0001 2.09 upa.3 PLAU 0.0002 1.41 CD68.2 CD68 0.0163 1.24 PF4.1 PF4 0.0003 1.34 CD44s.1 CD44_s <0.0001 1.66 PDCD1.1 PDCD1 <0.0001 1.40 CCNE2.2 CCNE2_2 <0.0001 1.50 PCSK6.1 PCSK6 0.0297 1.22 CCNE1.1 CCNE1 <0.0001 1.44 PCNA.2 PCNA 0.0025 1.33 CCNB1.2 CCNB1 <0.0001 2.28 NNMT.1 NNMT 0.0206 1.22 CCL5.2 CCL5 <0.0001 1.45 NME2.1 NME2 0.0124 1.25 CCL20.1 CCL20 0.0121 1.24 MYBL2.1 MYBL2 <0.0001 1.90 CAV2.1 CAV2 0.0003 1.35 MT1X.1 MT1X 0.0003 1.39 CA12.1 CA12 <0.0001 2.11 MMP9.1 MMP9 <0.0001 1.96 C3.1 C3 <0.0001 1.58 MMP7.1 MMP7 <0.0001 1.50 C1QB.1 C1QB 0.0032 1.31 MMP14.1 MMP14 <0.0001 1.50 BUB1.1 BUB1 <0.0001 2.25 Ki-67.2 MKI67 <0.0001 1.96 BRCA1.2 BRCA1 0.0006 1.35 mGST1.2 MGST1 <0.0001 1.63 SURV.2 BIRC5 <0.0001 2.11 cMet.2 MET 0.0357 1.22 cIAP2.2 BIRC3 <0.0001 1.47 MDK.1 MDK <0.0001 1.78 BCL2A1.1 BCL2A1 0.0004 1.34 MCM2.2 MCM2 0.0003 1.40 STK15.2 AURKA <0.0001 1.61 LRRC2.1 LRRC2 0.0114 1.22 PRO2000.3 ATAD2 0.0166 1.24 LOX.1 LOX <0.0001 1.99 APOL1.1 APOL1 <0.0001 1.54 LMNB1.1 LMNB1 <0.0001 2.04 APOC1.3 APOC1 0.0026 1.30 LIMK1.1 LIMK1 <0.0001 2.51 ANXA2.2 ANXA2 <0.0001 1.71 LGALS9.1 LGALS9 0.0136 1.25 ALOX5.1 ALOX5 0.0004 1.38 LGALS1.1 LGALS1 <0.0001 1.46 ADAM8.1 ADAM8 <0.0001 1.89 LAPTM5.1 LAPTM5 <0.0001 1.47 MRP2.3 ABCC2 0.0002 1.39 LAMB3.1 LAMB3 <0.0001 1.96 L1CAM.1 L1CAM <0.0001 1.43

(164) TABLE-US-00010 TABLE 6b Proxy genes for which increased expression is associated with the absence of necrosis (p-value ≤.05) Official CCF Necrosis Official CCF Necrosis Gene Symbol p-value OR Gene Symbol p-value OR YB-1.2 YBX1 0.0010 0.74 IGF2.2 IGF2 0.0117 0.79 XIAP.1 XIAP <0.0001 0.68 IGF1R.3 IGF1R <0.0001 0.38 WWOX.5 WWOX <0.0001 0.63 ID3.1 ID3 <0.0001 0.44 WISP1.1 WISP1 0.0002 0.71 ID2.4 ID2 <0.0001 0.68 VWF.1 VWF <0.0001 0.35 ID1.1 ID1 <0.0001 0.32 VHL.1 VHL 0.0086 0.78 ICAM2.1 ICAM2 <0.0001 0.47 VEGF.1 VEGFA <0.0001 0.50 HYAL2.1 HYAL2 <0.0001 0.28 VCAM1.1 VCAM1 <0.0001 0.55 HYAL1.1 HYAL1 <0.0001 0.39 USP34.1 USP34 <0.0001 0.64 HSPG2.1 HSPG2 <0.0001 0.33 UMOD.1 UMOD <0.0001 0.36 HSP90AB1.1 HSP90AB1 0.0004 0.73 UGCG.1 UGCG <0.0001 0.54 HSD11B2.1 HSD11B2 <0.0001 0.32 UBB.1 UBB <0.0001 0.48 Hepsin.1 HPN <0.0001 0.59 UBE1C.1 UBA3 <0.0001 0.59 HPCAL1.1 HPCAL1 <0.0001 0.68 TS.1 TYMS <0.0001 0.70 HMGB1.1 HMGB1 <0.0001 0.42 tusc4.2 TUSC4 <0.0001 0.68 HLA-DPB1.1 HLA-DPB1 0.0002 0.72 TUSC2.1 TUSC2 0.0462 0.83 HIF1AN.1 HIF1AN <0.0001 0.54 TSPAN7.2 TSPAN7 <0.0001 0.25 HDAC1.1 HDAC1 <0.0001 0.55 TSC2.1 TSC2 <0.0001 0.45 HAVCR1.1 HAVCR1 0.0012 0.76 TSC1.1 TSC1 <0.0001 0.45 HADH.1 HADH <0.0001 0.49 P53.2 TP53

<0.0001 0.61 GSTT1.3 GSTT1 0.0067 0.80 TOP2B.2 TOP2B <0.0001 0.69 GSTp.3 GSTP1 <0.0001 0.55 TNFSF12.1 TNFSF12 <0.0001 0.51 GSTM3.2 GSTM3 <0.0001 0.48 TRAIL.1 TNFSF10 <0.0001 0.68 GSTM1.1 GSTM1 <0.0001 0.54 TNFRSF11B.1 TNFRSF11B <0.0001 0.59 GRB7.2 GRB7 <0.0001 0.49 TNFRSF10D.1 TNFRSF10D <0.0001 0.58 GPX3.1 GPX3 <0.0001 0.59 DR5.2 TNFRSF10B 0.0001 0.71 GPC3.1 GPC3 0.0287 0.81 TNFAIP6.1 TNFAIP6 <0.0001 0.67 GJA1.1 GJA1 0.0004 0.74 TMEM47.1 TMEM47 <0.0001 0.29 GFRA1.1 GFRA1 0.0011 0.74 TMEM27.1 TMEM27 <0.0001 0.40 GCLC.3 GCLC <0.0001 0.50 TLR3.1 TLR3 <0.0001 0.64 GATM.1 GATM <0.0001 0.45 TIMP3.3 TIMP3 <0.0001 0.23 GATA3.3 GATA3 0.0159 0.79 TIMP2.1 TIMP2 <0.0001 0.52 GADD45B.1 GADD45B <0.0001 0.67 THBS1.1 THBS1 <0.0001 0.62 FOS.1 FOS <0.0001 0.56 TGFBR2.3 TGFBR2 <0.0001 0.34 FOLR1.1 FOLR1 <0.0001 0.59 TGFBR1.1 TGFBR1 <0.0001 0.63 FLT4.1 FLT4 <0.0001 0.27 TGFB2.2 TGFB2 <0.0001 0.55 FLT3LG.1 FLT3LG <0.0001 0.62 TGFb1.1 TGFB1 0.0036 0.76 FLT1.1 FLT1 <0.0001 0.32 TGFA.2 TGFA <0.0001 0.56 FILIP1.1 FILIP1 <0.0001 0.42 TEK.1 TEK <0.0001 0.23 FIGF.1 FIGF 0.0001 0.62 TCF4.1 TCF4 <0.0001 0.36 FHL1.1 FHL1 <0.0001 0.36 TAGLN.1 TAGLN <0.0001 0.50 FHIT.1 FHIT <0.0001 0.63 TACSTD2.1 TACSTD2 <0.0001 0.64 FH.1 FH <0.0001 0.65 SUCLG1.1 SUCLG1 <0.0001 0.50 FGFR2 isoform FGFR2 <0.0001 0.51 STK11.1 STK11 <0.0001 0.48 1.1 STAT5B.2 STAT5B <0.0001 0.36 FGFR1.3 FGFR1 <0.0001 0.55 STAT5A.1 STAT5A <0.0001 0.56 FGF2.2 FGF2 <0.0001 0.61 STAT3.1 STAT3 <0.0001 0.63 FGF1.1 FGF1 <0.0001 0.49 SPRY1.1 SPRY1 <0.0001 0.42 FDPS.1 FDPS <0.0001 0.43 SPAST.1 SPAST 0.0004 0.74 FBXW7.1 FBXW7 <0.0001 0.60 SPARCL1.1 SPARCL1 <0.0001 0.48 fas.1 FAS 0.0054 0.79 SPARC.1 SPARC <0.0001 0.54 ESRRG.3 ESRRG <0.0001 0.68 SOD1.1 SOD1 <0.0001 0.67 ERG.1 ERG <0.0001 0.34 SNRK.1 SNRK <0.0001 0.25 ERCC4.1 ERCC4 0.0197 0.81 SNAI1.1 SNAI1 0.0004 0.71 ERCC1.2 ERCC1 <0.0001 0.60 MADH4.1 SMAD4 <0.0001 0.33 ERBB4.3 ERBB4 0.0037 0.72 MADH2.1 SMAD2 <0.0001 0.40 ErbB3.1 ERBB3 <0.0001 0.59 SLC34A1.1 SLC34A1 <0.0001 0.47 HER2.3 ERBB2 <0.0001 0.40 SLC22A6.1 SLC22A6 <0.0001 0.52 EPHB4.1 EPHB4 <0.0001 0.44 SKIL.1 SKIL <0.0001 0.59 EPHA2.1 EPHA2 <0.0001 0.36 SHANK3.1 SHANK3 <0.0001 0.25 EPAS1.1 EPAS1 <0.0001 0.29 SGK.1 SGK1 <0.0001 0.54 ENPP2.1 ENPP2 <0.0001 0.50 FRP1.3 SFRP1 0.0053 0.77 ENPEP.1 ENPEP <0.0001 0.56 SEMA3F.3 SEMA3F <0.0001 0.43 CD105.1 ENG <0.0001 0.31 SELENBP1.1 SELENBP1 <0.0001 0.62 EMP1.1 EMP1 <0.0001 0.49 SDPR.1 SDPR <0.0001 0.28 EMCN.1 EMCN <0.0001 0.23 SDHA.1 SDHA <0.0001 0.47 ELTD1.1 ELTD1 <0.0001 0.59 SCNN1A.2 SCNN1A 0.0013 0.73 EIF2C1.1 EIF2C1 <0.0001 0.52 SCN4B.1 SCN4B <0.0001 0.35 EGR1.1 EGR1 <0.0001 0.54 S100A2.1 S100A2 0.0355 0.82 EGLN3.1 EGLN3 <0.0001 0.69 KIAA1303 RPTOR <0.0001 0.49 EGFR.2 EGFR <0.0001 0.70 raptor.1 EFNB2.1 EFNB2 <0.0001 0.34 RPS6KB1.3 RPS6KB1 <0.0001 0.55 EFNB1.2 EFNB1 <0.0001 0.41 RPS6KAI.1 RPS6KA1 0.0002 0.68 EEF1A1.1 EEF1A1 <0.0001 0.32 RPS23.1 RPS23 <0.0001 0.47 EDNRB.1 EDNRB <0.0001 0.31 ROCK2.1 ROCK2 <0.0001 0.39 EDN2.1 EDN2 <0.0001 0.51 ROCK1.1 ROCK1 <0.0001 0.35 EDN1 EDN1 <0.0001 0.41 RIPK1.1 RIPK1 <0.0001 0.45 endothelin.1 rhoC.1 RHOC 0.0001 0.70 EBAG9.1 EBAG9 0.0007 0.74 RhoB.1 RHOB <0.0001 0.41 DUSP1.1 DUSP1 <0.0001 0.65 ARHA.1 RHOA <0.0001 0.45 DPYS.1 DPYS <0.0001 0.66 RHEB.2 RHEB 0.0002 0.69 DPEP1.1 DPEP1 <0.0001 0.34 RGS5.1 RGS5 <0.0001 0.26 DLL4.1 DLL4 <0.0001 0.49 FLJ22655.1 RERGL <0.0001 0.26 DLC1.1 DLC1 <0.0001 0.36 NFKBp65.3 RELA <0.0001 0.71 DKFZP564O0823.1 DKFZP564O0823 <0.0001 0.35 RB1.1 RB1 <0.0001 0.54 DICER1.2 DICER1 <0.0001 0.46 RASSF1.1 RASSF1 <0.0001 0.63 DIAPH1.1 DIAPH1 <0.0001 0.63 RARB.2 RARB <0.0001 0.28 DIABLO.1 DIABLO 0.0002 0.72 RALBP1.1 RALBP1 <0.0001 0.52 DHPS.3 DHPS <0.0001 0.46 RAF1.3 RAF1 <0.0001 0.58 DET1.1 DET1 <0.0001 0.61 RAC1.3 RAC1 0.0118 0.80 DEFB1.1 DEFB1 <0.0001 0.69 PTPRG.1 PTPRG <0.0001 0.27 DDC.1 DDC <0.0001 0.51 PTPRB.1 PTPRB <0.0001 0.22 DCXR.1 DCXR 0.0061 0.78 PTN.1 PTN <0.0001 0.41 DAPK1.3 DAPK1 <0.0001 0.47 PTK2.1 PTK2 <0.0001 0.33 CYR61.1 CYR61 <0.0001 0.52 PTHR1.1 PTH1R <0.0001 0.32 CYP3A4.2 CYP3A4 0.0398 0.82 PTEN.2 PTEN <0.0001 0.44 CYP2C8v2.1 CYP2C8_21 0.0001 0.67 PSMA7.1 PSMA7 0.0173 0.81 CXCL12.1 CXCL12 <0.0001 0.53 PRSS8.1 PRSS8 <0.0001 0.62 CX3CR1.1 CX3CR1 <0.0001 0.60 PRKCH.1 PRKCH <0.0001 0.43 CX3CL1.1 CX3CL1 <0.0001 0.34 PRKCD.2 PRKCD <0.0001 0.68 CUL1.1 CUL1 <0.0001 0.62 PPP2CA.1 PPP2CA <0.0001 0.53 CUBN.1 CUBN <0.0001 0.39 PPARG.3 PPARG <0.0001 0.37 CTSH.2 CTSH 0.0018 0.77 PPAP2B.1 PPAP2B <0.0001 0.27 B-Catenin.3 CTNNB1 <0.0001 0.38 PMP22.1 PMP22 0.0155 0.81 A-Catenin.2 CTNNA1 <0.0001 0.58 PLG.1 PLG <0.0001 0.38 CTGF.1 CTGF <0.0001 0.64 PLAT.1 PLAT <0.0001 0.42 CSF1R.2 CSF1R 0.0308 0.83 PLA2G4C.1 PLA2G4C <0.0001 0.48 CSF1.1 CSF1 0.0002 0.73 PIK3CA.1 PIK3CA <0.0001 0.47 CRADD.1 CRADD <0.0001 0.60 PI3K.2 PIK3C2B <0.0001 0.42 COL4A2.1 COL4A2 <0.0001 0.51 PGF.1 PGF 0.0153 0.80 COL4A1.1 COL4A1 <0.0001 0.66 PFKP.1 PFKP <0.0001 0.70 COL18A1.1 COL18A1 <0.0001 0.50 CD31.3 PECAM1 <0.0001 0.34 CLU.3 CLU 0.0091 0.80 PDZK3.1 PDZK3 <0.0001 0.49 CLDN7.2 CLDN7 0.0029 0.76 PDZK1.1 PDZK1 <0.0001 0.45 CLDN10.1 CLDN10 <0.0001 0.48 PDGFRb.3 PDGFRB <0.0001 0.45 CLCNKB.1 CLCNKB 0.0001 0.61 PDGFD.2 PDGFD

<0.0001 0.33 CFLAR.1 CFLAR <0.0001 0.47 PDGFC.3 PDGFC <0.0001 0.53 CEACAM1.1 CEACAM1 <0.0001 0.43 PDGFB.3 PDGFB <0.0001 0.33 p27.3 CDKN1B 0.0002 0.73 PDGFA.3 PDGFA <0.0001 0.43 p21.3 CDKN1A <0.0001 0.65 PCK1.1 PCK1 <0.0001 0.44 CDH6.1 CDH6 <0.0001 0.67 PCCA.1 PCCA <0.0001 0.47 CDH5.1 CDH5 <0.0001 0.33 PARD6A.1 PARD6A 0.0045 0.77 CDH2.1 CDH2 0.0003 0.75 Pak1.2 PAK1 0.0003 0.74 CDH16.1 CDH16 <0.0001 0.51 PAH.1 PAH <0.0001 0.62 CDH13.1 CDH13 <0.0001 0.39 OGG1.1 OGG1 <0.0001 0.62 CD36.1 CD36 <0.0001 0.41 BFGF.3 NUDT6 <0.0001 0.45 CD34.1 CD34 <0.0001 0.34 NRG1.3 NRG1 0.0004 0.69 CD24.1 CD24 0.0148 0.81 NPR1.1 NPR1 <0.0001 0.36 CCND1.3 CCND1 <0.0001 0.51 NPM1.2 NPM1 <0.0001 0.55 MCP1.1 CCL2 <0.0001 0.68 NOTCH3.1 NOTCH3 <0.0001 0.40 CAT.1 CAT <0.0001 0.48 NOTCH2.1 NOTCH2 <0.0001 0.68 CASP10.1 CASP10 <0.0001 0.62 NOTCH1.1 NOTCH1 <0.0001 0.38 CALD1.2 CALD1 <0.0001 0.42 NOS3.1 NOS3 <0.0001 0.37 CACNA2D1.1 CACNA2D1 0.0006 0.74 NOS2A.3 NOS2 <0.0001 0.42 CA2.1 CA2 <0.0001 0.60 NOL3.1 NOL3 <0.0001 0.67 C7.1 C7 <0.0001 0.65 NFX1.1 NFX1 <0.0001 0.43 ECRG4.1 C2orf40 <0.0001 0.32 NFKBp50.3 NFKB1 <0.0001 0.56 C13orf15.1 C13orf15 <0.0001 0.31 NFATC2.1 NFATC2 <0.0001 0.67 BUB3.1 BUB3 <0.0001 0.65 NFAT5.1 NFAT5 <0.0001 0.55 BTRC.1 BTRC <0.0001 0.63 MYRIP.2 MYRIP <0.0001 0.36 BNIP3.1 BNIP3 0.0021 0.77 MYH11.1 MYH11 <0.0001 0.35 CIAP1.2 BIRC2 <0.0001 0.56 cMYC.3 MYC <0.0001 0.68 BIN1.3 BIN1 <0.0001 0.67 MVP.1 MVP <0.0001 0.66 BGN.1 BGN <0.0001 0.47 FRAP1.1 MTOR <0.0001 0.56 BCL2L12.1 BCL2L12 0.0374 0.82 MSH3.2 MSH3 <0.0001 0.47 Bclx.2 BCL2L1 <0.0001 0.60 MSH2.3 MSH2 <0.0001 0.51 Bcl2.2 BCL2 <0.0001 0.31 MMP2.2 MMP2 0.0229 0.82 BAG1.2 BAG1 <0.0001 0.42 STMY3.3 MMP11 <0.0001 0.66 BAD.1 BAD 0.0187 0.82 GBL.1 MLST8 0.0193 0.82 AXL.1 AXL 0.0077 0.79 MIF.2 MIF <0.0001 0.69 ATP6V1B1.1 ATP6V1B1 <0.0001 0.52 MICA.1 MICA <0.0001 0.52 ASS1.1 ASS1 <0.0001 0.61 MGMT.1 MGMT <0.0001 0.50 ARRB1.1 ARRB1 <0.0001 0.48 MCM3.3 MCM3 0.0105 0.78 ARHGDIB.1 ARHGDIB <0.0001 0.48 MCAM.1 MCAM <0.0001 0.41 ARF1.1 ARF1 0.0021 0.75 MARCKS.1 MARCKS 0.0259 0.82 AQP1.1 AQP1 <0.0001 0.28 ERK1.3 MAPK3 <0.0001 0.35 APOLD1.1 APOLD1 <0.0001 0.35 ERK2.3 MAPK1 <0.0001 0.61 APC.4 APC <0.0001 0.55 MAP4.1 MAP4 <0.0001 0.54 APAF1.2 APAF1 0.0264 0.82 MAP2K3.1 MAP2K3 <0.0001 0.52 ANXA4.1 ANXA4 0.0012 0.76 MAP2K1.1 MAP2K1 0.0172 0.81 ANXA1.2 ANXA1 0.0201 0.81 MAL2.1 MAL2 0.0267 0.82 ANTXR1.1 ANTXR1 <0.0001 0.58 MAL.1 MAL <0.0001 0.46 ANGPTL4.1 ANGPTL4 <0.0001 0.71 LTF.1 LTF 0.0038 0.74 ANGPTL3.3 ANGPTL3 0.0104 0.77 LRP2.1 LRP2 <0.0001 0.52 ANGPTL2.1 ANGPTL2 <0.0001 0.63 LMO2.1 LMO2 <0.0001 0.60 ANGPT2.1 ANGPT2 <0.0001 0.65 LDB2.1 LDB2 <0.0001 0.26 ANGPT1.1 ANGPT1 <0.0001 0.30 LDB1.2 LDB1 <0.0001 0.52 AMACR1.1 AMACR 0.0080 0.79 LAMA4.1 LAMA4 <0.0001 0.67 ALDOB.1 ALDOB <0.0001 0.40 KRT7.1 KRT7 <0.0001 0.56 ALDH6A1.1 ALDH6A1 <0.0001 0.38 K-ras.10 KRAS <0.0001 0.48 ALDH4.2 ALDH4A1 0.0001 0.71 KL.1 KL <0.0001 0.34 AKT3.2 AKT3 <0.0001 0.30 Kitlng.4 KITLG <0.0001 0.46 AKT2.3 AKT2 <0.0001 0.53 c-kit.2 KIT <0.0001 0.41 AKT1.3 AKT1 <0.0001 0.47 KDR.6 KDR <0.0001 0.28 AHR.1 AHR <0.0001 0.60 KCNJ15.1 KCNJ15 <0.0001 0.43 AGTR1.1 AGTR1 <0.0001 0.33 HTATIP.1 KAT5 <0.0001 0.27 AGT.1 AGT 0.0032 0.77 G-Catenin.1 JUP <0.0001 0.32 ADH6.1 ADH6 0.0011 0.71 AP-1 (JUN JUN <0.0001 0.64 ADH1B.1 ADH1B <0.0001 0.69 official).2 ADFP.1 ADFP 0.0001 0.73 JAG1.1 JAG1 <0.0001 0.23 ADD1.1 ADD1 <0.0001 0.33 ITGB5.1 ITGB5 <0.0001 0.64 ADAMTS9.1 ADAMTS9 <0.0001 0.69 ITGB3.1 ITGB3 0.0468 0.84 ADAMTS5.1 ADAMTS5 <0.0001 0.55 ITGB1.1 ITGB1 <0.0001 0.59 ADAMTS1.1 ADAMTS1 <0.0001 0.56 ITGA7.1 ITGA7 <0.0001 0.38 ADAM17.1 ADAM17 0.0009 0.76 ITGA6.2 ITGA6 <0.0001 0.40 ACE2.1 ACE2 <0.0001 0.45 ITGA5.1 ITGA5 0.0298 0.83 ACADSB.1 ACADSB <0.0001 0.46 ITGA4.2 ITGA4 <0.0001 0.61 BCRP.1 ABCG2 <0.0001 0.27 ITGA3.2 ITGA3 0.0018 0.76 MRP4.2 ABCC4 <0.0001 0.61 IQGAP2.1 IQGAP2 <0.0001 0.52 MRP3.1 ABCC3 0.0011 0.76 INSR.1 INSR <0.0001 0.38 MRP1.1 ABCC1 0.0008 0.75 IMP3.1 IMP3 <0.0001 0.53 ABCB1.5 ABCB1 <0.0001 0.59 IL6ST.3 IL6ST <0.0001 0.36 NPD009 ABAT 0.0001 0.70 IL15.1 IL15 <0.0001 0.67 (ABAT official).3 IGFBP6.1 IGFBP6 <0.0001 0.66 AAMP.1 AAMP <0.0001 0.62 A2M.1 A2M <0.0001 0.28

(165) TABLE-US-00011 TABLE 7a Proxy genes for which increased expression is associated with the presence of nodal invasion (p-value ≤.05) Official Nodal Invasion Official Nodal Invasion Gene Symbol p-value OR Gene Symbol p-value OR TUBB.1 TUBB2A 0.0242 2.56 IL-8.1 IL8 0.0019 3.18 C20 orf1.1 TPX2 0.0333 2.61 IL6.3 IL6 0.0333 2.31 TK1.2 TK1 0.0361 1.75 HSPA1A.1 HSPA1A 0.0498 2.11 SPHK1.1 SPHK1 0.0038 3.43 GSTp.3 GSTP1 0.0272 3.46 SLC7A5.2 SLC7A5 0.0053 4.85 GRB14.1 GRB14 0.0287 2.32 SILV.1 SILV 0.0470 1.54 GMNN.1 GMNN 0.0282 3.00 SELE.1 SELE 0.0311 1.93 ENO2.1 ENO2 0.0190 3.43 upa.3 PLAU 0.0450 2.78 CCNB1.2 CCNB1 0.0387 1.87 MMP9.1 MMP9 0.0110 2.65 BUB1.1 BUB1 0.0429 2.21 MMP7.1 MMP7 0.0491 2.34 BAG2.1 BAG2 0.0346 2.54 MMP14.1 MMP14 0.0155 3.21 ADAMTS1.1 ADAMTS1 0.0193 3.31 LAMB1.1 LAMB1 0.0247 3.04

(166) TABLE-US-00012 TABLE 7b Proxy genes for which increased expression is associated with the absence

of nodal invasion (p-value ≤.05) Official Nodal Invasion Official Nodal Invasion Gene Symbol p-value OR Gene Symbol p-value OR VWF.1 VWF 0.0221 0.42 HMGB1.1 HMGB1 0.0385 0.45 VCAM1.1 VCAM1 0.0212 0.42 HLA-DPB1.1 HLA-DPB1 0.0398 0.43 UBE1C.1 UBA3 0.0082 0.32 HADH.1 HADH 0.0093 0.33 tusc4.2 TUSC4 0.0050 0.28 GSTM1.1 GSTM1 0.0018 0.18 TSPAN7.2 TSPAN7 0.0407 0.43 GPX2.2 GPX2 0.0211 0.07 TSC1.1 TSC1 0.0372 0.38 GJA1.1 GJA1 0.0451 0.46 TMSB10.1 TMSB10 0.0202 0.43 GATM.1 GATM 0.0038 0.25 TMEM47.1 TMEM47 0.0077 0.32 GATA3.3 GATA3 0.0188 0.06 TMEM27.1 TMEM27 0.0431 0.41 FOLR1.1 FOLR1 0.0152 0.32 TLR3.1 TLR3 0.0041 0.32 FLT4.1 FLT4 0.0125 0.22 TIMP3.3 TIMP3 0.0309 0.40 FLT1.1 FLT1 0.0046 0.35 TGFBR2.3 TGFBR2 0.0296 0.35 FHL1.1 FHL1 0.0435 0.47 TGFB2.2 TGFB2 0.0371 0.30 FHIT.1 FHIT 0.0061 0.29 TGFA.2 TGFA 0.0025 0.32 fas.1 FAS 0.0163 0.39 TEK.1 TEK 0.0018 0.09 ErbB3.1 ERBB3 0.0145 0.33 TCF4.1 TCF4 0.0088 0.38 EPHA2.1 EPHA2 0.0392 0.37 STAT5A.1 STAT5A 0.0129 0.49 EPAS1.1 EPAS1 0.0020 0.28 SPRY1.1 SPRY1 0.0188 0.43 ENPEP.1 ENPEP 0.0002 0.34 SPARCL1.1 SPARCL1 0.0417 0.50 CD105.1 ENG 0.0112 0.38 SOD1.1 SOD1 0.0014 0.23 EMCN.1 EMCN 0.0022 0.19 SNRK.1 SNRK 0.0226 0.43 EIF2C1.1 EIF2C1 0.0207 0.33 MADH2.1 SMAD2 0.0098 0.37 EGLN3.1 EGLN3 0.0167 0.52 SLC22A6.1 SLC22A6 0.0051 0.00 EFNB2.1 EFNB2 0.0192 0.30 PTPNS1.1 SIRPA 0.0206 0.43 EFNB1.2 EFNB1 0.0110 0.37 SHANK3.1 SHANK3 0.0024 0.30 EDNRB.1 EDNRB 0.0106 0.36 SGK.1 SGK1 0.0087 0.35 EDN1 EDN1 0.0440 0.42 SELENBP1.1 SELENBP1 0.0016 0.29 endothelin.1 SCN4B.1 SCN4B 0.0081 0.08 DPYS.1 DPYS 0.0454 0.43 ROCK1.1 ROCK1 0.0058 0.41 DKFZP564O0823.1 DKFZP564O0823 0.0131 0.31 RhoB.1 RHOB 0.0333 0.43 DHPS.3 DHPS 0.0423 0.48 RGS5.1 RGS5 0.0021 0.31 DAPK1.3 DAPK1 0.0048 0.39 FLJ22655.1 RERGL 0.0009 0.01 CYP2C8v2.1 CYP2C8_21 0.0003 0.01 RB1.1 RB1 0.0281 0.48 CYP2C8.2 CYP2C8_2 0.0269 0.07 RASSF1.1 RASSF1 0.0004 0.23 CX3CR1.1 CX3CR1 0.0224 0.13 PTPRB.1 PTPRB 0.0154 0.40 CUBN.1 CUBN 0.0010 0.04 PTK2.1 PTK2 0.0158 0.38 CRADD.1 CRADD 0.0193 0.40 PTHR1.1 PTH1R <0.0001 0.01 CLDN10.1 CLDN10 0.0005 0.21 PRSS8.1 PRSS8 0.0023 0.10 CFLAR.1 CFLAR 0.0426 0.49 PRKCH.1 PRKCH 0.0475 0.48 CEACAM1.1 CEACAM1 0.0083 0.23 PPAP2B.1 PPAP2B 0.0110 0.40 CDKN2A.2 CDKN2A 0.0026 0.13 PLA2G4C.1 PLA2G4C 0.0002 0.03 p27.3 CDKN1B 0.0393 0.60 PI3K.2 PIK3C2B 0.0138 0.13 CDH5.1 CDH5 0.0038 0.33 PFKP.1 PFKP 0.0040 0.41 CDH13.1 CDH13 0.0203 0.39 CD31.3 PECAM1 0.0077 0.38 CD99.1 CD99 0.0173 0.38 PDGFD.2 PDGFD 0.0169 0.35 CD36.1 CD36 0.0015 0.29 PDGFC.3 PDGFC 0.0053 0.36 CD34.1 CD34 0.0250 0.38 PDGFB.3 PDGFB 0.0359 0.47 CD3z.1 CD247 0.0419 0.33 PCSK6.1 PCSK6 0.0103 0.34 CCND1.3 CCND1 0.0381 0.50 PCK1.1 PCK1 0.0003 0.02 CAT.1 CAT 0.0044 0.42 PCCA.1 PCCA 0.0074 0.12 CASP6.1 CASP6 0.0136 0.36 PARD6A.1 PARD6A 0.0243 0.21 CALD1.2 CALD1 0.0042 0.31 BFGF.3 NUDT6 0.0082 0.15 CA9.3 CA9 0.0077 0.48 NRG1.3 NRG1 0.0393 0.19 C13orf15.1 C13orf15 0.0152 0.39 NOS3.1 NOS3 0.0232 0.34 BUB3.1 BUB3 0.0157 0.26 NOS2A.3 NOS2 0.0086 0.11 BIN1.3 BIN1 0.0224 0.38 NFX1.1 NFX1 0.0065 0.35 Bclx.2 BCL2L1 0.0225 0.52 MYH11.1 MYH11 0.0149 0.36 Bcl2.2 BCL2 0.0442 0.45 cMYC.3 MYC 0.0472 0.46 AXL.1 AXL 0.0451 0.44 MUC1.2 MUC1 0.0202 0.26 ATP6V1B1.1 ATP6V1B1 0.0257 0.03 MIF.2 MIF 0.0145 0.39 ARRB1.1 ARRB1 0.0337 0.43 MICA.1 MICA 0.0015 0.15 ARHGDIB.1 ARHGDIB 0.0051 0.32 MGMT.1 MGMT 0.0414 0.50 AQP1.1 AQP1 0.0022 0.31 MAP2K1.1 MAP2K1 0.0075 0.40 APOLD1.1 APOLD1 0.0222 0.42 LMO2.1 LMO2 0.0127 0.07 APC.4 APC 0.0165 0.47 LDB2.1 LDB2 0.0048 0.33 ANXA5.1 ANXA5 0.0143 0.37 Kitlng.4 KITLG 0.0146 0.18 ANXA4.1 ANXA4 0.0019 0.30 KDR.6 KDR 0.0106 0.34 ANXA1.2 ANXA1 0.0497 0.39 ITGB1.1 ITGB1 0.0469 0.36 ANGPTL7.1 ANGPTL7 0.0444 0.16 ITGA7.1 ITGA7 0.0290 0.38 ANGPTL4.1 ANGPTL4 0.0197 0.55 ITGA6.2 ITGA6 0.0010 0.17 ANGPT1.1 ANGPT1 0.0055 0.15 ITGA4.2 ITGA4 0.0089 0.44 ALDOB.1 ALDOB 0.0128 0.02 INSR.1 INSR 0.0057 0.32 ALDH4.2 ALDH4A1 0.0304 0.33 IMP3.1 IMP3 0.0086 0.43 AGTR1.1 AGTR1 0.0142 0.11 IL6ST.3 IL6ST 0.0484 0.46 ADH6.1 ADH6 0.0042 0.03 IL15.1 IL15 0.0009 0.08 ADFP.1 ADFP 0.0223 0.47 IFI27.1 IFI27 0.0013 0.22 ADD1.1 ADD1 0.0135 0.42 HYAL2.1 HYAL2 0.0099 0.36 BCRP.1 ABCG2 0.0098 0.19 HYAL1.1 HYAL1 0.0001 0.08 MRP3.1 ABCC3 0.0247 0.44 Hepsin.1 HPN 0.0024 0.14 MRP1.1 ABCC1 0.0022 0.32 HPCAL1.1 HPCAL1 0.0024 0.38 NPD009 ABAT 0.0110 0.19 (ABAT official).3 A2M.1 A2M 0.0012 0.29

(167) TABLE-US-00013 TABLE 8a Genes for which increased expression is associated with lower risk of cancer recurrence after clinical/pathologic covariate adjustment (p < 0.05) Gene Official Symbol p-value HR ACE2.1 ACE2 0.0261 0.85 ADD1.1 ADD1 0.0339 0.85 ALDOB.1 ALDOB 0.0328 0.84 ANGPTL3.3 ANGPTL3 0.0035 0.79 APOLD1.1 APOLD1 0.0015 0.78 AQP1.1 AQP1 0.0014 0.79 BFGF.3 NUDT6 0.0010 0.77 CASP10.1 CASP10 0.0024 0.82 CAV2.1 CAV2 0.0191 0.86 CCL4.2 CCL4 0.0045 0.81 CCL5.2 CCL5 0.0003 0.78 CCR2.1 CCR2 0.0390 0.87 CCR4.2 CCR4 0.0109 0.82 CCR7.1 CCR7 0.0020 0.80 CD4.1 CD4 0.0195 0.86 CD8A.1 CD8A 0.0058 0.83 CEACAM1.1 CEACAM1 0.0022 0.81 CFLAR.1 CFLAR 0.0308 0.87 CTSS.1 CTSS 0.0462 0.87 CX3CL1.1 CX3CL1 0.0021 0.81 CXCL10.1 CXCL10 0.0323 0.86 CXCL9.1 CXCL9 0.0006 0.79 CXCR6.1 CXCR6 0.0469 0.88 DAPK1.3 DAPK1 0.0050 0.83 DDC.1 DDC 0.0307 0.86

DLC1.1 DLC1 0.0249 0.83 ECRG4.1 C2orf40 0.0244 0.84 EDNRB.1 EDNRB 0.0400 0.86 EMCN.1 EMCN <0.0001 0.68 EPAS1.1 EPAS1 0.0411 0.84 fas.1 FAS 0.0242 0.87 FH.1 FH 0.0407 0.88 GATA3.3 GATA3 0.0172 0.83 GZMA.1 GZMA 0.0108 0.84 HLA-DPB1.1 HLA-DPB1 0.0036 0.82 HSPG2.1 HSPG2 0.0236 0.84 ICAM2.1 ICAM2 0.0091 0.83 ICAM3.1 ICAM3 0.0338 0.87 ID1.1 ID1 0.0154 0.83 IGF1R.3 IGF1R 0.0281 0.85 IL15.1 IL15 0.0059 0.83 IQGAP2.1 IQGAP2 0.0497 0.88 KL.1 KL 0.0231 0.86 KLRK1.2 KLRK1 0.0378 0.87 LDB2.1 LDB2 0.0092 0.82 LRP2.1 LRP2 0.0193 0.86 LTF.1 LTF 0.0077 0.82 MAP4.1 MAP4 0.0219 0.84 MRP1.1 ABCC1 0.0291 0.87 NOS3.1 NOS3 0.0008 0.78 PI3K.2 PIK3C2B 0.0329 0.83 PLA2G4C.1 PLA2G4C 0.0452 0.87 PPAP2B.1 PPAP2B 0.0001 0.74 PRCC.1 PRCC 0.0333 0.88 PRKCB1.1 PRKCB 0.0353 0.87 PRKCH.1 PRKCH 0.0022 0.82 PRSS8.1 PRSS8 0.0332 0.87 PSMB9.1 PSMB9 0.0262 0.87 PTPRB.1 PTPRB 0.0030 0.80 RGS5.1 RGS5 0.0480 0.85 SDPR.1 SDPR 0.0045 0.80 SELE.1 SELE 0.0070 0.81 SGK.1 SGK1 0.0100 0.83 SHANK3.1 SHANK3 0.0311 0.84 SNRK.1 SNRK 0.0026 0.81 TEK.1 TEK 0.0059 0.78 TGFBR2.3 TGFBR2 0.0343 0.85 TIMP3.3 TIMP3 0.0165 0.83 TMEM27.1 TMEM27 0.0249 0.86 TSPAN7.2 TSPAN7 0.0099 0.83 UBB.1 UBB 0.0144 0.85 WWOX.5 WWOX 0.0082 0.83

(168) TABLE-US-00014 TABLE 8b Genes for which increased expression is associated with higher risk of cancer recurrence after clinical/pathologic covariate adjustment ($p < 0.05$) Gene Official Symbol p-value HR CIAP1.2 BIRC2 0.0425 1.14 BUB1.1 BUB1 0.0335 1.15 CCNB1.2 CCNB1 0.0296 1.14 ENO2.1 ENO2 0.0284 1.17 ITGB1.1 ITGB1 0.0402 1.16 ITGB5.1 ITGB5 0.0016 1.25 LAMB1.1 LAMB1 0.0067 1.20 MMP14.1 MMP14 0.0269 1.16 MMP9.1 MMP9 0.0085 1.19 PSMA7.1 PSMA7 0.0167 1.16 RUNX1.1 RUNX1 0.0491 1.15 SPHK1.1 SPHK1 0.0278 1.16 OPN, osteopontin.3 SPP1 0.0134 1.17 SQSTM1.1 SQSTM1 0.0347 1.13 C20 orf1.1 TPX2 0.0069 1.20 TUBB.1 TUBB2A 0.0046 1.21 VCAN.1 VCAN 0.0152 1.18

(169) TABLE-US-00015 TABLE 9 16 Significant Genes After Adjusting for Clinical/Pathologic Covariates and Allowing for an FDR of 10% Official Gene LR LR Symbol n Subset (Pathway) HR HR (95% CI) ChiSq p-value q-value EMCN 928 Angiogenesis 0.68 (0.57, 0.80) 19.87 <0.0001 0.0042 PPAP2B 928 Angiogenesis 0.74 (0.65, 0.85) 16.15 0.0001 0.0148 CCL5 928 Immune Response 0.78 (0.68, 0.89) 12.98 0.0003 0.0529 CXCL9 928 Immune Response 0.79 (0.70, 0.91) 11.74 0.0006 0.0772 NOS3 928 Angiogenesis 0.78 (0.68, 0.90) 11.32 0.0008 0.0774 NUDT6 926 Angiogenesis 0.77 (0.66, 0.90) 10.77 0.0010 0.0850 AQP1 928 Transport 0.79 (0.69, 0.91) 10.15 0.0014 0.0850 APOLD1 927 Angiogenesis 0.78 (0.68, 0.91) 10.07 0.0015 0.0850 ITGB5 928 Cell Adhesion/ 1.25 (1.09, 1.43) 9.92 0.0016 0.0850 Extracellular Matrix CCR7 928 Immune Response 0.80 (0.69, 0.92) 9.58 0.0020 0.0850 CX3CL1 926 Immune Response 0.81 (0.70, 0.92) 9.44 0.0021 0.0850 CEACAM1 928 Angiogenesis 0.81 (0.70, 0.93) 9.37 0.0022 0.0850 PRKCH 917 Angiogenesis 0.82 (0.72, 0.93) 9.36 0.0022 0.0850 CASP10 927 Apoptosis 0.82 (0.73, 0.93) 9.25 0.0024 0.0850 SNRK 928 Angiogenesis 0.81 (0.71, 0.92) 9.09 0.0026 0.0863 PTPRB 927 Angiogenesis 0.80 (0.69, 0.92) 8.83 0.0030 0.0933

(170) TABLE-US-00016 TABLE A Gene Accession Official Version SEQ ID SEQ ID SEQ ID Gene Num Sequence_ID Symbol ID F Primer Seq NO. R Primer Seq NO. Probe Seq NO. A-Catenin.2 NM_001903 NM_001903.1 CTNNA1 765 CGTTCCGATCCTCT 1 AGGTCCCTGTTG 733 ATGCCTACAGCACCC 1465 ATACTGCAT GCCTTATAGG TGATGTCGCA A2M.1 NM_000014 NM_000014.4 A2M 6456 CTCTCCCGCCTTCC 2 CCGTTTGCACAG 734 CGCTTGTTCCTTCTC 1466 TAGC ATGCAG CACTGGGAC AAMP.1 NM_001087 NM_001087.3 AAMP 5474 GTGTGGCAGGTGG 3 CTCCATCCACTC 735 CGCTTCAAAGGACCA 1467 ACACTAA CAGGTCTC GACCTCCTC ABCB1.5 NM_000927 NM_000927.2 ABCB1 3099 AAACACCACTGGAG 4 CAAGCCTGGAAC 736 CTCGCCAATGATGCT 1468 CATTGA CTATAGCC GCTCAAGTT ACADSB.1 NM_001609 NM_001609.3 ACADSB 6278 TGGCGGAGAACTA 5 AAGACAGCCCAG 737 CCTCCTGAAGCCTG 1469 GCCAT TCCTCAAAT CCATCATTGT ACE.1 NM_000789 NM_000789.2 ACE 4257 CCGCTGTACGAGG 6 CCGTGTCTGTGA 738 TGCCCCTCAGCAATGA 1470 ATTTCA AGCCGT AGCCTACAA ACE2.1 NM_021804 NM_021804.1 ACE2 6108 TACAATGAGAGGCT 7 TAATGGCCTCAG 739 CGACCTCAGATCTCC 1471 CTGGGC CTGCTTG AGCTTTCCC ADAM17.1 NM_003183 NM_003183.3 ADAM17 2617 GAAGTGCCAGGAG 8 CGGGCACTCACT 740 TGCTACTTGCAAAGG 1472 GCGATTA GCTATTACC CGTGTCCTACTGC ADAM8.1 NM_001109 NM_001109.2 ADAM8 3978 GTCACTGTGTCCAG 9 TGATGACCTGCT 741 TTCCCAGTTCCTGTC 1473 CCCA TTGGTGC TACACCCGG ADAMTS1.1 NM_006988 NM_006988.2 ADAMTS1 2639 GGACAGGTGCAAG 10 ATCTACAACCTT 742 CAAGCCAAAGGCATT 1474 CTCATCTG GGGCTGCAA GGCTACTTCTTCG ADAMTS2.1 NM_014244 NM_014244.1 ADAMTS2 3979 GAGAATGTCTGCC 11 ATCGTGGTATTC 743 TACCTCCAGCAGAAG 1475 GCTGG ATCGTGGC CCAGACACG ADAMTS4.1 NM_005099 NM_005099.3 ADAMTS4 2642 TTTGACAAGTGCAT 12 AATTTCCTGAAG 744 CTGCTTGCTGCAACC 1476 GGTGTG GAGCCTGA AGAACCGT ADAMTS5.1 NM_007038 NM_007038.1 ADAMTS5 2641 CACTGTGGCTCAC 13 GGAACCAAAGGT 745 ATTTACTTGGCCTCT 1477 GAAATCG CTCTTCACAGA

CCCATGACGATTCC ADAMTS8.1 NM_007037 NM_007037.2 ADAMTS8 2640 GCGAGTTCAAAGT 14 CACAGATGGCCA 746 CACACAGGGTGCCA 1478 GTTCGAG GTGTTTCT TCAATCACCT ADAMTS9.1 NM_182920 NM_182920.1 ADAMTS9 6109 GCACAGGTTACACA 15 CGACATTGGCAG 747 CCGGCTCCCGTTATA 1479 ACCCAA TCATCG GGGACATTC ADD1.1 NM_001119 NM_001119.3 ADD1 3980 GTCTACCCAGCAG 16 TCGTTCACAGGA 748 CATGTTTAAGGCAGC 1480 CTCCG GTCACCAT CATCCCTCC ADFP.1 NM_001122 NM_001122.2 PLIN2 4503 AAGACCATCACCTC 17 CAATTTGCGGCT 749 ATGACCAGTGCTCTG 1481 CGTGG CTAGCTTC CCCATCATC ADHIB.1 NM_000668 NM_000668.4 ADH1B 6325 AAGCCAACAAACCT 18 AAAATGCAAGAA 750 TTTCCTCAATGGCAA 1482 TCCTTC GTCACAGGAA AGGTGACACA ADH6.1 NM_000672 NM_000672.3 ADH6 6111 TGTTGGGGAGTAAA 19 AACGATTCCAGC 751 TCTTGTATCCCACCA 1483 CACTTGG CCCTTC TCTTGGGCC ADM.1 NM_001124 NM_001124.1 ADM 3248 TAAGCCCACAAGCAC 20 TGGGCGCCTAAA 752 CGAGTGGAAGTGCT 1484 ACGG TCCTAA CCCCACTTTC AGR2.1 NM_006408 NM_006408.2 AGR2 3245 AGCCAACATGTGAC 21 TCTGATCTCCAT 753 CAACACGTCACCACC 1485 TAATTGGA CTGCCTCA CTTTGCTCT AGT.1 NM_000029 NM_000029.2 AGT 6112 GATCCAGCCTCACT 22 CCAGTTGAGGGA 754 TGAGACCCTCCACCT 1486 ATGCCT GTTTTGCT TGTCCAGGT AGTR1.1 NM_000685 NM_000685.3 AGTR1 4258 AGCATTGATCGATA 23 CTACAAGCATTG 755 ATTGTTCACCCAATG 1487 CCTGGC TGCGTCG AAGTCCCGC AHR.1 NM_001621 NM_001621.2 AHR 3981 GCGGCATAGAGAC 24 ACATCTTGTGGG 756 CAGGCTAGCCAAAC 1488 CGACTT AAAGGCA GGTCCAACTC AIF1.1 NM_032955 NM_032955.1 AIF1 6452 GACGTTCAGCTACC 25 TCAGGATCATTTT 757 ATCTCTTGCCCAGCA 1489 CTGACTTT TAGGATGGC TCATCCTGA AKT1.3 NM_005163 NM_005163.1 AKT1 18 CGCTTCTATGGCG 26 TCCCGGTACACC 758 CAGCCCTGGACTAC 1490 CTGAGAT ACGTTCTT CTGCACTCGG AKT2.3 NM_001626 NM_001626.2 AKT2 358 TCCTGCCACCCTTC 27 GGCGGTAAATTC 759 CAGGTCACGTCCGA 1491 AAACC ATCATCGAA GGTCGACACA AKT3.2 NM_005465 NM_005465.1 AKT3 21 TTGTCTCTGCCTTG 28 CGAGCATTAGAT 760 TCACGGTACACAATC 1492 GACTATCTACA TCTCCAACTTGA TTTCCGGA ALDH4.2 NM_003748 NM_003748.2 ALDH4A1 2092 GGACAGGGTAAGA 29 AACCGGAAGAAG 761 CTGCAGCGTCAATCT 1493 CCGTGAT TCGATGAG CCGCTTG ALDH6A1.1 NM_005589 NM_005589.2 ALDH6A1 6114 GGCTCTTTCAACAG 30 GCATGCTCCACC 762 CAGCCACTTCTTGGC 1494 CAGTCC AGCTCT TTCTCCCAC AALDOA.1 NM_000034 NM_000034.2 ALDOA 3810 GCCTGTACGTGCC 31 TCATCGGAGCTT 763 TGCCAGAGCCTCAA 1495 AGCTC GATCTCG CTGTCTCTGC ALDOB.1 NM_000035 NM_000035.2 ALDOB 6321 CCCTCTACCAGAAG 32 TAACTTGATTCC 764 TCCCCTTTTCCTTGA 1496 GACAGC CACCACGA GGATGTTTCTG ALOX12.1 NM_000697 NM_000697.1 ALOX12 3861 AGTTCCTCAATGGT 33 AGCACTAGCCTG 765 CATGCTGTTGAGAC 1497 GCCAAC GAGGGC GCTCGACCTC ALOX5.1 NM_000698 NM_000698.2 ALOX5 4259 GAGCTGCAGGACT 34 GAAGCCTGAGGA 766 CCGCATGCCGTACA 1498 TCGTGA CTTGCG CGTAGACATC AMACR1.1 NM_014324 NM_014324.4 AMACR 3930 GGACAGTCAGTTTT 35 GACAGCCCAGAG 767 CAGTAACTCGGGGC 1499 AGGGTTGC ACCCAC CTGTTTCCC ANGPT1.1 NM_001146 NM_001146.3 ANGPT1 2654 TCTACTTGGGGTGA 36 CCTTTTTAAAGC 768 TCACGTGGCTCGAC 1500 CAGTGC CCGACAGT TATAGAAAACTCCA ANGPT2.1 NM_001147 NM_001147.1 ANGTP2 2655 CCGTGAAAGCTGC 37 TTGCAGTGGGAA 769 AAGCTGACACAGCC 1501 TCTGTAA GAACAGTC CTCCCAAGTG ANGPTL2.1 NM_012098 NM_012098.2 ANGPTL2 3982 GCCATCTGCGTCAA 38 TAGCTCCTGCTT 770 TCTCCAGAAGCACCT 1502 CTCC ATGCACTCG CAGGCTCCT ANGPTL3.3 NM_014495 NM_014495.2 ANGPTL3 6505 GTTGCGATTACTGG 39 TGCTTTGTGATC 771 CCAATGCAATCCCG 1503 CAATGT CCAAGTAGA GAAAACAAAG ANGPTL4.1 NM_016109 NM_016109.2 3237 ATGACCTCAGATGG 40 CCGGTTGAAGTC 772 CATCGTGGCGCCTC 1504 AGGCTG CACTGAG TGAATTACTG ANGPTL7.1 NM_021146 NM_021146.2 ANGPTL7 6115 CTGCACAGACTCCA 41 GCCATCCAGGTG 773 TCACCCAGGCGGTA 1505 ACCTCA CTTATTGT GTACACTCCA ANTXR1.1 NM_032208 NM_032208.1 ANTXR1 3363 CTCCAGGTGTACCT 42 GAGAAGGCTGG 774 AGCCTTCTCCCACAG 1506 CCAACC GAGACTCTG CTGCCTACA ANXA1.2 NM_000700 NM_000700.1 ANXA1 1907 GCCCCTATCCTACC 43 CCTTTAACCATTA 775 TCCTCGGATGTCGCT 1507 TTCAATCC TGGCCTTATGC GCCT ANXA2.2 NM_004039 NM_004039.1 ANXA2 2269 CAAGACACTAAGG 44 CGTGTCGGGCTT 776 CCACCACACAGGTA 1508 GCGACTACCA CAGTCAT CAGCAGCGCT ANXA4.1 NM_001153 NM_001153.2 ANXA4 3984 TGGGAGGGATGAA 45 CTCATACAGGTC 777 TGTCTCACGAGAGCA 1509 GGAAAT CTGGGCA TCGTCCAGA ANXZ5.1 NM_001154 NM_001154.2 ANXA5 3785 GCTCAAGCCTGGA 46 AGAACCACCAAC 778 AGTACCCTGAAGTGT 1510 AGATGAC ATCCGCT CCCCCACCA AP-1 (JUN

NM_002228 NM_002228.2 JUN 2157 GACTGCAAAGATG 47 TAGCCATAAGGT 779
CTATGACGATGCCCT 1511 official).2 GAAACGA CCGCTCTC CAACGCCTC AP1M2.1 NM_005498
NM_005498.3 AP1M2 5104 ACAACGACCGCAC 48 CTGAGGCGGTAT 780 CTTCATCCCGCCTGA 1512
CATCT GACATGAG TGGTGACTT APAF1.2 NM_181861 NM_181861.1 APAF1 4086
CACAAGGAAGAAG 49 CATCCTGGTTCA 781 TGCAATTCAGCAGAA 1513 CTGGTGA CCTTTCAA
GCTCTCCAAA APC.4 NM_000038 NM_000038.1 APC 41 GGACAGCAGGAAT 50 ACCCACTCGATT 782
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CCTGCACCTTCT 784 ACTGGCGCTGCATG 1516 GTTCG CCACCA TCTTCCAC APOL1.1 NM_003661
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NM_001658.2 ARF1 2776 CAGTAGAGATCCCC 57 ACAAGCACATGG 789 CTTGTCCTTGGGTCA 1521
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CCGCCGTGGACAC 75 TTGCCGTCAGAA 807 TGCCACTCGGAAAAA 1539 AGACT AACATGTCA
GACCTCTCGG BBC3.2 NM_014417 NM_014417.1 BBC3 574 CCTGGAGGGTCCT 76 CTAATTGGGCTC
808 CATCATGGGACTCCT 1540 GTACAAT CATCTCG GCCCTTACC Bcl2.2 NM_000633 NM_000633.1
BCL2 61 CAGATGGACCTAGT 77 CCTATGATTTAA 809 TTCCACGCCGAAGG 1541 ACCCACTGAGA
GGGCATTTTTCC ACAGCGAT BCL2A1.1 NM_04049 NM_04049.2 BCL2A1 6322 CCAGCCTCCATGTA
78 TGAAGCTGTTGA 810 CAGTCAAGCTCAGTG 1542 TCATCA GGCAATGT AGCATTCTCAGC
BCL2L12.1 NM_138639 NM_138639.1 BCL2L12 3364 AACCCACCCCTGTC 79 CTCAGCTGACGG 811
TCCGGGTAGCTCTC 1543 TTGG GAAAGG AAACTCGAGG Bclx.2 NM_001191 NM_001191.1 BCL2L1
83 CTTTTGTGGAACTC 80 CAGCGGTTGAAG 812 TTCGGCTCTCGGCT 1544 TATGGGAACA

CGTTCCT GCTGCA BCRP.1 NM_004827 NM_004827.1 AGCG2 364 TGTACTGGCGAAG 81 GCCACGTGATTC 813 CAGGGCATCGATCT 1545 AATATTTGGTAAA TTCCACAA CTCACCCTGG BFGF.3 NM_007083 NM_007083.1 NUDT6 345 CCAGGAAGAATGCT 82 TGGTGATGGGAG 814 TTCGCCAGGTCATTG 1546 TAAGATGTGA TTGTATTTTCAG AGATCCATCCA BGN.1 NM_001711 NM_001711.3 BGN 3391 GAGCTCCGCAAGG 83 CTTGTTGTTCAC 815 CAAGGGTCTCCAGC 1547 ATGAC CAGGACGA ACCTCTACGC BHLHB3.1 NM_030762 NM_030762.1 BHLHE41 6121 AGGAAGATCCCTC 84 TTGAACCTCCGT 816 AGGAAGCTCCCTGA 1548 GCAGC CCTTCG ATCCTTGCGT BIK.1 NM_001197 NM_001197.3 BIK 2281 ATTCCTATGGCTCT 85 GGCAGGAGTGAA 817 CCGGTTAACTGTGG 1549 GCAATTGTC TGGCTCTTC CCTGTGCCC BIN1.3 NM_004305 NM_004305.1 BIN1 941 CCTGCAAAAGGGA 86 CGTGGTTGACTC 818 CTTCGCCTCCCAGATG 1550 ACAAGAG TGATCTCG GCTCCC BLR1.1 NM_001716 NM_001716.3 CXCR5 6280 GACCAAGCAGGAA 87 AGCGCTGTTTCG 819 CCAGGGGCAGCTAC 1551 GCTCAGA GTCAGA CTGAACTCAA BNIP3.1 NM_004052 NM_004052.2 BNIP3 3937 CTGGACGGAGTAG 88 GGTATCTTGTGG 820 CTCTCACTGTGACAG 1552 CTCCAAG TGTCTGCG CCCACCTCG BRCA1.1 NM_007294 NM_007294.3 BRCA1 7481 TCAGGGGGCTAGA 89 CCATTCCAGTTG 821 CTATGGGCCCTTCAC 1553 AATCTGT ATCTGTGG CAACATGC BTRC.1 NM_033637 NM_033637.2 BTRC 2555 GTTGGGACACAGTT 90 TGAAGCAGTCAG 822 CAGTCGGCCCAGGA 1554 GGTCTG TTGTGCTG CGGTCTACT BUB1.1 NM_004336 NM_004336.1 BUB1 1647 CCGAGGTTAATCCA 91 AAGACATGGCGC 823 TGCTGGGAGCCTAC 1554 GCACGTA TCTCAGTTC ACTTGGCCC BUB3.1 NM_004725 NM_004725.1 BUB3 3016 CTGAAGCAGATGG 92 GCTGATTCCCAA 824 CCTCGCTTTGTTTAA 1556 TTCATCATT GAGTCTAACC CAGCCAGG c-kit.2 NM_000222 NM_000222.1 KIT 50 GAGGCAACTGCTTA 93 GGCACTCGGCTT 825 TTACAGCGACAGTCA 1557 TGGCTTAATTA GAGCAT TGGCCGCAT C13orf15.1 NM_014059 NM_014059.2 C13orf15 6122 TAGAATCTGCTGCC 94 CAAGGGCTGATT 826 TGCACTCAACCTTCT 1558 AGAGGG TTAAGGTGA ACCAGGCCA C1QA.1 NM_015991 NM_015991.2 C1QA 6123 CGGTCATCACCAAC 95 CGGGTACAGTGC 827 AGAACCGTACCAGAA 1559 CAGG AGACGA CCACTCCGG C1QB.1 NM_000491 NM_000491.3 C1QB 6124 CCAGTGGCCTCAC 96 CCCATGGGATCT 828 TCCCAGGAGGCGTC 1560 AGGAC TCATCATC TGACACAGTA C20 orf1.1 NM_012112 NM_012112.2 TPX2 1239 TCAGCTGTGAGCT 97 ACGGTCCTAGGT 829 CAGGTCCCATTGCC 1561 GCGGATA TTGAGGTTAAGA GGGCG C3.1 NM_000064 NM_000064.2 C3 6125 CGTGAAGGAGTGC 98 ACTCGGTGTCCCG 830 ACATCCCACCTGCAG 1562 AGAAAGA GGACTT ACCTCAGTG C3AR1.1 NM_004054 NM_004054.2 C3AR1 6126 AAGCCGCATCCCA 99 TGTTAAGTGCCC 831 CAACCCCCAGAGATT 1563 GACTT TTGCTGG CCGATTCAG C7.1 NM_000587 NM_000587.2 C7 6127 ATGTCTGAGTGTGA 100 AGGCCTTATGCT 832 ATGCTCTGCCCTCTG 1564 GGCGG GGTGACAG CATCTCAGA CA12.1 NM_001218 NM_001218.3 CA12 6128 CTCTCTGAAGGTGT 101 ACAGGAACTGAG 833 AGACACCAGTGCTTC 1565 CCTGGC GGGTGCT TCCAGGGCT CA2.1 NM_000067 NM_000067.1 CA2 5189 CAACGTGGAGTTTG 102 CTGTAAGTGCCA 834 CCTCCCTTGAGCACT 1566 ATGACTCT TCCAGGG GCTTTGTCC CA9.3 NM_001216 NM_001216.1 CA9 482 ATCCTAGCCCTGGT 103 CTGCCTTCTCAT 835 TTTGCTGTCACCAGC 1567 TTTTGG CTGCACAA GTCGC CACNA2D1.1 NM_000722 NM_000722.2 CACNA2D1 6129 CAAACATTAGCTGG 104 CAGCCAGTGGGT 836 CCATGGCATAACACT 1568 GCCTGT GCCTTA AAGGCGCAG CALD1.2 NM_004342 NM_004342.4 CALD1 1795 CACTAAGGTTTGAG 105 GCGAATTAGCCC 837 AACCCAAGCTCAAGA 1569 ACAGTTCCAGAA TCTACAACTGA CGCAGGACGAG CASP1.1 NM_033292 NM_033292.2 CASP1 6132 AGAAAGCCCACATA 106 TGTGGGATGTCT 838 CGCTTTCTGCTCTTC 1570 GAGAAGGA CCAAGAAA CACACCAGA CASP10.1 NM_001230 NM_001230.4 CASP10 6133 ACCTTTCTCTTGGC 107 GTGGGGACTGTC 839 TCTACTGCATCTGCC 1571 CGGAT CACTGC AGCCCTGAG CASP6.1 NM_032992 NM_032992.2 CASP6 6134 CCTCACACTGGTGA 108 AATTGCACTTGG 840 AAAGTCCACTCGGC 1572 ACAGGA GTCTTTGC GCTGAGAAAC Capase 3.1 NM_032991 NM_032991.2 CASP3 5963 TGAGCCTGAGCAG 109 CCTTCCTGCGTG 841 TCAGCCTGTTCCATG 1573 AGACATGA GTCCAT AAGGCAGAGC CAT.1 NM_001752 NM_001752.1 CAT 2745 ATCCATTCGATCTC 110 TCCGGTTTAAGA 842 TGGCCTCACAAGGA 1574 ACCAAGGT CCAGTTTACCA CTACCCTCTCATCC CAV1.1 NM_001753 NM_001753.3 CAV1 2557 GTGGCTCAACATTG 112 CAATGGCCTCCA 843 ATTTCAGCTGATCAG 1575 TGTTCC TTTACAG TGGGCCTCC CAV2.1 NM_198212 NM_198212.1 CAV2 6460 CTTCCCTGGGACG 112 CTCCTGGTCACC 844 CCCGTACTGTCATGC 1576 ACTTG CTTCTGG CTCAGAGCT CCL18.1 NM_002988 NM_002988.2 CCL18 3994 GCTCCTGTGCACAA 113 TGGAATCTGCCA 845 CAACAAAGAGCTCTG 1577 GTTGG GGAGGTA CTGCCTCGT CCL19.1 NM_006274 NM_006274.2

CCL19 4107 GAACGCATCATCCA 114 CCTCTGCACGGT 846 CGCTTCATCTTGGCT 1578 GAGACTG CATAGGTT GAGGTCCTC CCL20.1 NM_004591 NM_004591.1 CCL20 1998 CCATGTGCTGTACC 115 CGCCGCAGAGGT 847 CAGCACTGACATCAA 1579 AAGAGTTTG GGAGTA AGCAGCCAGGA CCL4.2 NM_002984 NM_002984.1 CCL4 4148 GGGTCCAGGAGTA 116 CCTTCCCCTGAAG 848 ACTGAACTGAGCTGC 1580 CGTGTATGAC ACTTCCTGTCT TCA CCL5.2 NM_002985 NM_002985.2 CCL5 4088 AGGTTCTGAGCTCT 117 ATGCTGACTTCC 849 ACAGAGCCCTGGCA 1581 GGCTTT TTCCTGGT AAGCCAAG CCNB1.2 NM_031966 NM_031966.1 CCNB1 619 TTCAGGTTGTTGCA 118 CATCTTCTTGGG 850 TGTCTCCATTATTGA 1582 GGAGAC CACACAAT TCGGTTCATGCA CCND1.3 NM_053056 NM001758.1 CCND1 88 GCATGTTCGTGGC 119 CGGTGTAGATGC 851 AAGGAGACCATCCC 1583 CTCTAAGA ACAGCTTCTC CCTGACGGC CCNE1.1 NM_001238 NM_001238.1 CCNE1 498 AAAGAAGATGATGA 120 GAGCCTCTGGAT 852 CAAACTCAACGTGCA 1584 CCGGGTTTAC GGTGCAAT AGCCTCGGA CCNE2 NM_057749 NM_057749var1 CCNE2 1650 GGTCACCAAGAAAC 121 TTCAATGATAATG 853 CCCAGATAATACAGG 1585 variant 1.1 ATCAGTATGAA CAAGGACTGATC TGGCCAACAATTCCT CCNE2.2 NM_057749 NM_057749.1 CCNE2 502 ATGCTGTGGCTCCT 122 ACCCAAATTGTG 854 TACCAAGCAACCTAC 1586 TCCTAACT ATATACAAAAAG ATGTCAAGAAAGCCC GTT CCR1.1 NM_001295 NM_001295.2 CCR1 6135 TCCAAGACCCAATG 123 TCGTAGGCTTTC 855 ACTCACCACACCTGC 1587 GGAA GTGAGGA AGCCTTCAC CCR2.1 NM_000648 NM_000648.1 4109 CTCGGGAATCCTG 124 GACTCTCACTGC 856 TCTTCTCGTTTCGAC 1588 AAACC CCTATGCC ACCGAAGCA CCR4.2 NM_005508 NM_005508.4 CCR4 6502 AGACCCTGGTGGA 125 AGAGTTTCTGTG 857 TCCTTCAGGACTGCA 1589 GCTAGAA GCCTGGAT CCTTTGAAAGA CCR5.1 NM_000579 NM_000579.1 CCR5 4119 CAGACTGAATGGG 126 CTGGTTTGTCTG 858 TGGAATAAGTACCTA 1590 GGTGG GAGAAGGC AGGCGCCCCC CCR7.1 NM_001838 NM_001838.2 CCR7 2661 GGATGACATGCACT 127 CCTGACATTTCC 859 CTCCCATCCCAGTG 1591 CAGCTC CTTGTCCT GAGCCAA CD105.1 NM_000118 NM_000118.1 ENG 486 GCAGGTGTCAGCA 128 TTTTTCCGCTGT 860 CGACAGGATATTGAC 1592 AGTATGATCAG GGTGATGA CACCGCCTCATT CD14.1 NM_000591 NM_000591.1 CD14 4341 GTGTGCTAGCGTA 129 GCATGGTGCCG 861 CAAGGAACTGACGC 1593 CTCCCG GTTATCT TCGAGGACCT CD18.2 NM_000211 NM_000211.1 ITGB2 49 CGTCAGGACCCAC 130 GGTTAATTGGTG 862 CGCGGCCGAGACAT 1594 CATGTCT ACCATCCTCAAGA GGCTTG CD1A.1 NM_001763 NM_001763.1 CD1A 4166 GGAGTGGAAGGAA 131 TCATGGGCGTAT 863 CGCACCATTCGGTCA 1595 CTGGAAA CTACGAAT TTTGAGG CD24.1 NM_013230 NM_013230.1 CD24 2364 TCCAACTAATGCCA 132 GAGAGAGTGAGA 864 CTGTTGACTGCAGG 1596 CCACCAA CCACGAAGAGAC GCACCACCA T CD274.2 NM_014143 NM_014143.2 CD274 4076 GCTGCATGATCAG 133 TGTTGTATGGGG 865 CACAGTAATTCGCTT 1597 CTATGGT CATTGACT GTAGTCGGCACC CD31.3 NM_000442 NM_000442.1 PECAM1 485 TGTATTTCAAGACC 134 TTAGCCTGAGGA 866 TTTATGAACCTGCCC 1598 TCTGTGCACTT ATTGCTGTGTT TGCTCCCACA CD34.1 NM_001773 NM_001773.1 CD34 3814 CCACTGCACACACC 135 CAGGAGTTTACC 867 CTGTTCTTGGGGCC 1599 TCAGA TGCCCCT CTACACCTTG CD36.1 NM_000072 NM_000072.2 CD36 6138 GTAACCCAGGACG 136 AAGGTTCGAAGA 868 CACAGTCTCTTTCCT 1600 CTGAGG TGGCACC GCAGCCCAA CD3z.1 NM_000734 NM_000734.1 CD247 64 AGATGAAGTGGAA 137 TGCCTCTGTAAT 869 CACCGCGGCCATCC 1601 GGCGCTT CGGCAACTG TGCA CD4.1 NM_000616 NM_000616.2 CD4 4168 GTGCTGGAGTCGG 138 TCCCTGCATTCA 870 CAGGTCCCTTGTCC 1602 GACTAAC AGAGGC CAAGTTCCAC CD44.1 NM_000610 NM_000610.3 CD44 4267 GGCACCACTGCTTA 139 GATGCTCATGGT 871 ACTGGAACCCAGAA 1603 TGAAGG GAATGAGG GCACACCCTC CD44s.1 M59040 M59040.1 1090 GACGAAGACAGTC 140 ACTGGGGTGGAA 872 CACCGACAGCACAG 1604 CCTGGAT TGTGTCTT ACAGAATCCC CD44v6.1 AJ251595v6 AJ251595v6 1061 CTCATACCAGCCAT 141 TTGGGTTGAAGA 873 CACCAAGCCCAGAG 1605 CCAATG AATCAGTCCC GACAGTTCCT CD53.1 NM_000560 NM_000560.3 CD53 6139 CGACAGCATCCAC 142 TGCAGAAATGAC 874 CACGCTGCCTTGGT 1606 CGTTAC TGGATGGA GCTATTGTCT CD68.2 NM_001251 NM_001251.1 CD68 84 TGGTTCCCAGCCCT 143 CTCCTCCACCCT 875 CTCCAAGCCCAGATT 1607 GTGT GGGTTGT CAGATTCGAGTCA CD82.3 NM_002231 NM_002231.2 CD82 316 GTGCAGGCTCAGG 144 GACCTCAGGGCG 876 TCAGCTTCTACAACT 1608 TGAAGTG ATTCATGA GGACAGACAACGCT G CD8A.1 NM_171827 NM_171827.1 CD8A 3804 AGGGTGAGGTGCT 145 GGGCACAGTATC 877 CCAACGGCAAGGGA 1609 TGAGTCT CCAGGTA ACAAGTACTTCT CD99.1 NM_002414 NM_002414.3 CD99 6323 GTTCCTCCCGGTAG 146 ACCATCACTGCC 878 TCCACCTGAAACGCC 1610 CTTTTCA TCCTTTTC ATCCG cdc25A.4 NM_001789 NM_001789.1 CDC25A 90 TCTTGCTGGCTACG 147

CTGCATTGTGGC 879 TGTCCCTGTTAGACG 1611 CCTCTT ACAGTTCTG TCCTCCGTCCATA CDC25B.1 NM_021873 NM_021874.1 CDC25B 389 AAACGAGCAGTTTG 148 GTTGGTGATGTT 880 CCTCACCGGCATAG 1612 CCATCAG CCGAAGCA ACTGGAAGCG CDH1.3 NM_004360 NM_004360.2 CDH1 11 TGAGTGTCCCCCG 149 CAGCCCGCTTTCA 881 TGCCAATCCCGATGA 1613 GTATCTTC GATTTTCAT AATTGGAAATTT CDH13.1 NM_001257 NM_001257.3 CDH13 6140 GCTACTTCTCCACT 150 CCTCTCTGTGGA 882 AGTCTGAATGCTGCC 1614 GTCCCG CCTGCCT ACAACCAGC CDH16.1 NM_004062 NM_004062.2 CDH16 4529 GACTGTCTGAATGG 151 CCAGGGGACTCA 883 CAGAGGCCAAGCTC 1615 CCCAG GATGGA CCAGCTAGAG CDH2.1 NM_001792 NM_001792.2 CDH2 3965 TGACCGATAAGGAT 152 GATCTCCGCCAC 884 ATACACCAGCCTGGA 1616 CAACCC TGATTCTG ACGCAGTGT CDH5.1 NM_001795 NM_001795.2 CDH5 6142 ACAGGAGACGTGT 153 CAGCAGTGAGGT 885 TATTCTCCCGGTCCA 1617 TCGCC GGTACTCTGA GCCTCTCAA CDH6.1 NM_004932 NM_004932.2 CDH6 3998 ACACAGGCGACATA 154 CTCGAAGGATGT 886 TTTCTTCCCTGTCCA 1618 CAGGC AAACGGGT GCCTCTTGG CDK4.1 NM_000075 NM_000075.2 CDK4 4176 CCTTCCCATCAGCA 155 TTGGGATGCTCA 887 CCAGTCGCCTCAGTA 1619 CAGTTC AAAGCC AAGCCACCT CDK6.1 NM_001259 NM_001259.5 CDK6 6143 AGTGCCCTGTCTCA 156 GCAGGTGGGAAT 888 TCTTTGCACCTTTCC 1620 CCCA CCAGGT AGGTCCTGG CDKN2A.2 NM_000077 NM_000077.3 CDKN2A 4278 AGCACTCACGCCCT 157 TCATGAAGTCGA 889 CGCAAGAAATGCCC 1621 AAGC CAGCTTCC ACATGAATGT CEACAM1.1 NM_001712 NM_001712.2 CEACAM1 2577 ACTTGCCTGTTCAG 158 TGGCAAATCCGA 890 TCCTTCCCACCCCCA 1622 AGCACTCA ATTAGAGTGA GTCCTGTC CEBPA.1 NM_004364 NM_004364.2 CEBPA 2961 TTGGTTTTGCTCGG 159 GTCTCAGACCCT 891 AAAATGAGACTCTCC 1623 ATACTTG TCCCCC GTCGGCAGC CENPF.1 NM_016343 NM_016343.2 CENPF 3251 CTCCCGTCAACAGC 160 GGGTGAGTCTGG 892 ACACTGGACCAGGA 1624 GTTC CCTTCA GTGCATCCAG CFLAR.1 NM_003879 NM_003879.3 VFLAR 6144 GGACTTTTGTCCAG 161 CGGCGCTTCTCT 893 CTCCTCCCGTGGTC 1625 TGACAGC CCTACA CTTGTTGTCT CGA (CHGA NM_001275 NM_001275.2 CHGA 1132 CTGAAGGAGCTCC 162 CAAAACCGCTGT 894 TGCTGATGTGCCCTC 1626 official).3 AAGACCT GTTTCTTC TCCTTGG Chk1.2 NM_001274 NM_001274.1 CHEK1 490 GATAAATTGGTACA 163 GGGTGCCAAGTA 895 CCAGCCCACATGTC 1627 AGGGATCAGCTT ACTGACTATTCA CTGATCATATGC Chk2.3 NM_007194 NM_007194.1 CHEK2 494 ATGTGGAACCCCCA 164 CAGTCCACAGCA 896 AGTCCCAACAGAAAC 1628 CCTACTT CGGTTATACC AAGAACTTCAGGCG CIAP1.2 NM_001166 NM_001166.2 BIRC2 326 TGCCTGTGGTGGG 165 GGAAAATGCCTC 897 TGACATAGCATCATC 1629 AAGCT CGGTGTT CTTTGGTTCCCAGTT cIAP2.2 NM_001165 NM_001165.2 BIRC3 79 GGATATTTCCGTGG 166 CTTCTCATCAAG 898 TCTCCATCAAATCCT 1630 CTCTTATTCA GCAGAAAAATCT GTAAACTCCAGAGCA T CLCNKB.1 NM_000085 NM_000085.1 CLCNKB 4308 GTGACCCTGAAGC 167 GGTTCAACAGCT 899 AGAGACTTCCCTGCA 1631 TGTCCC CAAAGAGGTT TGAGGCACA CLDN10.1 NM_182848 NM_182848.2 CLDN10 6145 GGTCTGTGGATGA 168 GATAGTAAAATG 900 TGGAAAGAACCCAAC 1632 ACTGCG CGGTCGGC GCGTTACCT CLDN7.2 NM_001307 NM_001307.3 CLDN7 2287 GGTCTGCCCTAGT 169 GTACCCAGCCTT 901 TGCACTGCTCTCCTG 1633 CATCCTG GCTCTCAT TTCCTGTCC CLU.3 NM_001831 NM_001831.1 CLU 2047 CCCCAGGATACCTA 170 TGCGGGACTTGG 902 CCCTTCAGCCTGCC 1634 CCACTACCT GAAAGA CCACCG CMet.2 NM_000245 NM_000245.1 MET 52 GACATTTCCAGTCC 171 CTCCGATCGCAC 903 TGCCTCTCTGCCCCA 1635 TGCAGTCA ACATTTGT CCCTTTGT cMYC.3 NM_002467 NM_002467.1 MYC 45 TCCCTCCACTCGGA 172 CGGTTGTTGCTG 904 TCTGACACTGTCCAA 1636 AGGACTA ATCTGTCTCA CTTGACCCTCTT COL18A1.1 NM_030582 NM_030582.3 COL18A1 6146 AGCTGCCATCACG 173 GTGGCTACTTGG 905 CGTGCTCTGCATTGA 1637 CCTAC AGGCAGTC GAACAGCTTC COL1A1.1 NM_000088 NM_000088.2 COL1A1 1726 GTGGCCATCCAGC 174 CAGTGGTAGGTG 906 TCCTGCGCCTGATGT 1638 TGACC ATGTTCTGGGA CCACCG COL1A2.1 NM_000089 NM_000089.2 COL1A2 1727 CAGCCAAGAACTG 175 AAACTGGCTGCC 907 TCTCCTAGCCAGACG 1639 GTATAGGAGCT AGCATTG TGTTTCTTGTCCTTG COL4A1.1 NM_001845 NM_001845.4 COL4A1 6147 ACAAAGGCCTCCCA 176 GAGTCCCAGGAA 908 CTCCTTTGACACCAG 1640 GGAT GACCTGCT GGATGCCAT COL4A2.1 NM_001846 NM_001846.2 COL4A2 6148 CAACCCTGGTGAT 177 CGCAGTGGTAGA 909 ACTATGCCAGCCGG 1641 GTCTGC GAGCCAGT AACGACAAGT COL5A2.2 NM_000393 NM_000393.3 COL5A2 6653 GGTCGAGGAACCC 178 GCCTGGAGGTCC 910 CCAGGAAATCCTGTA 1642 AAGGT AACTCTG GCACCAGGC COL7A1.1 NM_000094 NM_000094.2 COL7A1 4984 GGTGACAAAGGAC 179 ACCAGGCTCTCC 911 CTTGTCACCAGGGT 1643 CTCGG CTTGCT CCCCATTGTC COX2.1 NM_000963 NM_000963.1 PTGS2

71 TCTGCAGAGTTGGA 180 GCCGAGGCTTTT 912 CAGGATACAGCTCCA 1644 AGCACTCTA CTACCAGAA CAGCATCGATGTC CP.1 NM_000096 NM_000096.1 CP 3244 CGTGAGTACACAGA 181 CCAGGATGCCAA 913 TCTTCAGGGCCTCTC 1645 TGCCTCC GATGCT TCCTTTCGA CPB2.1 NM_001872 NM_001872.3 CPB2 9294 GGCACATACGGATT 182 CAGCGGCAAAAG 914 CGGAGCGTTACATCA 1646 CTTGCT CTTCTCTA AACCCACCT CRADD.1 NM_003805 NM_003805.3 CRADD 6149 GATGGTGCCTCCA 183 GAGTGAAAGTCA 915 CATGACTCAGGGAC 1647 GCAAC GGATTCAGCC ACACTCCCCA cripto (TDGF1 NM_003212 NM_003212.1 TDGF1 1096 GGGTCTGTGCCCC 184 TGACCGTGCCAG 916 CCTGGCTGCCCAAG 1648 official) ATGAC CATTTACA AAGTGTTCCCT CRP.1 NM_000567 NM_000567.2 CRP 4187 GACGTGAACCGACA 185 CTCCAGATAGGG 917 CTGTCAGAGGAGCC 1649 GGGTGT AGCTGGG CATCTCCCAT CSF1.1 NM_000757 NM_000757.3 CSF1 510 TGCAGCGGCTGAT 186 CAACTGTTCCTG 918 TCAGATGGAGACCTC 1650 TGACA CTCTACAAACTC GTGCCAAATTACA A CSFIR.2 NM_005211 NM_005211.1 CSF1R 2289 GAGCACAACCAAAC 187 CCTGCAGAGATG 919 AGCCACTCCCCACG 1651 CTACGA GGTATGAA CTGTTGT CSF2.1 NM_000758 NM_000758.2 CSF2 2779 GAACCTGAAGGACT 188 CTCATCTGGCCG 920 ATCCCCTTTGACTGC 1652 TTCTGCTTGT GTCTCACT TGGGAGCCAG CSF2R.2 NM_006140 NM_006140.3 CSF2RA 4477 TACCACACCCAGCA 189 CTAGAGGCTGGT 921 CGCAGATCCGATTTC 1653 TTCCTC GCCACTGT TCTGGGATC CSF3.2 NM_000759 NM_000759.1 CSF3 377 CCCAGGCCTCTGT 190 GGAGGACAGGA 922 TGCATTTCTGAGTTT 1654 GTCCTT GCTTTTTCTCA CATTCTCCTGCCTG CTGF.1 NM_001901 NM_001901.1 CTGF 2153 GAGTTCAAGTGCC 191 AGTTGTAATGGC 923 AACATCATGTTCTTC 1655 CTGACG AGGCACAG TTCATGACCTCGC CTSB.1 NM_001908 NM_001908.1 CTSB 382 GGCCGAGATCTAC 192 GCAGGAAGTCCG 924 CCCCGTGGAGGGAG 1656 AAAAACG AATACACA CTTTCTC CTSD.2 NM_001909 NM_001909.1 CTSD 385 GTACATGATCCCCT 193 GGGACAGCTTGT 925 ACCCTGCCCGCGAT 1657 GTGAGAAGGT AGCCTTTGC CACACTGA CTSH NM_004390 NM_004390.1 CTSH 845 GCAAGTTCCAACCT 194 CATCGCTTCCTC 926 TGGCTACATCCTTGA 1658 GGAAAG GTCATAGA CAAAGCCGA CTSL.2 NM_001912 NM_001912.1 CTSL1 446 GGGAGGCTTATCT 195 CCATTGCAGCCT 927 TTGAGGCCCAGAGC 1659 CACTGAGTGA TCATTGC AGTCTACCAGATTCT CTSL2.1 NM_001333 NM_001333.2 CTSL2 1667 TGTCTCACTGAGCG 196 ACCATTGCAGCC 928 CTTGAGGACGCGAA 1660 AGCAGAA CTGATTG CAGTCCACCA CTSS.1 NM_004079 NM_004079.3 CTSS 1799 TGACAACGGCTTTC 197 TCCATGGCTTTG 929 TGATAACAAGGGCAT 1661 CAGTACAT TAGGGATAGG CGACTCAGACGCT CUBN.1 NM_001081 NM_001081.2 CUBN 4110 GAGGCCGTTACTG 198 GAATCTCAGCGT 930 TGCCCCATCCTATCA 1662 TGGCA CAGGGC CATCCTTCA CUL1.1 NM_003592 NM_003592.2 CUL1 1889 ATGCCCTGGTAATG 199 GCGACCACAAGC 931 CAGCCACAAAGCCA 1663 TCTGCAT CTTATCAAG GCGTCATTGT CUL4A.1 NM_003589 NM_003589.1 CUL4A 2780 AAGCATCTTCCTGT 200 AATCCCATATCC 932 TATGTGCTGCAGAAC 1664 TCTTGGA CAGATGGA TCCACGCTG CX3CL1.1 NM_002996 NM_002996.3 CS3CL1 6150 GACCCTTGCCGTCT 201 GGAGTGTTCCTA 933 TAGAACCCAGCCCAT 1665 ACCTG GCACCTGG AAGAGGCCC CX3CR1.1 NM_001337 NM_001337.3 CX3CR1 4507 TTCCCAGTTGTGAC 202 GCTAAATGCAAC 934 ACTGAGGGCCAGCC 1666 ATGAGG CGTCTCAGT TCAGATCCT CXCL10.1 NM_001565 NM_001565.1 CXCL10 2733 GGAGCAAAATCGAT 203 TAGGGAAGTGAT 935 TCTGTGTGGTCCATC 1667 GCAGT GGGAGAGG CTTGGAAGC CXCL12.1 NM_000609 NM_000609.3 CXCL12 2949 GAGCTACAGATGC 204 TTTGAGATGCTT 936 TTCTTCGAAAGCCAT 1668 CCATGC GACGTTGG GTTGCCAGA CXCL14.1 NM_004887 NM_004887.3 CXCL14 3247 TGCGCCCTTTCCTC 205 CAATGCGGCATA 937 TACCCTTAAGAACGC 1669 TGTA TACTGGG CCCCTCCAC CXCL9.1 NM_002416 NM_002416.1 CXCL9 5191 ACCAGACCATTGTC 206 GTTACCAGAGGC 938 TGCTGGCTCTTTCCT 1670 TCAGAGC TAGCCAACA GGCTACTCC CXCR4.3 NM_003467 NM_003467.1 CXCR4 2139 TGACCGCTTCTACC 207 AGGATAAGGCCA 939 CTGAAACTGGAACAC 1671 CCAATG ACCATGATGT AACCACCCACAAG CXCR6.1 NM_006564 NM_006564.1 CXCR6 4002 CAGAGCCTGACGG 208 GCAGAGTGCAGA 940 CTGGTGAACCTACCC 1672 ATGTGT CAAACACC CTGGCTGAC CYP2C8.2 NM_000770 NM_000770.2 CYP2C8 505 CCGTGTTCAAGAG 209 AGTGGGATCACA 941 TTTTCTCAACTCCTC 1673 GAAGCTC GGGTGAAG CACAAGGCA CYP2C8v2.1 NM_030878 NM_030878.1 5423 GCTGTAGTGCACG 210 CAGTGGTCACTG 942 ATACAGTGACCTTGT 1674 AGATCCA CATGGG CCCCACCGG CYP3A4.2 NM_017460 NM_017460.3 CYP3A4 586 AGAACAAGGACAAC 211 GCAAACCTCATG 943 CACACCCTTTGGAAG 1675 ATAGATCCTTACAT CCAATGC TGGACCCAGAA AT CYR61.1 NM_001554 NM_001554.3 CYR61 2752 TGCTCATTCTTGAG 212 GTGGCTGCATTA 944 CAGCACCCTTGGCA 1676 GAGCAT GTGTCCAT

GTTTCGAAAT DAG1.1 NM_004393 NM_004393.2 CAG1 5968 GTGACTGGGCTCA 213 ATCCCACTTGTG 945 CAAGTCAGAGTTTCC 1677 TGCCT CTCCTGTC CTGGTGCCC DAPK1.3 NM_004938 NM_004938.1 DAPK1 636 CGCTGACATCATGA 214 TCTCTTTCAGCA 946 TCATATCCAAACTCG 1678 ATGTTCCT ACGATGTGTCTT CCTCCAGCCG DCBLD2.1 NM_080927 NM_080927.3 DCBLD2 3821 TCACCAGGGCAGG 215 GGTTGCATACTC 947 CATGCCTATGCTGAA 1679 AAGTTTA AGGCCC CCACTCCCA DCC.3 NM_005215 NM_005215.1 DCC 3763 AAATGTCCTCCTCG 216 TGAATGCCATCT 948 ATCACTGGAACTCCT 1680 ACTGCT TTCTTCCA CGGTCGGAC DCN.1 NM_001920 NM_001920.3 DCN 6151 GAAGGCCACTATCA 217 GCCTCTCTGTTG 949 CTGCTTGCACAAGTT 1681 TCCTCCT AAACGGTC TCCTGGGCT DCXR.1 NM_016286 NM_016286.2 DCXR 6311 CCATAGCGTCTACT 218 AGCTCTAGGGCC 950 TCAGCATGTCCAGG 1682 GCTCCA ATCACCT GCACCC DDC.1 NM_000790 NM_000790.3 DDC 5411 CAGAGCCCAGACA 219 CCACGTAATCCA 951 CCTCTCCTTCGGAAT 1683 CCATGA CCATCTCC TCACTTGCG DEFB1.1 NM_005218 NM_005218.3 DEFB1 4124 GATGGCCTCAGGT 220 TGCTGACGCAAT 952 CTCACAGGCCTTGG 1684 GGTAACT TGTAATGAT CCACAGATCT DET.1 NM_017996 NM_017996.2 DET1 2643 CTTGTGGAGATCAC 221 CCCGCCTGGATC 953 CTATGCCCGGGACT 1685 CCAATCAG TCAAACT CGGGCCT DHPS.3 NM_013407 NM_013407.1 DHPS 1722 GGGAGAACGGGAT 222 GCATCAGCCAGT 954 CTCATTGGGCACCA 1686 CAATAGGAT CCTCAAACT GCAGGTTTCC DIABLO.1 NM_019887 NM_019887.1 DIABLO 348 CACAATGGCGGCT 223 ACACAAACACTG 955 AAGTTACGCTGCGC 1687 CTGAAG TCTGTACCTGAA GACAGCCAA GA DIAPH1.1 NM_005219 NM_005219.2 DIAPH1 2705 CAAGCAGTCAAGG 224 AGTTTTGCTCGC 956 TTCTTCTGTCTCCCG 1688 AGAACCA CTCATCTT CCGCTTC DICER1.2 NM_177438 NM_177438.1 DICER1 1898 TCCAATTCCAGCAT 225 GGCAGTGAAGGC 957 AGAAAAGCTGTTTGT 1689 CACTGT GATAAAGT CTCCCCAGCA DKFZP564O0823. NM_015393 NM_015393.2 PARM1 3874 CAGCTACACTGTCG 226 ATGAGGCTGGAG 958 TGCTGAGCCTCCCA 1690 1 CAGTCC CTTGAGG CACTCATCTC DLC1.1 NM_006094 NM_006094.3 DLC1 3018 GATTCAGACGAGG 227 CACCTCTTGCTG 959 AAAGTCCATTTGCCA 1691 ATGAGCC TCCCTTTG CTGATGGCA DLL4.1 NM_019074 NM_019074.2 DLL4 5273 CACGGAGGTATAA 228 AGAAGGAAGGTC 960 CTACCTGGACATCCC 1692 GGCAGGAG CAGCCG TGCTCAGCC DPEP1.1 NM_004413 NM_004413.2 DPEP1 6295 GGACTCCAGATGC 229 TAAGCCCAGGCG 961 CACATGCAAGGACC 1693 CAGGA TCCTCT AGCATCTCCT DPYS.1 NM_001385 NM_001385.1 DPYS 6152 AAAGAATGGCACCA 230 AGTCGGGTGTTG 962 CACCATGTCATGGGT 1694 TGCAG AGGGGT CCACCTTTG DR4.2 NM_003844 NM_003844.1 TNFRSF10A 896 TGCACAGAGGGTG 231 TCTTCATCTGATT 963 CAATGCTTCCAACAA 1695 TGGGTTAC TACAAGCTGTAC TTTGTTTGCTTGCC ATG DR5.2 NM_003842 NM_003842.2 TNFRSF10B 902 CTCTGAGACAGTG 232 CCATGAGGCCCA 964 CAGACTTGGTGCCC 1696 CTTCGATGACT ACTTCCT TTTGACTCC DUSP1.1 NM_004417 NM_004417.2 DUSP1 2662 AGACATCAGCTCCT 233 GACAAACACCCT 965 CGAGGCCATTGACTT 1697 GGTTCA TCCTCCAG CATAGACTCCA DUSP9.1 NM_001395 NM_001395.1 DUSP9 6324 CGTCCTAATCAACG 234 CCCGCAAAGAAA 966 CGCTCGGAGCCTGC 1698 TGCCTA AAGTAACAG CTCTTC E2F1.3 NM_005225 NM_005225.1 E2F1 1077 ACTCCCTCTACCCT 235 CAGGCCTCAGTT 967 CAGAAGAACAGCTCA 1699 TGAGCA CCTTCAGT GGGACCCCT EBAG9.1 NM_004215 NM_004215.3 EBAG9 4151 CGCTCCTGTTTTTC 236 ACCGAAACTGGG 968 CAGTGGGTTTTGATT 1700 TCATCTGT TGATGG CCCACCATG ECRG4.1 NM_032411 NM_032411.1 C2orf40 3869 GCTCCTGCTCCTGT 237 TTTTGAAGCATC 969 ATTTCCACTTATGCC 1701 GCTG AGCTTGAGTT ACCTGGGCC EDG2.1 NM_001401 NM_001401.3 LPAR1 4673 ACGAGTCCATTGCC 238 GCTTGCTGACTG 970 CGAAGTGGAAAGCA 1702 TTCTTT TGTTCCAT TCTTGCCACA EDN1 NM_001955 NM_001955.1 EDN1 331 TGCCACCTGGACAT 239 TGGACCTAGGGC 971 CACTCCCGAGCACG 1703 endothelin.1 CATTTG TTCCAAGTC TTGTTCCGT DEN2.1 NM_001956 NM_001956.2 EDN2 2646 CGACAAGGAGTGC 240 CAGGCCGTAAGG 972 CCACTTGGACATCAT 1704 GTCTACTTCT AGCTGTCT CTGGGTGAACACTC EDNRA.2 NM_001957 NM_001957.1 EDNRA 3662 TTTCCTCAAATTTG 241 TTACACATCCAA 973 CCTTTGCCTCAGGG 1705 CCTCAAG CCAGTGCC CATCCTTTT EDNRB.1 NM_000115 NM_000115.1 EDNRB 3185 ACTGTGAACTGCCT 242 ACCACAGCATGG 974 TGCTACCTGCCCCTT 1706 GGTGC GTGAGAG TGTCATGTG EEF1A1.1 NM_001402 NM_001402.5 EEF1A1 5522 CGAGTGGAGACTG 243 CCGTTGTAACGT 975 CAAAGGTGACCACC 1707 GTGTTCTC TGACTGGA ATACCGGGTT EFNB1.2 NM_004429 NM_004429.3 DFNB1 3299 GGAGCCCGTATCC 244 GGATAGATCACC 976 CCCTCAACCCCAAGT 1708 TGGAG AAGCCCTTC TCCTGAGTG EFNB2.1 NM_004093 NM_004093.2 EFNB2 2567 TGACATTATCATCC 245 GTAGTCCCCGCT 977 CGGACAGCGTCTTC 1709

CGCTAAGGA GACCTTCTC TGCCCTCACT EGF.3 NM_001963 NM_001963.2 EGF 158 CTTTGCCTTGCTCT 246 AAATACCTGACA 978 AGAGTTTAACAGCCC 1710 GTCACAGT CCCTTATGACAA TGCTCTGGCTGACTT ATT EGFR.2 NM_005228 NM_005228.1 EGFR 19 TGTCGATGGACTTC 247 ATTGGGACAGCT 979 CACCTGGGCAGCTG 1711 CAGAAC TGGATCA CCAA EGLN3.1 NM_022073 NM_022073.2 RGLN3 2970 GCTGGTCCTCTACT 248 CCACCATTGCCT 980 CCGGCTGGGCAAAT 1712 GCGG TAGACCTC ACTACGTCAA EGR1.1 NM_001964 NM_001964.2 EGR1 2615 GTCCCCGCTGCAG 249 CTCCAGCTTAGG 981 CGGATCCTTTCCTCA 1713 ATCTCT GTAGTTGTCCAT CTCGCCCA EIF2C1.1 NM_012199 NM_012199.2 EIF2C1 6454 CCCTCACGGACTCT 250 TGGGTGACTTCC 982 CGTTCGCTTCACCAA 1714 CAGC ACCTTCA GGAGATCAA EIF4EBP1.1 NM_004095 NM_004095.2 EIF4EBP1 4275 GGCGGTGAAGAGT 251 TTGGTAGTGCTC 983 TGAGATGGACATTTA 1715 CACAGT CACACGAT AAGCACCAGCC ELTD1.1 NM_022159 NM_022159.3 ELTD1 6154 AGGTCTTGTGCAAG 252 AACCCCAAAGAT 984 CTCGCTCTTCTGTTC 1716 AGGAGC CCAGGTG CTTCTCGGC EMCN.1 NM_016242 NM_016242.2 EMCN 3875 AGGCACTGAGGGT 253 CACCGGCAAAAT 984 AATGCAAGCACTTCA 1717 GGAAA AATACTGGA GCAACCAGC EMP1.1 NM_001423 NM_001423.1 EMP1 986 GCTAGTACTTTGAT 254 GAACAGCTGGAG 986 CCAGAGAGCCTCCC 1718 GCTCCCTTGAT GCCAAGTC TGCAGCCA ENO2.1 NM_001975 NM_001975.2 ENO2 6155 TCCTTGGCTTACCT 255 AACCCCAATGAG 987 CTGTCTCTGCTCGCC 1719 GACCTC TAGGGCA CTCCTTTCT ENPEP.1 NM_001977 NM_001977.3 ENPEP 6156 CACCTACACGGAG 256 CCTGGCATCTGT 988 TCAAGAGCATAGTGG 1720 AACGGAC TGGTTCA CCACCGATC ENPP2.1 NM_006209 NM_006209.3 ENPP2 6174 CTCCTGCGCACTAA 257 TCCCTGGATAAT 989 TAACTTCCTCTGGCA 1721 TACCTTC TGGGTCTG TGGTTGGCC EPAS1.1 NM_001430 NM_001430.3 EPAS1 2754 AAGCCTTGGAGGG 258 TGCTGATGTTTT 990 TGTCGCCATCTTGG 1722 TTTCATTG CTGACAGAAAGA GTCACCACG EPB41L3.1 NM_012307 NM_012307.2 EPB41L3 4554 TCAGTGCCATACGC 259 CTTGGGCTCCAG 991 CTCTCCTTCCCTCTG 1723 TCTCAC GTAGCA GCTCTGTGC EPHA2.1 NM_004431 NM_004431.2 EPHA2 2297 CGCCTGTTCACCAA 260 GTGGCGTGCCTC 992 TGCGCCCGATGAGA 1724 GATTGAC GAAGTC TCACCG EPHB1.3 NM_004441 NM_004441.3 EPHB1 6508 CCTTGGGAGGGAA 261 GAAGTGAACTTG 993 ATGGCCTCTGGAGC 1725 GATCC CGGTAGGC TGTCCATCTC EPHB2.1 NM_004442 NM_004442.4 EPHB2 2967 CAACCAGGCAGCT 262 GTAATGCTGTCC 994 CACCTGATGCATGAT 1726 CCATC ACGGTGC GGACACTGC EPHB4.1 NM_004444 NM_004444.3 EPHB4 2620 TGAACGGGGTATC 263 AGGTACCTCTCG 995 CGTCCCATTTGAGCC 1727 CTCCTTA GTCAGTGG TGTCAATGT EPO.1 NM_000799 NM_000799.2 EPO 5992 CAGTGCCAGCAAT 264 CAAGTTGGCCCT 996 CTCTCTGGACAGTTC 1728 GACATCT GTGACAT CTCTGGCCC ErbB3.1 NM_001982 NM_001982.1 ERBB3 93 CGGTTATGTCATGC 265 GAACTGAGACCC 997 CCTCAAAGGTACTCC 1729 CAGATACAC ACTGAAGAAAGG CTCCTCCCGG ERBB4.3 NM_005235 NM_005235.1 ERBB4 407 TGGCTCTTAATCAG 266 CAAGGCATATCG 998 TGTCCCACGAATAAT 1730 TTTCGTTACCT ATCCTCATAAAG GCGTAAATTCTCCAG T ERCC1.2 NM_001983 NM_001983.1 ERCC1 869 GTCCAGGTGGATG 267 CGGCCAGGATAC 999 CAGCAGGCCCTCAA 1731 TGAAAGA ACATCTTA GGAGCTG ERCC4.1 NM_005236 NM_005236.1 ERCC4 5238 CTGCTGGAGTACG 268 GGGCGCACACTA 1000 CTGGTGCTGGAACT 1732 AGCGAC CTAGCC GCTCGACACT EREG.1 NM_001432 NM_001432.1 EREG 309 ATAACAAAGTGTAG 269 CACACCTGCAGT 1001 TTGTTTGCATGGACA 1733 CTCTGACATGAATG AGTTTTGACTCA GTGCATCTATCTGGT ERG.1 NM_004449 NM_004449.3 ERG 3884 CCAACACTAGGCTC 270 CCTCCGCCAGGT 1002 AGCCATATGCCTTCT 1734 CCCA CTTTAGT CATCTGGGC ERK1.3 NM_002746 Z11696.1 MAPK3 548 ACGGATCACAGTG 271 CTCATCCGTCGG 1003 CGCTGGCTCACCCC 1735 GAGGAAG GTCATAGT TACCTG ERK2.3 NM_002745 NM_002745.1 MAPK1 557 AGTTCTTGACCCCT 272 AAACGGCTCAAA 1004 TCTCCAGCCCGTCTT 1736 GGTCCT GGAGTCAA GGCTT ESPL1.3 NM_012291 NM_012291.1 ESPL1 2053 ACCCCCAGACCGG 273 TGTAGGGCAGAC 1005 CTGGCCCTCATGTC 1737 ATCAG TTCCTCAAACA CCCTTCACG ESRRG.3 NM_001438 NM_001438.1 ESRRG 2225 CCAGCACCATTGTT 274 AGTCTCTTGGGC 1006 CCCCAGACCAAGTG 1738 GAAGAT ATCGAGTT TGAATACATGCT F2.1 NM_000506 NM_000506.2 F2 2877 GCTGCATGTCTGG 275 CCTGACCGGGTG 1007 CCTCGGTAGTTCGTA 1739 AAGGTAACTG ATGTTCAC CCCAGACCCTCAG F3.1 NM_001993 NM_001993.2 F3 2871 GTGAAGGATGTGA 276 AACCGGTGCTCT 1008 TGGCACGGGTCTTC 1740 AGCAGACGTA CCACATTC TCCTACC FABP1.1 NM_001443 NM_001443.1 FABP1 6175 GGGTCCAAAGTGA 277 CCCTGTCATTGT 1009 ACATTCCTCCCCCAC 1741 TCCAAAA CTCCAGC CGTGAATTC FABP7.1 NM_001446 NM_001446.3 FABP7 4048 GGAGACAAAGTGG 278

CTCTTCTCCCAG 1010 TCTCAGCACATTCAA 1742 TCATCAGG CTGGAAACT GAACACGGAGA FAP.1 NM_004460 NM_004460.2 FAP 3403 CTGACCAGAACCAC 279 GGAAGTGGGTCA 1011 CGGCCTGTCCACGA 1743 GGCT TGTGGG ACCACTTATA fas.1 NM_000043 NM_000043.1 FAS 42 GGATTGCTCAACAA 280 GGCATTAACACT 1012 TCTGGACCCTCCTAC 1744 CCATGCT TTTGGACGATAA CTCTGGTTCTTACGT fasl.2 NM_000639 NM_000639.1 FSLG 94 GCACTTTGGGATTC 281 GCATGTAAGAAG 1013 ACAACATTCTCGGTG 1745 TTTCCATTAT ACCCTCACTGAA CCTGTAACAAAGA FBXW7.1 NM_033632 NM_033632.1 FBXW7 2644 CCCCAGTTTCAACG 282 GTTCCAGGAATG 1014 TCATTGCTCCCTAAA 1746 AGACTT AAAGCACA GAGTTGGCACTC FCER1G.2 NM_004106 NM_004106.1 FCER1G 4073 TGCCATCCTGTTTC 283 TGCCTTTCGCAC 1015 TTGTCCTCACCCTCC 1747 TGTATGGA TTGGATCT TCTACTGTCGACTG FCGR3S.1 NM_000569 NM_000569.4 FCGR3A 3080 GTCTCCAGTGGAA 284 AGGAATGCAGCT 1016 CCCATGATCTTCAAG 1748 GGGAAAA ACTCACTGG CAGGGAAGC FDPS.1 NM_002004 NM_002004.1 FDPS 516 GGATGATTACCTTG 285 TGCATTTGTTGT 1017 CAGTGTGACCGGCA 1749 ACCTCTTTGG CCTGGATGTC AAATTGGCAC FEN1.1 NM_004111 NM_004111.4 FEN1 3938 GTGGAGAAGGGTA 286 CTCATGGCAACC 1018 CGCTGAGAGACTCT 1750 CGCCAG AGTCCC GTTCTCCCTGG FGF1.1 NM_000800 NM_000800.2 FGF1 4561 GACACCGACGGGC 287 CAGCCTTTCCAG 1019 ACGGCTCACAGACA 1751 TTTTA GAACAAAC CCAAATGAGG FGF2.2 NM_002006 NM_002006.2 FGF2 681 AGATGCAGGAGAG 288 GTTTTGCAGCCT 1020 CCTGCAGACTGCTTT 1752 AGGAAGC TACCCAAT TTGCCCAAT FGF9.1 NM_002010 NM_002010.1 FGF9 6177 CACAGCTGCCATAC 289 AAGTAAGACTGC 1021 AGGCCACCCAGCCAG 1753 TTCGAC ACCCTCGC AATCCTGATA FGFR1.3 NM_023109 NM_023109.1 353 CACGGGACATTCAC 290 GGGTGCCATCCA 1022 ATAAAAAGACAACCA 1754 CACATC CTTCACA ACGGCCGACTGC FGFR2 NM_000141 NM_000141.2 FGFR2 2632 GAGGGACTGTTGG 291 GAGTGAGAATTC 1023 TCCCAGAGACCAAC 1755 isoform 1.1 CATGCA GATCCAAGTCTT GTTCAAGCAGTTG C FH.1 NM_000143 NM_000143.2 FH 4938 ATGGTTGCAGCCC 292 CAAAATGTCCAT 1024 ACAGTGACAGCAACA 1756 AAGTC TGCTGCC TGGTTCCCC FHIT.1 NM_002012 NM_002012.1 FHIT 871 CCAGTGGAGCGCT 293 CTCTCTGGGTCG 1025 TCGGCCACTTCATCA 1757 TCCAT TCTGAAACAA GGACGCAG FHL1.1 NM_001449 NM_001449.3 FHL1 4005 ATCCAGCCTTTGCC 294 CCTTGTAGCTGG 1026 TCCTATCTGCCACAC 1758 GAATA AGGGACC ATCCAGCGT FIGF.1 NM_004469 NM_004469.2 FIGF 3160 GGTTCCAGCTTTCT 295 GCCGCAGGTTCT 1027 ATTGGTGGCCACAC 1759 GTAGCTGT AGTTGCT CACCTCCTTA FILIP1.1 NM_015687 NM_015687.2 FILIP1 4510 ACACCGGTCACAAC 296 CTGGGATGACCC 1028 CCTGACACTGACTG 1760 GTCAT GTCTTG GGTTCCTCGA FKBP1A.1 NM_000801 NM_000801.2 FKBP1A 6330 CTGCCCTGACTGAA 297 TACGAGGAGAAA 1029 TCACTCAGCTTTGCT 1761 TGTGTT GGGGAAGA TCCGACACC FLJ22655.1 NM_024730 NM_024730.2 RERGL 3870 CTCCTTCACACAGA 298 AGGCAAACTGGG 1030 CACACTCACCCTAAC 1762 ACCTTTCA ATCGCT CTACTGGCGG FLT1.1 NM_002019 NM_002019.3 FLT1 6062 GGCCCTGAATCTA 299 TCCCACAGCAAT 1031 CTACAGCACCAAGA 1763 TCTTTG ACTCCGTA GCGACGTGTG FLT3LG.1 NM_001459 NM_001459.2 FLT3LG 6178 TGGGTCCAAGATG 300 GAAAGGCACATT 1032 AGTGTATCTCCGTGT 1764 CAAGG TGGTGACA TCACGCGCT FLT4.1 NM_002020 NM_002020.1 FLT4 2782 ACCAAGAAGCTGA 301 CCTGGAAGCTGT 1033 AGCCCGCTGACCAT 1765 GGACCTG AGCAGACA GGAAGATCT FN1.1 NM_002026 NM_002026.2 FN1 4528 GGAAGTGACAGAC 302 ACACGGTAGCCG 1034 ACTCTCAGGCGGTG 1766 GTGAAGGT GTCACT TCCACATGAT FOLR1.1 NM_016730 NM_016730.1 FOLR1 859 GAACGCCAAGCAC 303 CCAGGGTCGACA 1035 AAGCCAGGCCCCGA 1767 CACAAG CTGCTCAT GGACAAGTT FOS.1 NM_005252 NM_005252.2 FOS 2418 CGAGCCCTTTGATG 304 GGAGCGGGCTG 1036 TCCCAGCATCATCCA 1768 ACTTCCT TCTCAGA GGCCCAG FRAP1.1 NM_004958 NM_004958.2 MTOR 3095 AGCGCTAGAGACT 305 ATGATCCGGGAG 1037 CCTGACGGAGTCCC 1769 GTGGACC GCATAGT TGGATTTCAC FRP1.3 NM_003012 NM_003012.2 SFRP1 645 TTGGTACCTGTGG 306 CACATCCAAATG 1038 TCCCCAGGGTAGAAT 1770 GTTAGCA CAAACTGG TCAATCAGAGC FST.1 NM_006350 NM_006350.2 FST 2306 GTAAGTCGGATGA 307 CAGCTTCCTTCA 1039 CCAGTGACAATGCCA 1771 GCCTGTCTGT TGGCACACT CTTATGCCAGC FXD2.2 NM_001466 NM_001466.2 FZD2 3760 TGGATCCTCACCTG 308 GCGCTGCATGTC 1040 TGCGCTTCCACCTTC 1772 GTCG TACCAA TTCACTGTC G-Catenin.1 NM_002230 NM_002230.1 JUP 770 TCAGCAGCAAGGG 309 GGTGGTTTTCTT 1041 CGCCCGCAGGCCTC 1773 CATCAT GAGCGTGTACT ATCCT GADD45B.1 NM_015675 NM_015675.1 GADD45B 2481 ACCCTCGACAAGAC 310 TGGGAGTTCATG 1042 AACTTCAGCCCCAGC 1174 CACACT GGTACAGA TCCCAAGTC GAS2.1 NM_005256 NM_005256.2 GAS2 6451 AACATGTCATGGTC 311 GGGGTCGTGTTT 1043

CCTGCAAAAGTTTCC 1775 CGTGTG CAACAAAT CAGCCTCCT GATA3.3 NM_002051 NM_002051.1 GATA3 1 CAAAGGAGCTCACT 312 GAGTCAGAATGG 1044 TGTTCCAACCACTGA 1776 GTGGTGTCT CTTATTCACAGAT ATCTGGACC G GATM.1 NM_001482 NM_001482.2 GATM 6296 GATCTCGGCTTGG 313 GTAGCTGCCTGG 1045 AAAGTTCGCTGCACC 1777 ACGAAC GTGCTCT CATCCTGTC GBL.1 NM_022372 NM_022372.3 MLST8 6302 GCTGTCAATAGCAC 314 GGTCACCTCGTC 1046 CCCCCGTCAGATTCCC 1778 ACCAATG AGACATAGC GBP2.2 NM_004120 NM_004120.2 GBP2 2060 GCATGGGAACCAT 315 TGAGGAGTTTGC 1047 CCATGGACCAACTTC 1779 CAACCA CTTGATTCG ACTATGTGACAGAGC GCLC.3 NM_001498 NM_001498.1 GCLC 330 CTGTTGCAGGAAG 316 GTCAGTGGGTCT 1048 CATCTCCTGGCCCA 1780 GCATTGA CTAATAAAGAGA GCATGTT TGAG GCLM.2 NM_002061 NM_002061.1 GCLM 704 TGTAGAATCAAACT 317 CACAGAATCCAG 1049 TGCAGTTGACATGG 1781 CTTCATCATCAACT CTGTGCAACT CCTGTTCAGTCC AG GFRA1.1 NM_005264 NM_005264.3 GFRA1 6179 TCCGGGTTAAGAAC 318 GTGGCAAAACAT 1050 TTTCATTCTCAGACC 1782 AAGCC GAGTGGG CTGCTGGCC GJA1.1 NM_000165 NM_000165.2 GJA1 2600 GTTCACTGGGGGT 319 AAATACCAACAT 1051 ATCCCCTCCCTCTCC 1783 GTATGG GCACCTCTCTT ACCCATCTA GLYAT.1 NM_201648 NM_201648.2 GLYAT 6180 TACCATTGCAAGGT 320 GGATGCTGGGA 1052 AGGATTTCTCCAGCA 1784 GCCC GGCTCTT TCTGCAGCA GMNN.1 NM_015895 NM_015895.3 GMNN 3880 GTTCGCTACGAGG 321 TGCGTACCCACT 1053 CCTCTTGCCCACTTA 1785 ATTGAGC TCCTGC CTGGGTGGA GNAS.1 NM_000516 NM_000516.3 GNAS 2665 GAACGTGCCTGAC 322 ACTCCTTCATCC 1054 CCTCCCGAATTCTAT 1786 TTTGACTT TCCCACAG GAGCATGCC GPC3.1 NM_004484 NM_004484.2 GPC3 659 TGATGCGCCTGGA 323 CGAGGTTGTGAA 1055 AGCAGGCAACTCCG 1787 AACAGT AGGTGCTTATC AAGGACAACG GPX1.2 NM_000581 NM_000581.2 GPX1 2955 GCTTATGACCGACC 324 AAAGTTCCAGGC 1056 CTCATCACCTGGTCT 1788 CCAA AACATCGT CCGGTGTGT GPX2.2 NM_002083 NM_002083.1 GPX2 890 CACACAGATCTCCT 325 GGTCCAGCAGTG 1057 CATGCTGCATCCTAA 1789 ACTCCATCCA TCTCCTGAA GGCTCCTCAGG GPX3.1 NM_002084 NM_002084.3 GPS3 6271 GCTCTAGGTCCAAT 326 TGGAGGCAGTG 1058 ACTGATACCTCAACC 1790 TGTTCTGC GGAGATG TTGGGGCCA GRB14.1 NM_004490 NM_004490.1 SRG14 2784 TCCCACTGAAGCCC 327 AGTGCCCAGGCG 1059 CCTCCAAGCGAGTC 1791 TTTCAG TAAACATC CTTCTTCAACCG GRB7.2 NM_005310 NM_005310.1 GRB7 20 CCATCTGCATCCAT 328 GGCCACCAGGGT 1060 CTCCCCACCCTTGAG 1792 CTTGT ATTATCTG AAGTGCCT GRO1.2 NM_001511 NM_001511.1 CXCL1 86 CGAAAAGATGCTGA 329 TCAGGAACAGCC 1061 CTTCCTCCTCCCTTC 1793 ACAGTGACA ACCAGTGA TGGTCAGTTGGAT GSTM1.1 NM_000561 NM_000561.1 GSTM1 727 AAGCTATGAGGAAA 330 GGCCCAGCTTGA 1062 TCAGCCACTGGCTTC 1794 AGAAGTACACGAT ATTTTTCA TGTCATAATCAGGAG GSTM3.2 NM_000849 NM_000849.3 GSTM3 731 CAATGCCATCTTGC 331 TCCACTCGAAT 1063 CTCGCAAGCACAACA 1795 GCTACAT CTTTTCTTCTTCA TGTGTGGTGAGA GSTp.3 NM_000852 NM_000852.2 GSTP1 66 GAGACCCTGCTGT 332 GGTTGTAGTCAG 1064 TCCCACAATGAAGGT 1796 CCCAGAA CGAAGGAGATC CTTGCCTCCCT GSTT1.3 NM_000853 NM_000853.1 GSTT1 813 CACCATCCCCACCC 333 GGCCTCAGTGTG 1065 CACAGCCGCCTGAA 1797 TGTCT CATCATTCT AGCCACAAT GZMA.1 NM_6144 NM_6144.2 GZMA 4111 GAAAGAGTTTCCCT 334 TGCTTTTTCCGT 1066 AGCCACACGCGAAG 1798 ATCCATGC CAGCTGTAA GTGACCTTAA HADH.1 NM_005327 NM_005327.2 HADH 6181 CCACCAGACAAGA 335 CCACAAGTTTCA 1067 CTGGCCTCCATTTCT 1799 CCGATTC TGACAGGC TCAACCCAG HAVCR1.1 NM_012206 NM_012206.2 HAVCR1 6284 CCACCCAAGGTCA 336 GAACAGTGGTGC 1068 TCACAACTGTTCCAA 1800 CGACTAC TCGTTCG CCGTCACGA HDAC1.1 NM_004964 NM_004964.2 HDAC1 2602 CAAGTACCACAGC 337 GCTTGCTGTACT 1069 TTCTTGCGCTCCATC 1801 GATGACTACATTAA CCGACATGTT CGTCCAGA Hepsin.1 NM_002151 NM_002151.1 HPN 814 AGGCTGCTGGAGG 338 CTTCCTGCGGCC 1070 CCAGAGGCCGTTTC 1802 TCATCTC ACAGTCT TTGGCCG HER2.3 NM_004448 NM_004448.1 ERBB2 13 CGGTGTGAGAAGT 339 CCTCTCGCAAGT 1071 CCAGACCATAGCACA 1803 GCAGCAA GCTCCAT CTCGGGCAC HGD.1 NM_000187 NM_000187.2 HGD 6303 CTCAGGTCTGCCC 340 TTATTGGTGCTC 1072 CTGAGCAGCTCTCA 1804 CTACAAT CGTGGAC GGATCGGCTT HGF.4 M29145 457 CCGAAATCCAGATG 341 CCCAAGGAATGA 1073 CTCATGGACCCTGG 1805 ATGATG GTGGATTT TGCTACACG HGFAC.1 NM_001528 NM_001528.2 HGFAC 2704 CAGGACACAAGTG 342 GCAGGGAGCTG 1074 CGCTCACGTTCTCAT 1806 CCAGATT GAGTAGC CCAAGTGG HIF1A.3 NM_001530 NM_001530.1 HIF1A 399 TGAACATAAAGTCT 343 TGAGGTTGGTTA 1075 TTGCACTGCACAGG 1807 GCAACATGGA CTGTTGGTATCA CCACATTCAC TATA HIFAN.1 NM_17902 NM_17902.2 HIF1AN 7211 TGTTGGCCAGGTC 344 GCATCATAGGGC 1076 CTCTAGCCAGTTAGC 1808

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CCATGTC AGCACTGCA p53.2 NM_000546 NM_000546.2 TP53 59 CTTTGAACCCTTGC 510 CCCGGGACAAAG 1242 AAGTCCTGGGTGCTT 1974 TTGCAA CAAATG CTGACGCAC PAH.1 NM_000277 NM_000277.1 PAH 6199 TGGCTGATTCCATT 511 CACATTCTGTCC 1243 ATCCTTTGCAGTGCC 1975 AACAGTGA ATGGCTTTAC CTCCAGAAA PAI1.3 NM_000602 NM_000602.1 SERPINE1 36 CCGCAACGTGGTTT 512 TGCTGGGTTTCT 1244 CTCGGTGTTGGCCA 1976 TCTCA CCTCCTGTT TGCTCCAG Pak1.2 NM_002576 NM_002576.3 PAK1 3421 GAGCTGTGGGTTG 513 CCATGCAAGTTT 1245 ACATCTGTCAAGGAG 1977 TTATGGA CTGTCACC CCTCCAGCC PARD6A.1 NM_016948 NM_016948.2 PARD6A 4514 GATCCTCGAGGTC 514 ACCATCATGTCC 1246 TCCAAGGTCTTCCCG 1978 AATGGC GTCACTTG GCTACTTCA PBOB1.1 NM_021635 NM_021635.1 PBOV1 3936 GCAAAGCCTTTCCA 515 GGCTGGGCTTAA 1247 TGGTAGCAGAATTGC 1979 GAAAAA ACAGTCAT CTTTTCAACCA PCCA.1 NM_000282 NM_000282.2 PCCA 6250 GGTGAAATCTGTGC 516 ATTCCAGCTCCA 1248 TCCCCTTCTCCAACT 1980 ACTGTCA CGAGCA GTGTCTCCA PCK1.1 NM_002591 NM_002591.2 PCK1 6251 CTTAGCATGGCCCA 517 CTTCCGGAACCA 1249 CAGCCAAACTGCCC 1981 GCAC GTTGACA AAGATCTTCC PNCA.2 NM_002592 NM_002592.1 PCNA 148 GAAGGTGTTGGAG 518 GGTTTACACCGC 1250 ATCCCAGCAGGCCT 1982 GCACTCAAG TGGAGCTAA CGTTGATGAG PCSK6.1 NM_002570 NM_002570.3 PCSK6 6282 ACCTTGAGTAGCAG 519 GTTGCTGGAGCC 1251 CACACCTTCCTCAGA 1983 AGGCCC ATTTCAC ATGGACCCC PDCD1.1 NM_005018 NM_005018.2 PDCD1 6286 GACAACGCCACCTT 520 GGCTCATGCGGT 1252 TCTCCAACACATCGG 1984 CACC ACCAGT AGAGCTTCG PDE4DIP.1 NM_014644 NM_014644.4 PDE4DIP 6417 GCTTCGTCTTGCTG 521 AGCTTCATTGGA 1253 TCGCGCAGTCTCTCT 1985 TGAGAG GGAGAGGA AAGTCATGATCTC PDGFA.3 NM_002607 NM_002607.2 PDGFA 56 TTGTTGGTGTGCCC 522 TGGGTTCTGTCC 1254 TGGTGGCGGTCACT 1986 TGGTG AAACACTGG CCCTCTGC PDGFB.3 NM_002608 NM_002608.1 PDGFB 67 ACTGAAGGAGACC 523 TAAATAACCCTG 1255 TCTCCTGCCGATGC 1987 CTTGGAG CCCACACA CCCTAGG PDGFC.3 NM_016205 NM_016205.1 PDGFC 29 AGTTACTAAAAAAT 524 GTCGGTGAGTGA 1256 CCCTGACACCGGTC 1988 ACCACGAGGTCCTT TTTGTGCAA TTTGGTCTCAACT PDGFD.2 NM_025208 NM_025208.2 PDGFD 31 TATCGAGAGGCAGGT 525 TAACGCTTGGCA 1257 TCCAGGTCAACTTTT 1989 CATACCA TCATCATT GACTTCCGGT PDGFRa.2 NM_006206 NM_006206.2 PDGFRA 24 GGGAGTTTCCAAG 526 CTTCAACCACCT 1258 CCCAAGACCCGACC 1990 AGATGGA TCCCAAAC AAGCACTAG PDGFRb.3 NM_002609 NM_002609.2 PDGFRB 464 CCAGCTCTCCTTCC 527 GGGTGGCTCTCA 1259 ATCAATGTCCCTGTC 1991 AGCTAC CTTAGCTC CGAGTGCTG PDZK1.1 NM_002614 NM_002614.3 PDZK1 6319 AATGACCTCCACCT 528 CGCAGGAAGAAG 1260 TGCCCTTCTTGCTTG 1992 TCAACC CCATAGTT GACAGTTTACA PDZK3.1 NM_015022 NM_015022.2 3885 GAGCTGAGAGCCT 529 CTCGGCCCTGCT 1261 CTCGCTGCAGAGCT 1993 TGAGCAT GAGTAA TGTCAAGGTC PF4.1 NM_002619 NM_002619.1 PF4 6326 GCAGTGCCTGTGT 530 GGCCTTGATCAC 1262 TCCGTCCCAGGCAC 1994 GTGAAG CTCCAG ATCACC PFKP.1 NM_002627 NM_002627.3 PFKP 6252 AGCTGATGCCGCA 531 GGTGCTCCACGT 1263 CAGATCCCTGATGTC 1995 TACATT TGGACT GAAGGGCTC PFN2.1 NM_053024 NM_053024.1 PFN2 3426 TCTATACGTCGATG 532 GCCGACAGCCAC 1264 CTCCCCACCTTGACT 1996 GTGACTGC ATTGTAT CTTTGTCCG PGF.1 NM_002632 NM_002632.4 PGF 4026 GTGGTTTTCCCTCG 533 AGCAAGGGAACA 1265 ATCTTCTCAGACGTC 1997 GAGC GCCTCAT CCGAGCCAG PI3K.2 NM_002646 NM_002646.2 PIK3C2B 368 TGCTACCTGGACA 534 AGGCCGTCCTTC 1266 TCCTCCTGAAACGAG 1998 GCCCG AGTAACCA CTGTGTCTGACTT PI3KC2A.1 NM_002645 NM_002645.2 PI3KC2A 6064 ATACCAATCACCGC 535 CACACTAGCATT 1267 TGTGCTGTGACTGG 1999 ACAAACC TTCTCCGCATA ACTTAACAAATAGCC PIK3CA.1 NM_006218 NM_006218.1 PIK3CA 2962 GTGATTGAAGAGCA 536 GTCCTGCGTGGG 1268 TCCTGCTTCTCGGGA 2000 TGCCAA AATAGC TACAGACCA PLA2G4C.1 NM_003706 NM_003706.1 PLA2G4C 6249 CCCTTTCCCCAAGT 537 AGGATGTAGCAG 1269 CCTTGGACCACAAAT 2001 AGAAGAG CTGGCG CCAGCTCAG PLAT.1 NM_033011 NM_033011.2 PLAT 6459 GATTTGCTGGGAA 538 TAGCTGATGCCC 1270 TAGATACCAGGGCC 2002 GTGCTGT TGGTCC ACGTGCTACG PLAUR.3 NM_002659 NM_002659.1 PLAUR 708 CCCATGGATGCTC 539 CCGGTGGCTACC 1271 CATTGACTGCCGAG 2003 CTCTGAA AGACATTG GCCCCATG PLG.1 NM_000301 NM_000301.1 PLG 6310 GGCAAAATTTCCAA 540 ATGTATCCATGA 1272 TGCCAGGCCTGGGA 2004 GACCAT GCGTGTGG CTCTCA PLN.1 NM_002667 NM_002667.2 PLN 3886 TGATGCTTCTCTGA 541 CCTGTCTGCATG 1273 AGATCTGCAGCTTGC 2005 AGTTCTGC GGATGAC CACATCAGC PLOD2.1 NM_000935 NM_000935.2 PLOD2 3820 CAGGGAGGTGGTT 542 TCTCCCAGGATG 1274 TCCAGCCTTTTCGTG 2006 GCAAAT CATGAAG

GTGACTCAA PLP1.1 NM_000533 NM_000533.3 PLP1 4027 AGAACAGACTGGC 543 CAGAGGGCCATC 1275 CACCATTAGCCACCA 2007 CTGAGGA TCAGGTT GCAACTGCT PMP22.1 NM_000304 NM_000304.2 PMP22 6254 CCATCTACACGGTG 544 TGTAGGCGAAAC 1276 AATCCGAGTTGAGAT 2008 AGGCA CGTAGGA GCCACTCCG PPAP2B.1 NM_003713 NM_003713.3 PPAP2B 6457 ACAAGCACCATCCC 545 CACGAAGAAAAC 1277 ACCAGGGCTCCTTG 2009 AGTGA TATGCAGCAG AGCAAATCCT PPARG.3 NM_005037 NM_005037.3 PPARG 1086 TGACTTTATGGAGC 546 GCCAAGTCGCTG 1278 TTCCAGTGCATTGAA 2010 CCAAGTT TCATCTAA CTTCACAGCA PPP1R3C.1 NM_005398 NM_005398.4 PPP1R3C 6320 TTCCTTCCCTCTCA 547 CACAGCTTTCCA 1279 CCTTCCTCAACTTTT 2011 ATCCAC TCACCATC CCTTGCCCA PPP2CA.1 NM_002715 NM_002715.2 PPP2CA 3879 GCAATCATGGAACT 548 ATGTGGCTCGCC 1280 TTTCTTGCAGTTTGA 2012 TGACGA TCTACG CCCAGCACC PRCC.1 NM_005973 NM_005973.4 PRCC 6002 GAGGAAGAGGAGG 549 CAGGGAGAGAAG 1281 CTACATCTGGGCCC 2013 CGGTG CGAACAA GCTTTAGGG PRKCA.1 NM_002737 NM_002737.1 PRKCA 2626 CAAGCAATGCGTCA 550 GTAAATCCGCCC 1282 CAGCCTCTGCGGAA 2014 TCAATGT CCTCTTCT TGGATCACACT PRLCB1.1 NM_002738 NM_002738.5 PRLCB1 3739 GACCCAGCTCCACT 551 CCCATTCACGTA 1283 CCAGACCATGGACC 2015 CCTG CTCCATCA GCCTGTACTT PRKCD.2 NM_006254 NM_006254.1 PRKCD 626 CTGACACTTGCCG 552 AGGTGGTCCTTG 1284 CCCTTTCTCACCCAC 2016 CAGAGAA GTCTGGAA CTCATCTGCAC PRKCH.1 NM_006255 NM_006255.3 PRKCH 4370 CTCCACCTATGAGC 553 CACACTTTCCCT 1285 TCCTGTTAACATCCC 2017 GTCTGTC CCTTTTGG AAGCCCACA PRO2000.3 NM_014109 NM_014109.2 ATAD2 1666 ATTGGAAAAACCTC 554 TCGGTATCTTGG 1286 CCCAACATATTTTAT 2018 GTCACC TCTTGCAG AGTGGCCCAGC PROM1.1 NM_006017 NM_006017.1 PROM1 4516 CTATGACAGGCAG 555 CTCCAACCATGA 1287 ACCCGAGGCTGTGT 2019 CCACC GGAAGACG CTCCAACAC PROM2.1 NM_144707 NM_144707.1 PROM2 5108 CTTCAGCGCATCCA 556 CCATGCTGGTCT 1288 CTTCCTCGTTCAGAT 2020 CTACC TCACCAC CCAGAGGCC PRPS2.1 NM_00103901 NM_00103901 PRPS2 4694 CACTGCACCAAGAT 557 ATTGTGTGTCCT 1289 TTGACATTTCCATGA 2021 TCAGGT TCGGATTG TCTTGGCCG PRSS8.1 NM_002773 NM_002773.2 PRSS8 4742 GTACACTCTGGCCT 558 CCACACGAGGCT 1290 TCCTGGATCCAAAGC 2022 CCAGCTA GGAGTT AAGGTGACA PSMA7.1 NM_002792 NM_002792.2 PSMA7 6255 GCCAAACTGCAGG 559 AGGCCATGCAGA 1291 CCAAAGCACAGATCT 2023 ATGAAAG CGTTGT TCCGCACTG PSMB8.1 NM_148919 NM_148919.3 PSMB8 6461 CAGTGGCTATCGG 560 TAAGCAATAGCC 1292 CAAGGTCATAGGCCT 2024 CCTAATC CTGCGG CTTCAGGGC PSMB9.1 NM_148954 NM_148954.2 PSMB9 6462 GGGGTGTCATCTA 561 CATTGCCCAAGA 1293 TGGTCCACACCGGC 2025 CCTGGTC TGACTCG AGCTGTAATA PTEN.2 NM_000314 NM_000314.1 PTEN 54 TGGCTAAGTGAAGA 562 TGCACATATCATT 1294 CCTTTCCAGCTTTAC 2026 TGACAATCATG ACACCAGTTCGT AGTGAATTGCTGCA PTGIS.1 NM_000961 NM_000961.3 PTGIS 5429 CCACACTGGCATCT 563 GCCCATGGGATG 1295 CCTTCTCCAGGGACA 2027 CCCT AGAAACT GAAGCAGGA PTHR1.1 NM_000316 NM_000316.1 PTHR1 2375 CGAGGTACAAGCT 564 GCGTGCCTTTCG 1296 CCAGTGCCAGTGTC 2028 GAGATCAAGAA CTTGAA CAGCGGCT PTK2.1 NM_005607 NM_005607.3 PTK2 2678 GACCGGTCGAATG 565 CTGGACATCTCG 1297 ACCAGGCCCGTCAC 2029 ATAAGGT ATGACAGC ATTCTCGTAC PTK2B.1 NM_004103 NM_004103.3 PTK2B 2883 CAAGCCCAGCCGA 566 GAACCTGGAACT 1298 CTCCGCAAACCAACC 2030 CCTAAG GCAGCTTTG TCCTGGCT PTN.1 NM_002825 NM_002825.5 PTN 3964 CCTTCCAGTCCAAA 567 CCCCTCTCTCCA 1299 TTCCTCTGCTCTGGG 2031 AATCCC CTTTGGAT GCTCTCTTG PTPNS1.1 NM_080792 NM_080792.1 SIRPA 2896 CTCCAGCTAGCACT 568 TTTCAAGATTGC 1300 TCTCAGTAATTTACA 2032 AAGCAACATC ACGTTTCACAT GGCGTCCACAG PTPRB.1 NM_002837 NM_002837.2 PTPRB 3881 GATATGCGGTGAG 569 CTGGCCACACCG 1301 ATGCACACAGACTCA 2033 GAACAGC TATAGTGA TCCGCCACT PTPRC.1 NM_080921 NM_080921.2 PTPRC 6450 TGGCCGTCATGG 570 GGACATCTTTTG 1302 CAACATCTCCAAAAG 2034 AAGA TGCTGGTTG CCCGAGTGC PTPRG.1 NM_002841 NM_002841.2 PTPRG 2682 GGACAGCGACAAA 571 GGACTCGGAACA 1303 CACCATTAGCCATGT 2035 GACTTGA GGTAAAGG CTCACCCGA PTTG1.2 NM_004219 NM_004219.2 PTTG1 1724 GGCTACTCTGATCT 572 GCTTCAGCCCAT 1304 CACACGGGTGCCTG 2036 ATGTTGATAAGGAA CCTTAGCA GTTCTCCA PVALB.1 NM_002854 NM_002854.2 PVALB 4316 AAACCAAGATGCTG 573 CAGCCACCAGAG 1305 TTTTGCCGTCCCCAT 2037 ATGGCT TGGAGAA CTTTGTCTC PXDN.1 NM_012293 NM_012293.1 PXDN 6257 GCTGCTCAAGCTG 574 ACCCACGATCTT 1306 ACTGGGACGGCGAC 2038 AACCC CCTGGTC ACCATCTACT RAC1.3 NM_006908 NM_006908.3 RAC1 2698 TGTTGTAAATGTCT 575 TTGAGCAAAGCG 1307 CGTTCTTGGTCCTGT 2039 CAGCCCC

TACAAAGG CCCTTGGA RAD51.1 NM_002875 NM_002875.2 RAD51 3976 AGACTACTCGGGT 576 AGCATCCGCAGA 1308 CTTTCAGCCAGGCA 2040 CGAGGTG AACCTG GATGCACTTG RAF1.3 NM_002880 NM_002880.1 RAF1 2130 CGTCGTATGCGAG 577 TGAAGGCGTGAG 1309 TCCAGGATGCCTGTT 2041 AGTCTGT GTGTAGAA AGTTCTCAGCA RALBP1.1 NM_006788 NM_006788.2 RALBP1 2105 GGTGTCAGATATAA 578 TTCGATATTGCC 1310 TGCTGTCCTGTCGG 2042 ATGTGCAAATGC AGCAGCTATAAA TCTCAGTACGTTCA RARB.2 NM_016152 NM_016152.2 RARB 687 TGCCTGGACATCCT 579 AAGGCCGTCTGA 1311 TGCACCAGGTATACC 2043 GATTCT GAAAGTCA CCAGAACAAGA RASSF1.1 NM_007182 NM_007182.4 RASSF1 6658 AGGGCACGTGAAG 580 AAAGAGTGCAAA 1312 CACCACCAAGAACTT 2044 TCATTG CTTGCGG TCGCAGCAG RB1.1 NM_000321 NM_000321.1 RB1 956 CGAAGCCCTTACAA 581 GGACTCTTCAGG 1313 CCCTTACGGATTCCT 2045 GTTTCC GGTGAAAT GGAGGGAAC RBM35A.1 NM_017697 NM_017697.2 ESRP1 5109 TGGTTTTGAATCAC 582 CTCTGTCCGCAG 1314 CGCCCATCAGGAGA 2046 CAGGG ACTTCATCT TGCCTTTATC REG4.1 NM_032044 NM_032044.2 REG4 3226 TGCTAACTCCTGCA 583 TGCTAGGTTTCC 1315 TCCTCTTCCTTTCTG 2047 CAGCC CCTCTGAA CTAGCCTGGC RET.1 NM_020630 NM_020630.3 RET 5001 GCCTGTGCAGTTCT 584 GGAAGGGCAGA 1316 AACATCAGCGTGGC 2048 TGTGC CCCTCAC CTACAGGCTC RGS1.1 NM_002922 NM_002922.3 RGS1 6258 TGCCCTGTAAAGCA 585 CTCGAGTGCGGA 1317 AGCAGCATCTGAATG 2049 GAAGAGAT AGTCAATA CACAAATGCT RGS5.1 NM_003617 NM_003617.1 RGS5 2004 TTCAAACGGAGGCT 586 GAAGGTTCCACC 1318 AATATTGACCACTTC 2050 CCTAAAGAG AGGTTCTTCA ACTAAGGACATCACA RHEB.2 NM_005614 NM_005614.3 RHEB 6609 GATGATTGAGAACA 587 GCTCCCAAGACT 1319 TGTCACTGTCCTAGA 2051 GCCTTGC CTGACACA ACACCCTGGAGTT RhoB.1 NM_004040 NM_004040.2 RHOB 2951 AAGCATGAACAGGA 588 CCTCCCCAAGTC 1320 CTTTCCAACCCCTGG 2052 CTTGACC AGTTGC GGAAGACAT rhoC.1 NM_175744 NM_175744.1 RHOC 773 CCCGTTCGGTCTG 589 GAGCACTCAAGG 1321 TCCGGTTCGCCATGT 2053 AGGAA TAGCCAAAGG CCCG RIPK1.1 NM_003804 NM_003804.3 RIPK1 6259 AGTACCTTCAAGCC 590 AAGTCCCTGGGA 1322 CAGCCACAGAACAG 2054 GGTCAA ACTGTGC CCTGGTTCAC RND3.1 NM_005168 NM_005168.3 RND3 5381 TCGGAATTGGACTT 591 CTGGTTACTCCC 1323 TTTTAAGCCTGACTC 2055 GGGAG CTCCAACA CTCACCGCG ROCK1.1 NM_005406 NM_005406.1 ROCK1 2959 TGTGCACATAGGAA 592 GTTTAGCACGCA 1324 TCACTCTCTTTGCTG 2056 TGAGCTTC ATTGCTCA GCCAACTGC ROCK2.1 NM_004850 NM_004850.3 ROCK2 2992 GATCCGAGACCCT 593 AGGACCAAGGAA 1325 CCCATCAACGTGGA 2057 CGCTC TTTAAGCCA GAGCTTGCT RPLP1.1 NM_213725 NM_213725.1 RPLP1 5478 CAAGGTGCTCGGT 594 GTCGCCGGATGA 1326 CCTCACCCCAACGC 2058 CCTTC AGTGAG AGCCTTAGCT RPS23.1 NM_001025 NM_001025.1 RPS23 3320 GTTCTGGTTGCTG 595 CCTTAAAGCGGA 1327 ATCACCAACAGCATG 2059 GATTTGG CTCCAGG ACCTTTGCG RPS27A.1 NM_02954 NM_02954.3 RPS27A 6329 CTTACGGGGAAGA 596 TCCTGGATCTTG 1328 TCGTATCCGAGGGTT 2060 CCATCAC GCCTTTAC CAACCTCG RPS6KA1.1 NM_002953 NM_002953.3 RPS6KA1 3865 GCTCATGGAGCTA 597 CGGCTGAAGTCC 1329 ACCCGGAGAATGGA 2061 GTGCCTC AGCTTCT CAGACCTCAG RPS6KB1.3 NM_003161 NM_003161.1 RPS6KB1 928 GCTCATTATGAAAA 598 AAGAAACAGAAG 1330 CACACCAACCAATAA 2062 ACATCCCAAAC TTGTCTGGCTTT TTTCGCATT CT RRM1.2 NM_001033 NM_001033.1 RRM1 1000 GGGCTACTGGCAG 599 CTCTCAGCATCG 1331 CATTGGAATTGCCAT 2063 CTACATT GTACAAGG TAGTCCCAGC RRM2.1 NM_001034 NM_001034.1 RRM2 2546 CAGCGGGATTAAAC 600 ATCTGCGTTGAA 1332 CCAGCACAGCCAGT 2064 AGTCCT GCAGTGAG TAAAAGATGCA RUNX1.1 NM_001754 NM_001754.3 RUNX1 6067 AACAGAGACATTGC 601 GTGATTTGCCCA 1333 TTGGATCTGCTTGCT 2065 CAACCA GGAAAGTTT GTCCAAACC S100A1.1 NM_006271 NM_006271.1 S100A1 2851 TGGACAAGGTGAT 602 AGCACCACATAC 1334 CCTCCCCGTCTCCAT 2066 GAAGGAG TCCTGGAA TCTCGTCTA S100A10.1 NM_002966 NM_002966.1 S100A10 3579 ACACCAAAATGCCA 603 TTTATCCCCAGC 1335 CACGCCATGGAAAC 2067 TCTCAA GAATTTGT CATGATGTTT S100A2.1 NM_005978 NM_005978.2 S100A2 2369 TGGCTGTGCTGGT 604 TCCCCCTTACTC 1336 CACAAGTACTCCTGC 2068 CACTACCT AGCTTGAACT CAAGAGGGCGAC SAA2.2 NM_030754 NM_030754.2 SAA2 6655 CTACAGCACAGATC 605 TGCTGACACTCA 1337 AGCTTCTCACGGGC 2069 AGCACCA GGACCAAG CTGGTTTTCT SCN4B.1 NM_174934 NM_174934.3 SCN4B 7223 GCCTTCCTGGAGTA 606 GTGGCCCAATTC 1338 TGCTCCCTATGCCTT 2070 CCCG CCCAAGT TCCAAGCAT SCNN1A.2 NM_001038 NM_001038.4 SCNN1A 3263 ATCAACATCCTGTC 607 GAAGTTGCCCAG 1339 AGAGACTCTGCCATC 2071 GAGGCT CGTGTC CCTGGAGGA SDHA.1 NM_004168 NM_004168.1 SDHA 5443 GCAGAACTGAAGAT 608 CCCTTTCCAAAC 1340

CTGTCCACCAAATGC 2072 GGGAAGAT TTGAGGC ACGCTGATA SDPR.1 NM_004657 NM_004657.4 SDPR 3877 ACCAGCACAAGATG 609 GGTCATTCTGGA 1341 CGGAGCCCTCCAAA 2073 GAGCA TGCCCTT CTGATCTGTC SELE.1 NM_000450 NM_000450.1 SELE 5383 ACACTGGTCTGGC 610 AAAGTCCAGCTA 1342 CCTGTGAAGCTCCCA 2074 CTGCTAC CCAAGGGAA CTGAGTCCA SELENBP1.1 NM_003944 NM_003944.2 SELENBP1 6200 GGTACCAGCCTCG 611 CCATCTCGTAAG 1343 TGCCCACTCAGTGCT 2075 ACACAA ACATTGGGA GATCATGAC SELL.1 NM_000655 NM_000655.3 SELL 5483 TGCAACTGTGATGT 612 CCTCCAAAGGCT 1344 CACAAACTGACACTG 2076 GGGG CACACTG GGGCCCATA SELPG.1 NM_003006 NM_003006.3 SELPG 6316 TGGCCACTATCTTC 613 CTAATTACGCAC 1345 CACTGTGGTGCTGG 2077 TTCGTG GGGGTACA CGGTCC SEMA3B.1 NM_004636 NM_004636.1 SEMA3B 1013 GCTCCAGGATGTG 614 ACGTGGAGAAGA 1346 TCGCGGGACCACCG 2078 TTTCTGTTG CGGCATAGA GACC SEMA3C.1 NM_006379 NM_006379.2 SEMA3C 5409 ATGGCCATTCCTGT 615 GTCTCACATCTT 1347 CCTCCGTTTCCCAGT 2079 TCCAG GTCTTCGGC TGGGTAGAA SEMA3F.3 NM_004186 NM_004186.1 SEMA3F 1008 CGCGAGCCCCTCA 616 CACTCGCCGTTG 1348 CTCCCCACAGCGCA 2080 TTATACA ACATCCT TCGAGGAA SEMA5B.1 NM_ NM_ SEMA5B 5003 CTCGAGGACAGCT 617 TCACATTCCGCA 1349 AGCCTCTGGACCCA 2081 001031702 001031702 CCAACAT CAGGAC GAACATCACC SERPINA5.1 NM_000624 NM_000624.3 SERPINA5 5201 CAGCATGGTAGTG 618 CTTTGTGGCACT 1350 AGGTCCAGAGTCCT 2082 GCAAAGA GAGCTGG GGCCCTTGAT SFN.1 NM_006142 NM_006142.3 SFN 3580 GAGAGAGCCAGTC 619 AGGCTGCCATGT 1351 CTGCTCTGCCAGCTT 2083 TGATCCA CCTCATA GGCCTTC SGK.1 NM_005627 NM_005627.2 SGK 2960 TCCGCAAGACACCT 620 TGAAGTCATCCT 1352 TGTCCTGTCCTTCTG 2084 CCTG TGGCCC CAGGAGGC SHANK3.1 XM_037493 XM_037493.5 3887 CTGTGCCCTCTACA 621 GGACATCCCTGT 1353 AGCTGTGCTCGTGT 2085 ACCAGG TAGCTCCA CCTGCTCTTC SHC1.1 NM_003029 NM_003029.3 SHC1 2342 CCAACACCTTCTTG 622 CTGTTATCCCAA 1354 CCTGTGTTCTTGCTG 2086 GCTTCT CCCAAACC AGCACCCTC SILV.1 NM_006928 NM_006928.3 SILV 4113 CCGCATCTTCTGCT 623 ACTCAGACCTGC 1355 TTGGTGAGAATAGCC 2087 CTTGT TGCCCA CCCTCCTCA SKIL.1 NM_005414 NM_005414.2 SKIL 5272 AGAGGCTGAATATG 624 CTATCGGCCTCA 1356 CCAATCTCTGCCTCA 2088 CAGGACA GCATGG GTTCTGCCA SLC13A3.1 NM_022829 NM_022829.4 SLC13A3 6202 CTTGCCCTCCAACA 625 AGCCCACTGAGG 1357 CCCCCAGTACTTCCT 2089 AGGTC AAGAGGA CGACACCAA SLC16A3.1 NM_004207 NM_004207.1 SLC16A3 4569 ATGCGACCCACGT 626 AATCAGGGAGGA 1358 CCCCGCCAGGATGA 2090 CTACAT GGTGAGC ACACGTAC SLC22A3.1 NM_021977 NM_021977.2 SLC22A3 6506 ATCGTCAGCGAGTT 627 CAGGATGGCTTG 1359 CAGCATCCACGCATT 2091 TGACCT GGTGAG GACACAGAC SLC22A6.1 NM_153277 NM_153277.1 SLC22A6 6463 TCCGCCACCTCTTC 628 GACCAGCCCATA 1360 CTCTCCATGCTGTGG 2092 CTCT GTATGCAAAG TTTGCCACT SLC2A1.1 NM_006516 NM_006516.1 SLC2A1 2966 GCCTGAGTCTCCT 629 AGTCTCCACCCT 1361 ACATCCCAGGCTTCA 2093 GTGCC CAGGCAT CCCTGAATG SLC34A1.1 NM_003052 NM_003052.3 SLC34A1 6203 GCTGAGACCCACT 630 AGCCTCTCTCCG 1362 TCCTGGGCACCCAC 2094 GACCTG TAGGACAA TATGAGGTCT SLC7A5.2 NM_003486 NM_003486.4 SLC7A5 3268 GCGCAGAGGCCAG 631 AGCTGAGCTGTG 1363 AGATCACCTCCTCGA 2095 TTAAA GGTTGC ACCCACTCC SLC9A1.1 NM_003047 NM_003047.2 SLC9A1 5385 CTTCGAGATCTCCC 632 AGTGGGGATCAC 1364 CCTTCTGGCCTGCCT 2096 TCTGGA ATGGAAAC CATGAAGAT SLIT2.2 NM_004787 NM_004787.1 SLIT2 3708 TTTACCGATGCACC 633 TGCAGGCATGA 1365 CACAGTCCTGCCCCT 2097 TGTCC ATTGGG TGAAACCAT SNAI1.1 NM_005985 NM_005985.2 SNAI1 2205 CCCAATCGGAAGC 634 GTAGGGCTGCTG 1366 TCTGGATTAGAGTCC 2098 CTAACTA GAAGGTAA TGCAGCTCGC SNRK.1 NM_017719 NM_017719.4 SNRK 6710 GAGGAAAAGTCAG 635 GCCGGCTTTCAG 1367 CCAGCTGCAGTAGTT 2099 GGCCG AATCATC CGGAGACCA SOD1.1 NM_000454 NM_000454.3 SOD1 2742 TGAAGAGAGGCAT 636 AATAGACACATC 1368 TTTGTCAGCAGTCAC 2100 GTTGGAG GGCCACAC ATTGCCCAA SP3.1 NM_00101731 NM_00101731 SP3.1 5430 TCAAGAGTCTCAGC 637 CCATGGATTGTC 1369 CAGTCAAGCCCAAAT 2101 AGCCAA TGTGGTGT TGTGCAAGG SPARC.1 NM_003118 NM_003118.1 SPARC 2378 TCTTCCCTGTACAC 638 AGCTCGGTGTGG 1370 TGGACCAGCACCCC 2102 TGGCAGTTC GAGAGGTA ATTGACGG SPARCL1.1 NM_004684 NM_004684.2 SPARCL1 3904 GGCACAGTGCAAG 639 GATTGAGCTCTC 1371 ACTTCATCCCAAGCC 2103 TGATGA TCGGCCT AGGCCTTTC SPAST.1 NM_014946 NM_014946.3 SPAST 4033 CCTGAGTTGTTCAC 640 ATTCCCAGGTGG 1372 TAACAGCCCTCTGGC 2104 AGGGC ACCAAAG AGGAGCTCT SPHK1.1 NM_021972 NM_021972.2 SPHK1 4178 GGCAGCTTCCTTGA 641 GCAGTTGGTCAG 1373 TGGTGACCTGCTCAT 2105 ACCAT GAGGTCTT AGCCAGCAT SPRY1.1

AK026960 AK026960.1 SPRY1 1051 CAGACCAGTCCCT 642 CCTTCAAGTCAT 1374 CTGGGTCCGGATTG 2106 GGTCATAGG CCACAATCAGTT CCCTTTCAG SQSTM1.1 NM_003900 NM_003900.3 SQSTM1 4662 GGACCCGTCTACA 643 GGGTCCAGAGAG 1375 CAGTCCCTACAGATG 2107 GGTGAAC CTTGGC CCAGAATCCG STAT1.3 NM_007315 NM_007315.1 STAT1 530 GGGCTCAGCTTTCA 644 ACATGTTCAGCT 1376 TGGCAGTTTTCTTCT 2108 GAAGTG GGTCCACA GTCACCAAAA STAT3.1 NM_003150 NM_003150.1 STAT3 537 TCACATGCCACTTT 645 CTTGCAGGAAGC 1377 TCCTGGGAGAGATT 2109 GGTGTT GGCTATAC GACCAGCA STAT5A.1 NM_003152 NM_003152.1 STAT5A 403 GAGGCGCTCAACA 646 GCCAGGAACACG 1378 CGGTTGCTCTGCACT 2110 TGAAATTC AGGTTCTC TCGGCCT STAT5B.2 NM_012448 NM_012448.1 STAT5B 857 CCAGTGGTGGTGA 647 GCAAAAGCATTG 1379 CAGCCAGGACAACA 2111 TCGTTCA TCCCAGAGA ATGCGACGG STC2.1 NM_003714 NM_003714.2 STC2 6507 AAGGAGGCCATCA 648 AGATGGAGCACA 1380 TTCTGCTCACACTGA 2112 CCCAC GGCTTCC ACCTGCACG STK11.1 NM_000455 NM_000455.3 STK11 3383 GGACTCGGAGACG 649 GGGATCCTTCGC 1381 TTCTTGAGGATCTTG 2113 CTGTG AACTTCTT ACGGCCCTC STK15.2 NM_003600 NM_003600.1 AURKA 341 CATCTTCCAGGAG 650 TCCGACCTTCAA 1382 CTCTGTGGCACCCT 2114 GACCACT TCATTTCA GGACTACCTG STK4.1 NM_006282 NM_006282.2 STK4 5424 GAGCCATCTTCCTG 651 CTGAGGTGCAAC 1383 ACCTCTTTCCCTCAG 2115 CAACTT CCAGTCA ATGGGGAGC STMY3.3 NM_005940 NM_005940.2 MMP11 741 CCTGGAGGCTGCA 652 TACAATGGCTTT 1384 ATCCTCCTGAAGCCC 2116 ACATACC GGAGGATAGCA TTTTCGCAGC SUCLG1.1 NM_003849 NM_003849.2 SUCLG1 6205 CCAAGCCTGTAGT 653 CGGCATGACCCA 1385 CCCAGGAGGAGCAG 2117 GTCCTTCA TTCTTC TTAAACCAGC SULT1C2.1 NM_001056 NM_001056.3 SULT1C2 6206 GGGACCCAAAGCA 654 AGCACTGTTTCA 1386 TTCCCATGAACTGCA 2118 TGAAAT TCCACCTTC TCACCTTCC SURV.2 NM_001168 NM_001168.1 BIRC5 81 TGTTTTGATTCCCG 655 CAAAGCTGTCAG 1387 TGCCTTCTTCCTCCC 2119 GGCTTA CTCTAGCAAAAG TCACTTCTCACCT TACSTD2.1 NM_002353 NM_002353.2 TACSTD2 6335 ATCACCAACCGGA 656 AAGCTCGGTTCC 1388 CCCCCAGTTCCTTGA 2120 GAAAGTC TTTCTCAA TCTCCACC TAGLN.1 NM_003186 NM_003186.3 TAGLN 6073 GATGGAGCAGGTG 657 AGTCTGGAACAT 1389 CCCATAGTCCTCAGC 2121 GCTCAGT GTCAGTCTTGAT CGCCTTCAG G TAP1.1 NM_000593 NM_000593.5 TAP1 3966 GTATGCTGCTGAAA 658 TCCCACTGCTTA 1390 CACCAGCTGCCCAC 2122 GTGGGAA CAGCCC CAATGTAGAG TCF4.1 NM_003199 NM_003199.1 TCF4 4097 CACACCCTGGAATG 659 ATGTGGCAACTT 1391 CGCATCGAATCACAT 2123 GGAG GGACCCT GGGACAGAT TCOF1.2 NM_0010086 NM_001008656 TCOF1 6719 AGCGAGGATGAGG 660 CACCACATTGGT 1392 TCCCCGCTACACAGT 2124 ACGTG TCTGATGC GCTTGACTC TEK.1 NM_000459 NM_000459.1 TEK 2345 ACTTCGGTGCTACT 661 CCTGGGCCTTGG 1393 AGCTCGGACCACGT 2125 TAACAACTTACATC TGTTGAC CTGCTCCCTG TERT.1 NM_003219 NM_003219.1 992 GACATGGAGAACAA 662 GAGGTGTCACCA 1394 ACCAAACGCAGGAG 2126 GCTGTTTGC ACAAGAAATCAT CCAGCCCG TFAP2B.1 NM_003221 NM_003221.3 TFAP2B 6207 CGTGTGACGTGCG 663 CCACACGCTCTC 1395 ATGGACGCGCCTTG 2127 AGAGA AGGACC CTCTTACTGT TFAP2C.1 NM_003222 NM_003222.3 TFAP2C 4663 CATGCCTCACCAGA 664 CTGTCTGATCGT 1396 CTGGTCGTCGACATT 2128 TGGA GCAGCAAC CTGCACCTC TFPI.1 NM_006287 NM_006287.4 TFPI 6270 CCGAATGGTTTCCA 665 TTGCGGAGTCAG 1397 ATGGAACCCAGCTCA 2129 GGTG GGAGTTA ATGCTGTGA TGFA.2 NM_003236 NM_003236.1 TGFA 161 GGTGTGCCACAGA 666 ACGGAGTTCTTG 1398 TTGGCCTGTAATCAC 2130 CCTTCCT ACAGAGTTTTGA CTGTGCAGCCTT TGFb1.1 NM_000660 NM_000660.3 TGFB1 4041 CTGTATTTAAGGAC 667 TGACACAGAGAT 1399 TCTCTCCATCTTTAA 2131 ACCCGTGC CCGCAGTC TGGGGCCCC TGFB2.2 NM_003238 NM_003238.1 TGFB2 2017 ACCAGTCCCCCAG 668 CCTGGTGCTGTT 1400 TCCTGAGCCCGGG 2132 AAGACTA GTAGATGG AAGTCCC TGFBI.1 NM_000358 NM_000358.1 TGFBI 3768 GCTACGAGTGCTG 669 AGTGGTAGGGCT 1401 CCTTCTCCCCAGGG 2133 TCCTGG GCTGGAC ACCTTTTCAT TGFBR1.1 NM_004612 NM_004612.1 TGFBR1 3385 GTCATCACCTGGC 670 GCAGACGAAGCA 1402 AGCAATGACAGCTG 2134 CTTGG CACTGGT CCAGTTCCAC TGFBR2.3 NM_003242 NM_003242.2 TGFBR2 864 AACACCAATGGGTT 671 CCTCTTCATCAG 1403 TTCTGGGCTCCTGAT 2135 CCATCT GCCAAACT TGCTCAAGC THBD.1 NM_000361 NM_000361.2 THBD 4050 AGATCTGCGACGG 672 GGAAATGACATC 1404 CACCTAATGACAGTG 2136 ACTGC GGCAGC CGCTCCTCG THBS1.1 NM_0003246 NM_0003246.1 THBS1 2348 CATCCGCAAAGTGA 673 GTACTGAACTCC 1405 CCAATGAGCTGAGG 2137 CTGAAGAG GTTGTGATAGCA CGGCCTCC TAG TIMP1.1 NM_003254 NM_003254.2 TIMP1 6075 TCCCTGCGGTCCC 674 GTGGGAACAGG 1406 ATCCTGCCCGGAGT 2138 AGATAG GTGGACACT

GGAAGCTGAAGC TIMP2.1 NM_003255 NM_003255.2 TIMP2 606 TCACCCTCTGTGAC 675 TGTGGTTCAGGC 1407 CCCTGGGACACCCT 2139 TTCATCGT TCTTCTTCTG GAGCACCA TIMP3.3 NM_000362 NM_000362.2 TIMP3 593 CTACCTGCCTTGCT 676 ACCGAAATTGGA 1408 CCAAGAACGAGTGT 2140 TTGTGA GAGCATGT CTCTGGACCG TKI1.2 NM_003258 NM_003258.1 TKI1 264 GCCGGGAAGACCG 677 CAGCGGCACCAG 1409 CAAATGGCTTCCTCT 2141 TAATTGT GTTCAG GGAAGGTCCCA TLR3.1 NM_003265 NM_003265.2 TLR3 6289 GGTTGGGCCACCT 678 CCATTCCTGGCC 1410 CTTGCCCAATTTCAT 2142 AGAAGT TGTGAG TAAGGCCCA TMEM27.1 NM_020665 NM_020665.2 TMEM27 3878 CCCTGAAAGAATGT 679 TCTGCACCTGGT 1411 TCTGGTGACTGCCAT 2143 TGTGGC TGACAGAG TCATGCTGA TMEM47.1 NM_031442 NM_031442.3 TMEM47 6713 GGATTCCACTGTTA 680 GCAAATAACCAA 1412 CCGCCTGCTTATCCT 2144 GAGCCCTT CAGCCAATG ACCCAATGA TMSB10.1 NM_021103 NM_021103. TMSB10 6076 GAAATCGCCAGCTT 681 CTCGGCAGGGT 1413 CGTCTCCGTTTTCTT 2145 CGATAA GTTCTTCT CAGCTTGGC TNF.1 NM_000594 NM_000594.1 TNF 2852 GGAGAAGGGTGAC 682 TGCCCAGACTCG 1414 CGCTGAGATCAATCG 2146 CGACTCA GCAAAG GCCCGACTA TNFAIP3.1 NM_006290 NM_006290.2 TNFAIP3 6290 ATCGTCTTGGCTGA 683 GTGGAATGGCTC 1415 CAACCCACGCGACTT 2147 GAAAGG TGGCTTC GTGTGTCTT TNFAIP6.1 NM_007115 NM_007115.2 TNFAIP6 6291 AGGAGTGAAAGAT 684 CTGTAAAGACGC 1416 ATTGCTACAACCCAC 2148 GGGATGC CACCACAC ACGCAAAGG TNFRSF10C.3 NM_003841 NM_003841.2 TNFRSF10C 6652 GGAGTTTGACCAG 685 CTCTGTCCCCAG 1417 AACGGTAGGAAGCG 2149 AGATGCAA AGTTCCC CTCCTTCACC TNFRSF10D.1 NM_003840 NM_003840.3 TNFRSF10D 4406 CCTCTCGCTTCTGG 686 GCTCAGGAATCT 1418 AGGCATCCCAGGGA 2150 TGGTC CTGCCCTA CTCAGTTCAC TNFRSF11B.1 NM_002546 NM_002546.2 TNFRSF11B 4675 TGGCGACCAAGAC 687 GGGAAAGTGGTA 1419 AGGGCCTAATGCAC 2151 ACCTT CGTCTTTGAG GCACTAAAGC TNFRSF1A.1 NM_001065 NM_001065.2 TNFRSF1A 4943 ACTGCCCTGAGCC 688 GTCAGGCACGGT 1420 TGCCAGACAGCTATG 2152 CAAAT GGAGAG GCCTCTCAC TNFSF12.1 NM_003809 NM_003809.2 TNFSF12 2987 TAGGCCAGGAGTT 689 CACAGGGAATTC 1421 TTGTCTTGTTTCTCG 2153 CCCAA TCAAGGGA CCCCTCACA TNFSF13B.1 NM_006573 NM_006573.3 TNFSF13B 4944 CCTACGCCATGGG 690 TCGAAACAAAGT 1422 TCCCCAAAGACATGG 2154 ACATC CACCAGACTC ACCTTCTTCC TNFSF7.1 NM_001252 NM_001252.2 CD70 4101 CCAACCTCACTGG 691 ACCCACTGCACT 1423 TGCCTTCCCGAAACA 2155 GACACTT CCAAAGAA CTGATGAGA TNIP2.1 NM_024309 NM_024309.2 TNIP2 3872 CATGTCAGAAAGG 692 GCGACCTTTTCC 1424 AATCCTACTTTGAGC 2156 GCCCGA TCCAGTT CCGTTCCCG TOP2A.4 NM_001067 NM_001067.1 TOP2A 74 AATCCAAGGGGGA 693 GTACAGATTTTG 1425 CATATGGACTTTGAC 2157 GAGTGAT CCCGAGGA TCAGCTGTGGC TOP2B.2 NM_001068 NM_001068.1 TOP2B 75 TGTGGACATCTTCC 694 CTAGCCCGACCG 1426 TTCCCTACTGAGCCA 2158 CCTCAGA GTTCGT CCTTCTCTG TP.3 NM_001953 NM_001953.2 TYMP 91 CTATATGCAGCCAG 695 CCACGAGTTTCT 1427 ACAGCCTGCCACTCA 2159 AGATGTGACA TACTGAGAATGG TCACAGCC TRAIL.1 NM_003810 NM_003810.1 TNFSF10 898 CTTCACAGTGCTCC 696 CATCTGCTTCAG 1428 AAGTACACGTAAGTT 2160 TGCAGTCT CTCGTTGGT ACAGCCACACA TS.1 NM_001071 NM_001071.1 TYMS 76 GCCTCGGTGTGCC 697 CGTGATGTGCGC 1429 CATCGCCAGCTACG 2161 TTTCA AATCATG CCCTGCTC TSC1.1 NM_000368 NM_000368.3 TSC1 6292 TCACCAAATCTCAG 698 GTGTCAGCATAA 1430 TTTCCTCATCGTTCA 2162 CCCG GGGCTGGT GCCGATGTC TSC2.1 NM_000548 NM_000548 TSC2 5132 CACAGTGGCCTCTT 699 CAGGAAACGCTC 1431 TACCAGTCCAGCTGC 2163 TCTCCT CTGTGC CAAGGACAG TSPAN7.2 NM_004615 NM_004615.3 TSPAN7 6721 ATCACTGGGGTGAT 700 GGGAGATATAGG 1432 AAGTTTGCCCCAGAC 2164 CCTGC TGCCCAGAG TCCAACAGC TSPAN8.1 NM_004616 NM_004616.2 TSPAN8 6317 CAGAAATCTCTGCA 701 AATCCAGATGCC 1433 TGCTCCAGAGCATAT 2165 GGCAAGT GTGAATTT TGCAGGACA TUBB.1 NM_001069 NM_001069.1 TUBB2A 2094 CGAGGACGAGGCT 702 ACCATGCTTGAG 1434 TCTCAGATCAATCGT 2166 TAAAAAC GACAACAG GCATCCTTAGTGAA TUSC2.1 NM_007275 NM_007275.1 TUSC2 6208 CACCAAGAACGGG 703 CGATGCCCTGAG 1435 TCTTATGCACTCGCC 2167 CAGAA GAATCA TCAGCTTGG tusc4.2 NM_006545 NM_006545.4 TUSC4 3764 GGAGGAGCTAAAT 704 CCTTCAAGTGGA 1436 ACTCATCAATGGGCA 2168 GCCTCAG TGGTGTTG GAGTGCACC TXLNA.1 NM_175852 NM_175852.3 TXLNA 6209 GCCAGAACGGCTC 705 ATGTCTTCCAGT 1437 TCCTCAGAGACATCA 2169 AGTCT TGGCGG CGAAGGGCC UBB.1 NM_018955 NM_018955.1 UBB 3303 GAGTCGACCCTGC 706 GCGAATGCCATG 1438 AATTAACAGCCACCC 2170 ACCTG ACTGAA CTCAGGCG UBE1C.1 NM_003968 NM_003968.3 UBA3 2575 GAATGCACGCTGG 707 CTGGTAGCCTGG 1439 AATTTTCCCATGTGC 2171 AACTTTA

GCATAGA ACCATTGCA UBE2C.1 NM_007019 NM_007019.2 UBE2C 2550 TGTCTGGCGATAAA 708 ATGGTCCCTACC 1440 TCTGCCTTCCCTGAA 2172 GGGATT CATTTGAA TCAGACAACC UBE2T.1 NM_014176 NM_014176.1 UBE2T 3882 TGTTCTCAAATTGC 709 AGAGGTCAACAC 1441 AGGTGCTTGGAGAC 2173 CACCAA AGTTGCGA CATCCCTCAA UGCG.1 NM_003358 NM_003358.1 UGCG 6210 GGCAACTGACAAAC 710 AGGATCTACCCC 1442 CAAGCTCCCAGGTG 2174 AGCCTT TTTCAGTGG TCTCTCTTCTGA UMOD.1 NM_003361 NM_003361.2 UMOD 6211 GCGTGGACCTGGA 711 TTACGCAGCTGC 1443 CCATTCCTGGAGCTC 2175 TGAGT TGTTGG ACAACTGCT upa.3 NM_002658 NM_002658.1 PLAU 89 GTGGATGTGCCCT 712 CTGCGGATCCAG 1444 AAGCCAGGCGTCTA 2176 GAAGGA GGTAAGAA CACGAGAGTCTCAC USP34.1 NM_014709 NM_014709.2 USP34 4040 AAGCTGTGATGGC 713 GGAATGGCCACA 1445 TCCCAGGACCCTGA 2177 CAAGC ACTGAGA GGTTGCTTTA VCAM1.1 NM_001078 NM_001078.2 VCAM1 1220 TGGCTTCAGGAGC 714 TGCTGTCGTGAT 1446 CAGGCACACACAGG 2178 TGAATACC GAGAAAATAGTG TGGGACACAAAT VCAN.1 NM_004385 NM_004385.2 VCAN 5979 CCTGCTACACAGCC 715 AGAAAGCGCCTG 1447 CCCACTGTGGAAGA 2179 AACAAG AGGTCC CAAAGAGGCC VDR.2 NM_000376 NM_000376.1 VDR 971 GCCCTGGATTTCAG 716 AGTTACAAGCCA 1448 CAAGTCTGGATCTG 2180 AAAGAG GGGAAGGA GGACCCTTTCC VEGF.1 NM_003376 NM_003376.3 VEGFA 7 CTGCTGTCTTGGGT 717 GCAGCCTGGGAC 1449 TTGCCTTGCTGCTCT 2181 GCATTG CACTTG ACCTCCACCA VEGFB.1 NM_003377 NM_003377.2 VEGFB 964 TGACGATGGCCTG 718 GGTACCGGATCA 1450 CTGGGCAGCACCAA 2182 GAGTGT TGAGGATCTG GTCCGGA VHL.1 NM_000551 NM_000551.2 VHL 4102 CGGTTGGTGACTT 719 AAGACTTGTCCC 1451 ATGCCTCAGTCTTCC 2183 GTCTGC TGCCTCAC CAAAGCAGG VIM.3 NM_003380 NM_003380.1 VIM 339 TGCCCTTAAAGGAA 720 GCTTCAACGGCA 1452 ATTTCACGCATCTGG 2184 CCAATGA AAGTTCTCTT CGTTCCA VTCN1.1 NM_024626 NM_024626.2 VTCN1 4754 ACAGTGGTCTGGG 721 GCTCAAAGCTGG 1453 CGAGAAGTTGGCTC 2185 CATCC TATTGGAGAC CCTGGTCAAC VTN.1 NM_000638 NM_000638.2 VTN 4502 AGTCAATCTTCGCA 722 GTACTGAGCGAT 1454 TGGACACTGTGGAC 2186 CACGG GGAGCGT CCTCCCTACC VWF.1 NM_000552 NM_000552.3 VWF 6212 TGAAGCACAGTGC 723 CCAGTCTCCCAT 1455 CTCCATGTCACTGTG 2187 CCTCTC TCACCGT CAGCTCGAC WIF.1 NM_007191 NM_007191.3 WIF 6077 AACAAGCTGAGTGC 724 CACTCGCAGATG 1456 TACAAAAGCCTCCAT 2188 CCAGG CGTCTTT TTCGGCACC WISP1.1 NM_003882 NM_003882.2 WISP1 603 AGAGGCATCCATGA 725 CAAACTCCACAG 1457 CGGGCTGCATCAGC 2189 ACTTCACA TACTTGGGTTGA ACACGC WT1.1 NM_000378 NM_000378.3 WT1 6458 TGTACGGTCGGCA 726 TTATTGCAGCCT 1458 CAGTGAGAAACGCC 2190 TCTGAG GGGTAAGC CCTTCATGTG WWOX.5 NM_016373 NM_016373.1 WWOX 974 ATCGCAGCTGGTG 727 AGCTCCCTGTTG 1459 CTGCTGTTTACCTTG 2191 GGTGTAC CATGGACTT GCGAGGCCTTTC XDH.1 NM_000379 NM_000379.3 XDH 5089 TGGTGGCAGACAT 728 GCCACAACTGTC 1460 TGAAGCCAACCTTGT 2192 CCCTT CCAGTCTT ATCTGGCCA XIAP.1 NM_001167 NM_001167.1 XIAP 80 GCAGTTGGAAGAC 729 TGCGTGGCACTA 1461 TCCCCAAATTGCAGA 2193 ACAGGAAAGT TTTTCAAGA TTTATCAACGGC XPNPEP2.2 NM_003399 NM_003399.5 XPNPEP2 6503 CACCCTGCACTGAA 730 AAGGAGGATGAA 1462 CCTGCTGGCCCATT 2194 CATACC TGCAAAGG GCCTAGAA YB-1.2 NM_004559 NM_004559.1 YBX1 395 AGACTGTGGAGTTT 731 GGAACACCACCA 1463 TTGCTGCCTCCGCA 2195 GATGTTGTTGA GGACCTGTAA CCCTTTTCT ZHX.1 NM_014943 NM_014943.3 ZHX2 6215 GAGTACGACCAGTT 732 TCTCCTTGAACC 1464 ATCTCAGTTCGGACC 2196 AGCGGC AACGCAC AGGCCAGTC

(171) TABLE-US-00017 TABLE B Target Sequence SEQ ID Gene Length Amplicon Sequence NO. A-Catenin.2 78 CGTTCCGATCCTCTATACTGCATCCCAGGCATGCCTACAGCACCCTGATGTCGCAGCCTATAAGGCCAA 2197 CAGGGACCT A2M.1 66 CTCTCCCGCCTTCCTAGCTGTCCCAGTGGAGAAGGAACAAGCGCCTCACTGCATCTGTGCAAACGG 2198 AAMP.1 66 GTGTGGCAGGTGGACACTAAGGAGGAGGTCTGGTCCTTTGAAGCGGGAGACCTGGAGTGGATGGAG 2199 ABCB1.5 77 AAACACCACTGGAGCATTGACTACCAGGCTCGCCAATGATGCTGCTCAAGTTAAAGGGGCTATAGGTTC 2200 CAGGCTTG ACADSB.1 68 TGGCGGAGAACTAGCCATCAGCCTCCTGAAGCCTGCCATCATTGTTAATTTGAGGACTGGGCTGTCTT 2201 ACE.1 67 CCGCTGTACGAGGATTTCACTGCCCTCAGCAATGAAGCCTACAAGCAGGACGGCTTCACAGACACGG

2202 ACE2.1 66
TACAATGAGaGGCTCTGGGCTTGGGAAAGCTGGAGATGTGAGGTCGGCAAGCAGCTGAGGCCATTA
2203 ADAM17.1 73
GAAGTGCCAGGAGGCGATTAATGCTACTTGCAAAGGCGTGTCCTACTGCACAGGTAATAGCAGTGAGTG
2204 CCCG ADAM8.1 67
GTCACTGTGTCCAGCCCACCCTTCCCAGTTCCTGTCTACACCCGGCAGGCACCAAAGCAGGTCATCA
2205 ADAMTS1.1 73
GGACAGGTGCAAGCTCATCTGCCAAGCCAAAGGCATTGGCTACTTCTTCGTTTTGCAGCCCAAGGTTGT
2206 AGAT ADAMTS4.1 66
GAGAATGTCTGCCGCTGGGCCTACCTCCAGCAGAAGCCAGACACGGGCCACGATGAATACCACGAT
2207 ADAMTS4.1 71
TTTGACAAGTGCATGGTGTGCGGAGGGGACGGTTCTGGTTGCAGCAAGCAGTCAGGCTCCTTCAGGAA
2208 ATT ADAMTS5.1 79
CACTGTGGCTCACGAAATCGGACATTTACTTGGCCTCTCCCATGACGATTCCAAATTCTGTGAAGAGACC
2209 TTTGGTTGC ADAMTS8.1 72
GCGAGTTCAAAGTGTTCGAGGCCAAGGTGATTGATGGCACCCTGTGTGGGCCAGAAACACTGGCCATCT
2210 GTG ADAMTS9.1 66
GCACAGGTTACACAACCCAACAGAATGTCCCTATAACGGGAGCCGGCGCGATGACTGCCAATGTCG
2211 ADD1.1 74
GTCTACCCAGCAGCTCCGCAAGGAGGGATGGCTGCCTTAAACATGAGTCTTGGTATGGTGACTCCTGTG
2212 AACGA ADFP.1 67
AAGACCATCACCTCCGTGGCCATGACCAGTGCTCTGCCCATCATCCAGAAGCTAGAGCCGCAAATTG
2213 ADH1B.1 84
AAGCCAACAAACCTTCCTTCTTAACCATTCTACTGTGTCACCTTTGCCATTGAGGAAAAATATTCCTGTGA
2214 CTTCTTGCATTTT ADH6.1 68
TGTTGGGGAGTAAACACTTGGACCTCTTGTATCCCACCATCTTGGGCCATGAAGGGGCTGGAATCGTT
2215 ADM.1 75
TAAGCCACAAGCACACGGGGCTCCAGCCCCCCCGAGTGGAAGTGCTCCCCACTTTCTTTAGGATTTAGG
2216 CGCCCA AGR2.1 70
AGCCAACATGTGACTAATTGGAAGAAGAGCAAAGGGTGGTGACGTGTTGATGAGGCAGATGGAGATCAG
2217 A AGT.1 73
GATCCAGCCTCACTATGCCTCTGACCTGGACAAGGTGGAGGGTCTCACTTTCCAGCAAAACTCCCTCAA
2218 CTGG AGTR1.1 67
AGCATTGATCGATACCTGGCTATTGTTCACCCAATGAAGTCCCGCCTTCGACGCACAATGCTTGTAG
2219 AHR.1 69
GCGGCATAGAGACCGACTTAATACAGAGTTGGACCGTTTGGCTAGCCTGCTGCCTTTCCCACAAGATGT
2220 AIF1.1 71
GACGTTCAGCTACCCTGACTTTCTCAGGATGATGCTGGGCAAGAGATCTGCCATCCTAAAAATGATCCTG
2221 A AKT1.3 71
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2222 GGA AKT2.3 71
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2223 AKT3.2 75
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2225 ALDH6A1.1 66
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2229 ALOX5.1 66
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2231 TC ANGPT.1 71
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2236 ANGPTL7.1 67
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2243 official).2 ACCTTATGGCTA AP1M2.1 67
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2244 APAF1.2 66
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2245 APC.4 69
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2247 TG APOE.1 75
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2258 ASS1.1 85
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2384 AGATGAG CSF2RA.2 67
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2394 TC CUL1.1 71
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2406 G CYP3A4.2 79
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2416 DET1.1 70
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2417 GG DHPS.3 78
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2423 DLL4.1 67
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2424 DPEP1.1 72
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2588 IQGAP2.1 68
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2589 ISG20.1 70
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2591 AAGCTTC ITGA4.2 66
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2592 ITGA5.1 75
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2595 ATGGAAGTTCT ITGAV.1 79
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2601 K-ras.10 71
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2603 KDR.6 68
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2604 Ki-67.2 80
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2607 Kiting.4 79
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2609 TAG KLK3.1 66
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2619 LAMC2.2 80
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2913 GC VEGFB.1 71
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2916 AGC VTCN1.1 70
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Claims

1. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising: measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of: a set of genes consisting of one or more angiogenesis genes selected from: ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; and a set of genes consisting of one or more immune response genes selected from: CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1; and a set of genes consisting of one or more cell cycle genes selected from: BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; and optionally EIF4EBP1.
2. The method of claim 1, wherein the renal cancer is renal cell carcinoma (RCC).
3. The method of claim 2, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
4. The method of claim 1, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
5. The method of claim 1, wherein the renal tumor sample is paraffin-embedded and fixed.
6. The method of claim 1, wherein the panel consists of: at least two angiogenesis genes, at least two immune response genes, and at least two cell cycle genes, and optionally EIF4EBP1.
7. The method of claim 1, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.
8. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising: measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of: a set of genes consisting of one or more angiogenesis genes selected from: ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; and a set of genes consisting of one or more immune response genes selected from: CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1; and a set of genes consisting of one or more cell cycle genes selected from: BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; and a set of genes consisting of one or more transport genes selected from: AQP1, SGK1, and both AQP1 and SGK1; and a set of genes consisting of one or more cell adhesion/extracellular matrix genes selected from: ITGB1, A2M, ITGB5, LAMB1, LOX, MMP14, TGFBR2, TIMP3, and TSPAN7; and optionally EIF4EBP1.
9. The method of claim 8, wherein the renal cancer is renal cell carcinoma (RCC).
10. The method of claim 9, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
11. The method of claim 8, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
12. The method of claim 8, wherein the renal tumor sample is obtained from a biopsy.
13. The method of claim 8, wherein the renal tumor sample is paraffin-embedded and fixed.
14. The method of claim 8, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.
15. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising: measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of: at least five genes selected from the group consisting of: EMCN, PPAP2B, CCL5, CXCL9, NOS3, NUDT6, AQP1, APOLD1, ITGB5, CCR7, CX3CL1, CEACAM1, PRKCH, CASP10, SNRK, and PTPRB.
16. The method of claim 15, wherein the renal cancer is renal cell carcinoma (RCC).
17. The method of claim 16, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
18. The method of claim 15, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
19. The method of claim 15, wherein the renal tumor sample is paraffin-embedded and fixed.
20. The method of claim 15, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.